

R Notebook

```
library(latex2exp)
library(imageseg)
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
## Warning: package 'lattice' was built under R version 4.3.3
```

```
library(tools)
library(xfun)
```

```
## Warning: package 'xfun' was built under R version 4.3.3
```

```
##
## Attaching package: 'xfun'
```

```
## The following objects are masked from 'package:tools':
##
##   file_ext, Rcmd
```

```
## The following objects are masked from 'package:base':
##
##   attr, isFALSE
```

```
library(EBImage)
library(stringi)
```

```
## Warning: package 'stringi' was built under R version 4.3.3
```

```
library(ROCR)
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':  
##  
##   cov, smooth, var
```

```
library(reticulate)
```

```
## Warning: package 'reticulate' was built under R version 4.3.3
```

```
library(tensorflow)
```

```
## Warning: package 'tensorflow' was built under R version 4.3.3
```

```
##  
## Attaching package: 'tensorflow'
```

```
## The following object is masked from 'package:caret':  
##  
##   train
```

```
library(keras)
```

```
## Warning: package 'keras' was built under R version 4.3.3
```

```
##  
## Attaching package: 'keras'
```

```
## The following object is masked from 'package:EBImage':  
##  
##   normalize
```

```
library(imager)
```

```
## Warning: package 'imager' was built under R version 4.3.3
```

```
## Loading required package: magrittr
```

```
##  
## Attaching package: 'imager'
```

```
## The following object is masked from 'package:magrittr':  
##  
##   add
```

```
## The following object is masked from 'package:PROC':  
##  
##   ci
```

```
## The following objects are masked from 'package:EBImage':  
##  
##   channel, dilate, display, erode, resize, watershed
```

```
## The following objects are masked from 'package:stats':  
##  
##   convolve, spectrum
```

```
## The following object is masked from 'package:graphics':  
##  
##   frame
```

```
## The following object is masked from 'package:base':  
##  
##   save.image
```

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:imager':  
##  
##   where
```

```
## The following object is masked from 'package:EBImage':  
##  
##   combine
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)  
library(purrr)
```

```
##  
## Attaching package: 'purrr'
```

```
## The following object is masked from 'package:magrittr':  
##  
##   set_names
```

```
## The following object is masked from 'package:EBImage':  
##  
##   transpose
```

```
## The following object is masked from 'package:caret':  
##  
##   lift
```

```
library(skimr)  
library(tm)
```

```
## Warning: package 'tm' was built under R version 4.3.3
```

```
## Loading required package: NLP
```

```
## Warning: package 'NLP' was built under R version 4.3.3
```

```
##  
## Attaching package: 'NLP'
```

```
## The following object is masked from 'package:ggplot2':  
##  
##   annotate
```

```
library(RWeka)  
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.3.3
```

```
library(scales)
```

```
##  
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:purrr':  
##  
##   discard
```

```
library(cutpointr)
```

```
##
## Attaching package: 'cutpointr'
```

```
## The following objects are masked from 'package:PROC':
##
## auc, roc
```

```
## The following objects are masked from 'package:caret':
##
## precision, recall, sensitivity, specificity
```

```
library(magick)
```

```
## Warning: package 'magick' was built under R version 4.3.3
```

```
## Linking to ImageMagick 6.9.12.98
## Enabled features: cairo, freetype, fftw, ghostscript, heic, lcms, pango, raw, rsvg, webp
## Disabled features: fontconfig, x11
```

```
# Set the working directory
setwd("C:/Ngoc/MastersThesis/Draft")

main_folder <- "cell_images"
class_folders <- list.dirs(main_folder, full.names = FALSE) # -> class_folders[1] is an empty string
class_folders <- stri_remove_empty(class_folders, na_empty = TRUE)
```

A/ Find training and test set AUC for a built - from - scratch CNN model:

1. Delete folder 'cell_images/cell_images':

```
##### Turn to comments from the second run onwards #####
# unlink('cell_images/cell_images', recursive = TRUE)
```

Unlike some preprocessing steps in tabular data (like normalization / standardization, filling in missing values, ...), format conversion and resizing do not involve calculations and transformations based on data distributions that may affect the learning process of the model. These two steps are just used to ensure consistent input format and size for the model to work properly on the dataset. Therefore, instead of:

- Checking for any inconsistencies in the format of the images based on just the training set, then based on the result, convert all images from the training, validation and test set to the same format,
- As well as finding the range of image height and width from just the training set, then based on the result, resize all images to the same size,

We can:

- Check for any inconsistencies in the format of the images based on the whole dataset, then based on the result, convert all images from the training, validation and test set to the same format,

- As well as finding the range of image height and width from the whole dataset, then based on the result, resize all images to the same size.

For this approach, we only need to apply resizing and format conversion steps once at the beginning then save the processed images for future code runs, instead of having to re - run these steps every time we split the train - validation - test set differently. This is practical and helps simplify the coding process.

2. Check for any inconsistencies in the format of the images:

```
# ##### Turn to comments from the second run onwards #####
# num_gray = 0
# num_rgb = 0
# num_other = 0
#
# for (class_folder in class_folders){
#   directory <- file.path(main_folder, class_folder)
#   files <- list.files(directory, full.names = TRUE, pattern = "\\.*png$")
#   for (file in files){
#     tryCatch(
#       {
#         img <- image_read(file)
#         info <- image_info(img)
#         if (info$colorspace == "Gray"){
#           num_gray = num_gray + 1
#         }
#         else if (info$colorspace == "sRGB"){
#           num_rgb = num_rgb + 1
#         }
#         else {
#           num_other = num_other + 1
#           cat("Image format: ", info$colorspace, "\n", file, "\n\n")
#         }
#       },
#       error = function(e)
#       {
#         cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#   }
# }
#
# cat("Number of Grayscale images:", num_gray, "\n")
# cat("Number of RGB images:", num_rgb, "\n")
# cat("Number of other types of images:", num_other, "\n")
#
# # -> The dataset contains ... images in RGB format
```

3. Find the range of image height and width in the dataset:

```

# ##### Turn to comments from the second run onwards #####
# min_width <- Inf
# max_width <- 0
# min_height <- Inf
# max_height <- 0
# widths <- list()
# heights <- list()
#
# for (class_folder in class_folders){
#   directory <- file.path(main_folder, class_folder)
#   files <- list.files(directory, full.names = TRUE, pattern = "\\\\.png$")
#   for (file in files){
#     tryCatch(
#       {
#         img <- load.image(file)
#         width <- dim(img)[1]
#         height <- dim(img)[2]
#         widths <- c(widths, width)
#         heights <- c(heights, height)
#
#         if (width < min_width){
#           min_width <- width
#         }
#         if (width > max_width){
#           max_width <- width
#         }
#         if (height < min_height){
#           min_height <- height
#         }
#         if (height > max_height){
#           max_height <- height
#         }
#       },
#       error = function(e)
#       {
#         cat("An exception occurred:", e$message, "\n", file, "\n\n")
#       })
#     }
#   }
#
#   cat("The range of image width:", min_width, " - ", max_width, "pixels \n")
#   cat("The range of image height:", min_height, " - ", max_height, "pixels \n")

```

```

# # Distribution of image widths and heights
# hist(unlist(widths), main = "Distribution of Image Widths",
#       xlab = "Width (pixels)", ylab = "Frequency", col = "lightblue",
#       xlim = c(0, 400), ylim = c(0, 14000), border = "black")
#
# hist(unlist(heights), main = "Distribution of Image Heights",
#       xlab = "Height (pixels)", ylab = "Frequency", col = "lightblue",
#       xlim = c(0, 400), ylim = c(0, 14000), border = "black")

```

4. Resize all images into size 50 x 50 pixels:

```
##### Turn to comments from the second run onwards #####
# resize <- function(directory)
# {
#   files <- list.files(directory, full.names = TRUE)
#   for (file in files)
#   {
#     tryCatch(
#       {
#         img <- load.image(file)
#         img_resized <- EBImage::resize(img, 50, 50, filter = "bilinear")
#         save.image(img_resized, file)
#       },
#       error = function(e)
#       {
#         cat("An exception occurred:", e$message, "\n")
#       })
#   }
# }
#
# for (class_folder in class_folders) {
#   resize(file.path(main_folder, class_folder))
# }
```

5. Get the list of images and the list of labels:

```
# Get the list of image paths and the list of categorical labels
image_paths <- list()
labels <- list()
for (class_folder in class_folders) {
  images <- list.files(file.path(main_folder, class_folder), full.names = TRUE, pattern =
"\\.png$")
  image_paths <- c(image_paths, images)
  labels <- c(labels, rep(class_folder, length(images)))
}

# Load the images
X <- lapply(unlist(image_paths), load.image)

# Label encoding
y <- ifelse(unlist(labels) == "Parasitized", 1, 0)
```

6. Split training - test - validation sets:

```

set.seed(1)

# Shuffle X and labels in the same order to make sure the label values appear in a random order
shuffle_indices <- sample(1:length(X))
X <- X[shuffle_indices]
y <- y[shuffle_indices]

# Split train : test : val as 70 : 20 : 10
train_indices_temp <- createDataPartition(y, p = 0.8, list = FALSE)
test_indices <- setdiff(seq_along(y), train_indices_temp)
# -> seq_along(x) creates a sequence of integers with the same length as x

X_train_temp <- X[train_indices_temp]
y_train_temp <- y[train_indices_temp]
X_test <- X[test_indices]
y_test <- y[test_indices]

val_indices <- createDataPartition(y_train_temp, p = 0.125, list = FALSE)
train_indices <- setdiff(seq_along(y_train_temp), val_indices)

X_train <- X_train_temp[train_indices]
y_train <- y_train_temp[train_indices]
X_val <- X_train_temp[val_indices]
y_val <- y_train_temp[val_indices]
# (train: 0.875 x 0.8 = 0.7 ; val: 0.125 x 0.8 = 0.1)

# Examine the size of the training, validation, and test sets
cat("Train set's size: X_train = ", length(X_train), ", y_train = ", length(y_train), "\n")

```

```
## Train set's size: X_train = 19292 , y_train = 19292
```

```
cat("Validation set's size: X_val = ", length(X_val), ", y_val = ", length(y_val), "\n")
```

```
## Validation set's size: X_val = 2756 , y_val = 2756
```

```
cat("Test set's size: X_test = ", length(X_test), ", y_test = ", length(y_test), "\n")
```

```
## Test set's size: X_test = 5510 , y_test = 5510
```

```

# Check the pixel value range of training images
cat("Pixel value range of training images: [", min(unlist(X_train)), ",", max(unlist(X_train)), "] \n")

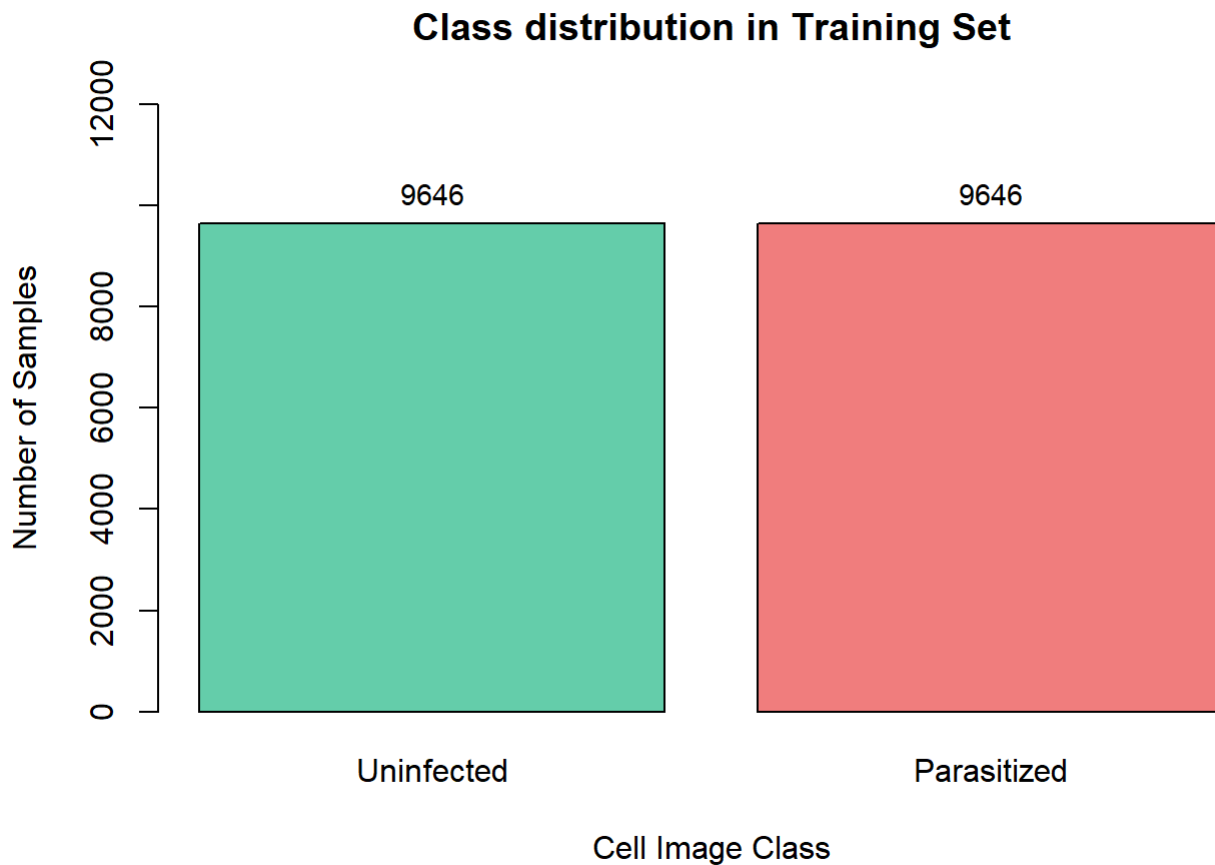
```

```
## Pixel value range of training images: [ 0 , 1 ]
```

```
# -> All training images are already normalized
```

7. Check for class distribution in training set:

```
count <- as.numeric(table(y_train))
custom_labels <- ifelse(names(table(y_train)) == "1", "Parasitized", "Uninfected")
bp <- barplot(table(y_train), names.arg = custom_labels, main = "Class distribution in Training Set", col = c("aquamarine3", "lightcoral"), ylim = c(0, 12000), xlab = "Cell Image Class", ylab = "Number of Samples")
text(x = bp, y = count, label = count, cex = 0.9, pos = 3, col = "black")
```



```
# -> There is no class imbalance
```

8. Apply other preprocessing steps and show some image samples:

All training images are already normalized, so there is no need to normalize the training, validation and test set.

```

## One - hot encode the labels (not needed when using sigmoid activation function in last FC layer in binary classification)
y_train_ <- keras::to_categorical(y_train)
y_val_ <- keras::to_categorical(y_val)
y_test_ <- keras::to_categorical(y_test)

# Reshape the images
xtr <- do.call(rbind, X_train)
X_train_ <- array_reshape(xtr, c(dim(xtr)[1], 50, 50, 3), order = "F")
xv <- do.call(rbind, X_val)
X_val_ <- array_reshape(xv, c(dim(xv)[1], 50, 50, 3), order = "F")
xte <- do.call(rbind, X_test)
X_test_ <- array_reshape(xte, c(dim(xte)[1], 50, 50, 3), order = "F")
# order = "C" means elements should be read in row-major order, with the last index changing fastest; "F" means elements should be read in column-major order, with the first index changing fastest.

# Show some samples in the dataset
labels_tr <- c()
labels_te <- c()
labels_tr <- ifelse(y_train == 1, "Parasitized", "Uninfected")
labels_te <- ifelse(y_test == 1, "Parasitized", "Uninfected")

# Display the 1080th image from the training set
plot(as.cimg(X_train_[1080,,]))

```

```

## Warning in as.cimg.array(X_train_[1080, , , ]): Assuming third dimension
## corresponds to colour

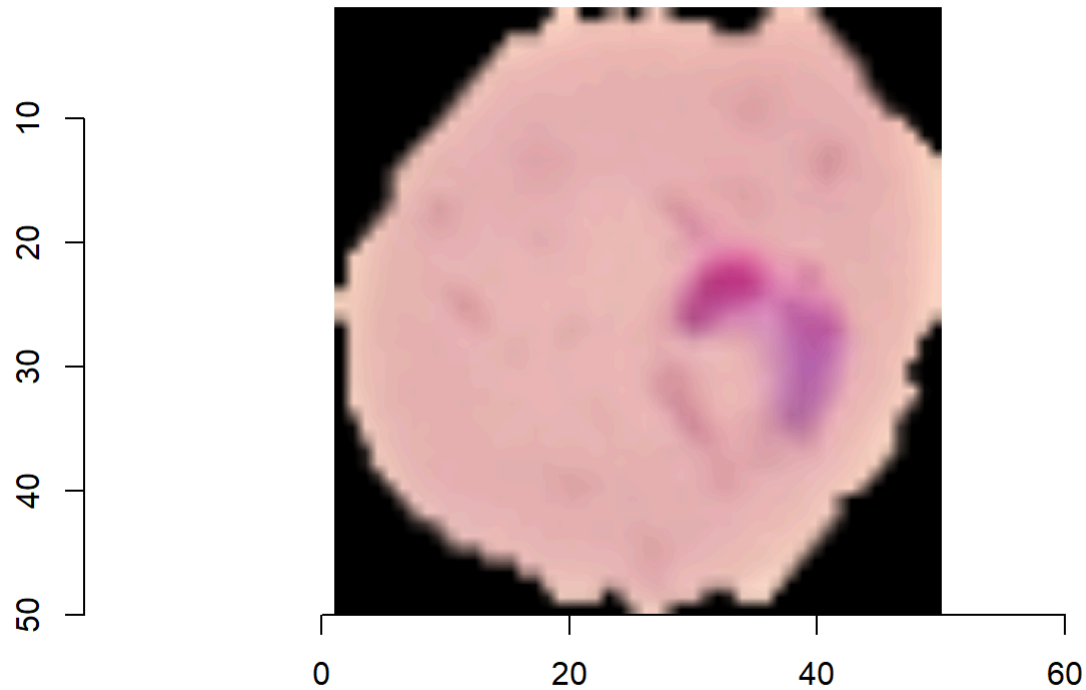
```

```

title(main = 'Sample from Training Set', sub = paste0("Class: ", labels_tr[1080]))

```

Sample from Training Set



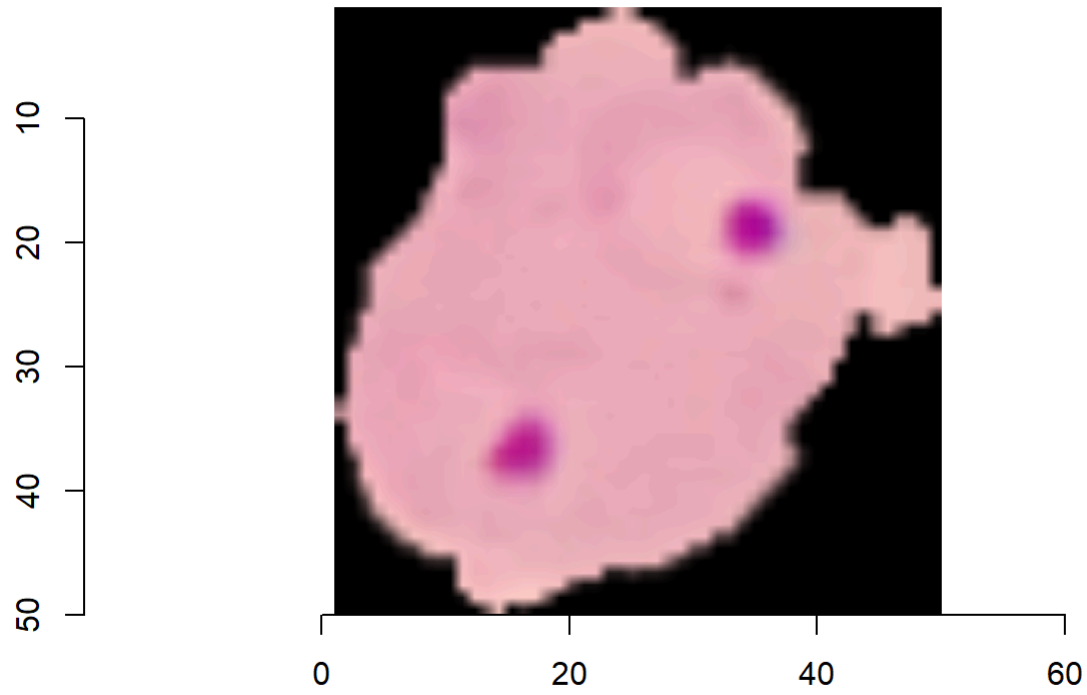
Class: Parasitized

```
# Display the 320th image from the test set  
plot(as.cimg(X_test_[320,,,]))
```

```
## Warning in as.cimg.array(X_test_[320, , , ]): Assuming third dimension  
## corresponds to colour
```

```
title(main = 'Sample from Test Set', sub = paste0("Class: ", labels_te[320]))
```

Sample from Test Set



Class: Parasitized

9. Build the CNN model:

```

# Initialize the Sequential model
my_model <- keras_model_sequential()

# Block 1
my_model %>%
  layer_conv_2d(filters = 16, input_shape = c(50, 50, 3), kernel_size = c(3, 3), strides = c
(1, 1), padding = 'same', name = 'Block_1_Conv_1') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_conv_2d(filters = 32, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', nam
e = 'Block_1_Conv_2') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bloc
k_1_MaxPool') %>%
  layer_dropout(rate = 0.2)

# Block 2
my_model %>%
  layer_conv_2d(filters = 64, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', nam
e = 'Block_2_Conv_1') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_conv_2d(filters = 128, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', na
me = 'Block_2_Conv_2') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bloc
k_2_MaxPool') %>%
  layer_dropout(rate = 0.2)

# Block 3
my_model %>%
  layer_conv_2d(filters = 256, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', na
me = 'Block_3_Conv_1') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_conv_2d(filters = 512, kernel_size = c(3, 3), strides = c(1, 1), padding = 'same', na
me = 'Block_3_Conv_2') %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_max_pooling_2d(pool_size = c(2, 2), strides = c(2, 2), padding = 'same', name = 'Bloc
k_3_MaxPool') %>%
  layer_dropout(rate = 0.2)

# Flatten
my_model %>%
  layer_flatten()

# Fully connected (FC) Layer 1
my_model %>%
  layer_dense(units = 128) %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_dropout(rate = 0.4)

```

```
# Fully connected (FC) Layer 2
my_model %>%
  layer_dense(units = 100) %>%
  layer_batch_normalization() %>%
  layer_activation('relu') %>%
  layer_dropout(rate = 0.4)

# Fully connected (FC) Layer 3
my_model %>%
  layer_dense(units = 1) %>%
  layer_activation('sigmoid')

# Model Summary
summary(my_model)
```

```
## Model: "sequential"
##
## Layer (type)                Output Shape                Param #    Trainable
## =====
## Block_1_Conv_1 (Conv2D)      (None, 50, 50, 16)         448        Y
## batch_normalization_1 (BatchN (None, 50, 50, 16)         64          Y
## ormalization)
## activation_1 (Activation)    (None, 50, 50, 16)         0           Y
## Block_1_Conv_2 (Conv2D)      (None, 50, 50, 32)         4640        Y
## batch_normalization (BatchNor (None, 50, 50, 32)         128          Y
## malization)
## activation (Activation)      (None, 50, 50, 32)         0           Y
## Block_1_MaxPool (MaxPooling2D (None, 25, 25, 32)         0           Y
## )
## dropout (Dropout)           (None, 25, 25, 32)         0           Y
## Block_2_Conv_1 (Conv2D)      (None, 25, 25, 64)         18496        Y
## batch_normalization_3 (BatchN (None, 25, 25, 64)         256          Y
## ormalization)
## activation_3 (Activation)    (None, 25, 25, 64)         0           Y
## Block_2_Conv_2 (Conv2D)      (None, 25, 25, 128)        73856        Y
## batch_normalization_2 (BatchN (None, 25, 25, 128)        512          Y
## ormalization)
## activation_2 (Activation)    (None, 25, 25, 128)        0           Y
## Block_2_MaxPool (MaxPooling2D (None, 13, 13, 128)        0           Y
## )
## dropout_1 (Dropout)         (None, 13, 13, 128)        0           Y
## Block_3_Conv_1 (Conv2D)      (None, 13, 13, 256)        295168        Y
## batch_normalization_5 (BatchN (None, 13, 13, 256)        1024          Y
## ormalization)
## activation_5 (Activation)    (None, 13, 13, 256)        0           Y
## Block_3_Conv_2 (Conv2D)      (None, 13, 13, 512)        1180160        Y
## batch_normalization_4 (BatchN (None, 13, 13, 512)        2048          Y
## ormalization)
## activation_4 (Activation)    (None, 13, 13, 512)        0           Y
## Block_3_MaxPool (MaxPooling2D (None, 7, 7, 512)         0           Y
## )
## dropout_2 (Dropout)         (None, 7, 7, 512)         0           Y
## flatten (Flatten)           (None, 25088)              0           Y
## dense (Dense)                (None, 128)                3211392        Y
## batch_normalization_6 (BatchN (None, 128)              512          Y
## ormalization)
## activation_6 (Activation)    (None, 128)                0           Y
## dropout_3 (Dropout)         (None, 128)                0           Y
## dense_1 (Dense)              (None, 100)                12900          Y
## batch_normalization_7 (BatchN (None, 100)              400          Y
## ormalization)
## activation_7 (Activation)    (None, 100)                0           Y
## dropout_4 (Dropout)         (None, 100)                0           Y
## dense_2 (Dense)              (None, 1)                  101          Y
## activation_8 (Activation)    (None, 1)                  0           Y
## =====
## Total params: 4,802,105
## Trainable params: 4,799,633
## Non-trainable params: 2,472
##
```

10. Compile the model:

```
my_model %>%  
  compile(loss = 'binary_crossentropy', optimizer = optimizer_sgd(learning_rate = 0.007, momentum = 0.9), metrics = list('AUC'))
```

11. Train the model:

```
##### Turn to comments from the second run onwards #####  
# Create a callback ModelCheckpoint. This callback will save the model weights when finding one better than the current best  
checkpoint_filepath <- "malaria_best_model_weights.h5"  
mc <- callback_model_checkpoint(filepath = checkpoint_filepath, monitor = "val_loss", save_best_only = TRUE, save_weights_only = TRUE, save_freq = "epoch", verbose = 0)  
  
# Train the model  
history <- my_model %>%  
  fit(X_train_, y_train, batch_size = 32, epochs = 20, verbose = 1, validation_data = list(X_val_, y_val), callbacks = list(mc))
```

```
## Epoch 1/20
##
## 1/603 [.....] - ETA: 46:45 - loss: 0.7467 - auc: 0.5813
## 4/603 [.....] - ETA: 10s - loss: 0.7899 - auc: 0.5288
## 8/603 [.....] - ETA: 9s - loss: 0.7859 - auc: 0.5714
## 12/603 [.....] - ETA: 9s - loss: 0.7982 - auc: 0.5504
## 16/603 [.....] - ETA: 9s - loss: 0.7795 - auc: 0.5721
## 20/603 [.....] - ETA: 9s - loss: 0.7759 - auc: 0.5815
## 24/603 [>.....] - ETA: 9s - loss: 0.7793 - auc: 0.5764
## 28/603 [>.....] - ETA: 9s - loss: 0.7621 - auc: 0.5897
## 32/603 [>.....] - ETA: 9s - loss: 0.7552 - auc: 0.5967
## 36/603 [>.....] - ETA: 9s - loss: 0.7538 - auc: 0.5977
## 40/603 [>.....] - ETA: 8s - loss: 0.7464 - auc: 0.6017
## 44/603 [=>.....] - ETA: 8s - loss: 0.7333 - auc: 0.6153
## 48/603 [=>.....] - ETA: 8s - loss: 0.7226 - auc: 0.6284
## 52/603 [=>.....] - ETA: 8s - loss: 0.7190 - auc: 0.6359
## 56/603 [=>.....] - ETA: 8s - loss: 0.7007 - auc: 0.6552
## 60/603 [=>.....] - ETA: 8s - loss: 0.6885 - auc: 0.6694
## 64/603 [==>.....] - ETA: 8s - loss: 0.6713 - auc: 0.6865
## 67/603 [==>.....] - ETA: 8s - loss: 0.6584 - auc: 0.6992
## 71/603 [==>.....] - ETA: 8s - loss: 0.6418 - auc: 0.7157
## 74/603 [==>.....] - ETA: 8s - loss: 0.6293 - auc: 0.7277
## 78/603 [==>.....] - ETA: 8s - loss: 0.6133 - auc: 0.7431
## 82/603 [===>.....] - ETA: 8s - loss: 0.5952 - auc: 0.7592
## 86/603 [===>.....] - ETA: 8s - loss: 0.5816 - auc: 0.7714
## 90/603 [===>.....] - ETA: 8s - loss: 0.5689 - auc: 0.7824
## 94/603 [===>.....] - ETA: 8s - loss: 0.5551 - auc: 0.7948
## 98/603 [===>.....] - ETA: 8s - loss: 0.5408 - auc: 0.8065
## 102/603 [====>.....] - ETA: 8s - loss: 0.5305 - auc: 0.8149
## 106/603 [====>.....] - ETA: 7s - loss: 0.5217 - auc: 0.8227
## 109/603 [====>.....] - ETA: 7s - loss: 0.5126 - auc: 0.8291
## 112/603 [====>.....] - ETA: 7s - loss: 0.5068 - auc: 0.8334
## 115/603 [====>.....] - ETA: 7s - loss: 0.5002 - auc: 0.8379
## 118/603 [====>.....] - ETA: 7s - loss: 0.4945 - auc: 0.8418
## 122/603 [=====>.....] - ETA: 7s - loss: 0.4844 - auc: 0.8486
## 126/603 [=====>.....] - ETA: 7s - loss: 0.4768 - auc: 0.8540
## 130/603 [=====>.....] - ETA: 7s - loss: 0.4706 - auc: 0.8586
## 134/603 [=====>.....] - ETA: 7s - loss: 0.4629 - auc: 0.8635
## 138/603 [=====>.....] - ETA: 7s - loss: 0.4541 - auc: 0.8691
## 141/603 [=====>.....] - ETA: 7s - loss: 0.4504 - auc: 0.8715
## 145/603 [=====>.....] - ETA: 7s - loss: 0.4443 - auc: 0.8754
## 149/603 [=====>.....] - ETA: 7s - loss: 0.4399 - auc: 0.8787
## 153/603 [=====>.....] - ETA: 7s - loss: 0.4333 - auc: 0.8824
## 157/603 [=====>.....] - ETA: 7s - loss: 0.4277 - auc: 0.8857
## 161/603 [=====>.....] - ETA: 7s - loss: 0.4221 - auc: 0.8888
## 165/603 [=====>.....] - ETA: 7s - loss: 0.4170 - auc: 0.8917
## 168/603 [=====>.....] - ETA: 6s - loss: 0.4117 - auc: 0.8946
## 172/603 [=====>.....] - ETA: 6s - loss: 0.4071 - auc: 0.8970
## 176/603 [=====>.....] - ETA: 6s - loss: 0.4036 - auc: 0.8989
## 180/603 [=====>.....] - ETA: 6s - loss: 0.3989 - auc: 0.9013
## 184/603 [=====>.....] - ETA: 6s - loss: 0.3957 - auc: 0.9028
## 188/603 [=====>.....] - ETA: 6s - loss: 0.3903 - auc: 0.9055
## 192/603 [=====>.....] - ETA: 6s - loss: 0.3869 - auc: 0.9071
## 196/603 [=====>.....] - ETA: 6s - loss: 0.3821 - auc: 0.9096
## 200/603 [=====>.....] - ETA: 6s - loss: 0.3779 - auc: 0.9116
```

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## 204/603 [=====>.....] - ETA: 6s - loss: 0.3743 - auc: 0.9134
## 208/603 [=====>.....] - ETA: 6s - loss: 0.3727 - auc: 0.9146
## 212/603 [=====>.....] - ETA: 6s - loss: 0.3705 - auc: 0.9158
## 215/603 [=====>.....] - ETA: 6s - loss: 0.3686 - auc: 0.9168
## 219/603 [=====>.....] - ETA: 6s - loss: 0.3649 - auc: 0.9185
## 223/603 [=====>.....] - ETA: 6s - loss: 0.3616 - auc: 0.9201
## 227/603 [=====>.....] - ETA: 6s - loss: 0.3593 - auc: 0.9212
## 231/603 [=====>.....] - ETA: 5s - loss: 0.3559 - auc: 0.9227
## 235/603 [=====>.....] - ETA: 5s - loss: 0.3537 - auc: 0.9236
## 239/603 [=====>.....] - ETA: 5s - loss: 0.3504 - auc: 0.9250
## 243/603 [=====>.....] - ETA: 5s - loss: 0.3471 - auc: 0.9263
## 247/603 [=====>.....] - ETA: 5s - loss: 0.3433 - auc: 0.9280
## 250/603 [=====>.....] - ETA: 5s - loss: 0.3409 - auc: 0.9290
## 254/603 [=====>.....] - ETA: 5s - loss: 0.3391 - auc: 0.9297
## 258/603 [=====>.....] - ETA: 5s - loss: 0.3385 - auc: 0.9301
## 262/603 [=====>.....] - ETA: 5s - loss: 0.3362 - auc: 0.9311
## 266/603 [=====>.....] - ETA: 5s - loss: 0.3344 - auc: 0.9318
## 270/603 [=====>.....] - ETA: 5s - loss: 0.3329 - auc: 0.9324
## 274/603 [=====>.....] - ETA: 5s - loss: 0.3303 - auc: 0.9335
## 278/603 [=====>.....] - ETA: 5s - loss: 0.3284 - auc: 0.9342
## 281/603 [=====>.....] - ETA: 5s - loss: 0.3263 - auc: 0.9350
## 285/603 [=====>.....] - ETA: 5s - loss: 0.3241 - auc: 0.9359
## 289/603 [=====>.....] - ETA: 5s - loss: 0.3225 - auc: 0.9365
## 293/603 [=====>.....] - ETA: 4s - loss: 0.3207 - auc: 0.9371
## 296/603 [=====>.....] - ETA: 4s - loss: 0.3193 - auc: 0.9376
## 300/603 [=====>.....] - ETA: 4s - loss: 0.3175 - auc: 0.9383
## 304/603 [=====>.....] - ETA: 4s - loss: 0.3150 - auc: 0.9392
## 308/603 [=====>.....] - ETA: 4s - loss: 0.3145 - auc: 0.9396
## 312/603 [=====>.....] - ETA: 4s - loss: 0.3126 - auc: 0.9403
## 315/603 [=====>.....] - ETA: 4s - loss: 0.3110 - auc: 0.9409
## 319/603 [=====>.....] - ETA: 4s - loss: 0.3095 - auc: 0.9414
## 322/603 [=====>.....] - ETA: 4s - loss: 0.3081 - auc: 0.9420
## 326/603 [=====>.....] - ETA: 4s - loss: 0.3061 - auc: 0.9427
## 330/603 [=====>.....] - ETA: 4s - loss: 0.3047 - auc: 0.9432
## 334/603 [=====>.....] - ETA: 4s - loss: 0.3029 - auc: 0.9439
## 337/603 [=====>.....] - ETA: 4s - loss: 0.3014 - auc: 0.9445
## 341/603 [=====>.....] - ETA: 4s - loss: 0.2996 - auc: 0.9451
## 344/603 [=====>.....] - ETA: 4s - loss: 0.2983 - auc: 0.9456
## 348/603 [=====>.....] - ETA: 4s - loss: 0.2970 - auc: 0.9460
## 352/603 [=====>.....] - ETA: 4s - loss: 0.2961 - auc: 0.9463
## 356/603 [=====>.....] - ETA: 3s - loss: 0.2948 - auc: 0.9468
## 359/603 [=====>.....] - ETA: 3s - loss: 0.2943 - auc: 0.9469
## 363/603 [=====>.....] - ETA: 3s - loss: 0.2928 - auc: 0.9474
## 367/603 [=====>.....] - ETA: 3s - loss: 0.2908 - auc: 0.9481
## 371/603 [=====>.....] - ETA: 3s - loss: 0.2905 - auc: 0.9483
## 375/603 [=====>.....] - ETA: 3s - loss: 0.2895 - auc: 0.9486
## 379/603 [=====>.....] - ETA: 3s - loss: 0.2880 - auc: 0.9491
## 383/603 [=====>.....] - ETA: 3s - loss: 0.2864 - auc: 0.9496
## 387/603 [=====>.....] - ETA: 3s - loss: 0.2852 - auc: 0.9500
## 391/603 [=====>.....] - ETA: 3s - loss: 0.2832 - auc: 0.9507
## 395/603 [=====>.....] - ETA: 3s - loss: 0.2826 - auc: 0.9509
## 399/603 [=====>.....] - ETA: 3s - loss: 0.2810 - auc: 0.9514
## 403/603 [=====>.....] - ETA: 3s - loss: 0.2795 - auc: 0.9519
## 407/603 [=====>.....] - ETA: 3s - loss: 0.2779 - auc: 0.9524
## 411/603 [=====>.....] - ETA: 3s - loss: 0.2762 - auc: 0.9530
## 415/603 [=====>.....] - ETA: 3s - loss: 0.2746 - auc: 0.9536
```

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## 419/603 [=====>.....] - ETA: 2s - loss: 0.2740 - auc: 0.9538
## 423/603 [=====>.....] - ETA: 2s - loss: 0.2722 - auc: 0.9544
## 427/603 [=====>.....] - ETA: 2s - loss: 0.2713 - auc: 0.9546
## 431/603 [=====>.....] - ETA: 2s - loss: 0.2699 - auc: 0.9550
## 435/603 [=====>.....] - ETA: 2s - loss: 0.2685 - auc: 0.9555
## 439/603 [=====>.....] - ETA: 2s - loss: 0.2688 - auc: 0.9553
## 442/603 [=====>.....] - ETA: 2s - loss: 0.2680 - auc: 0.9556
## 446/603 [=====>.....] - ETA: 2s - loss: 0.2678 - auc: 0.9555
## 449/603 [=====>.....] - ETA: 2s - loss: 0.2664 - auc: 0.9560
## 453/603 [=====>.....] - ETA: 2s - loss: 0.2656 - auc: 0.9562
## 457/603 [=====>.....] - ETA: 2s - loss: 0.2642 - auc: 0.9566
## 461/603 [=====>.....] - ETA: 2s - loss: 0.2633 - auc: 0.9569
## 465/603 [=====>.....] - ETA: 2s - loss: 0.2630 - auc: 0.9570
## 469/603 [=====>.....] - ETA: 2s - loss: 0.2619 - auc: 0.9573
## 472/603 [=====>.....] - ETA: 2s - loss: 0.2617 - auc: 0.9573
## 476/603 [=====>.....] - ETA: 2s - loss: 0.2606 - auc: 0.9577
## 480/603 [=====>.....] - ETA: 1s - loss: 0.2606 - auc: 0.9578
## 484/603 [=====>.....] - ETA: 1s - loss: 0.2599 - auc: 0.9580
## 488/603 [=====>.....] - ETA: 1s - loss: 0.2589 - auc: 0.9582
## 491/603 [=====>.....] - ETA: 1s - loss: 0.2582 - auc: 0.9584
## 495/603 [=====>.....] - ETA: 1s - loss: 0.2576 - auc: 0.9586
## 499/603 [=====>.....] - ETA: 1s - loss: 0.2566 - auc: 0.9589
## 503/603 [=====>.....] - ETA: 1s - loss: 0.2558 - auc: 0.9592
## 506/603 [=====>.....] - ETA: 1s - loss: 0.2554 - auc: 0.9592
## 510/603 [=====>.....] - ETA: 1s - loss: 0.2547 - auc: 0.9594
## 513/603 [=====>.....] - ETA: 1s - loss: 0.2542 - auc: 0.9596
## 517/603 [=====>.....] - ETA: 1s - loss: 0.2532 - auc: 0.9599
## 521/603 [=====>.....] - ETA: 1s - loss: 0.2523 - auc: 0.9602
## 525/603 [=====>....] - ETA: 1s - loss: 0.2513 - auc: 0.9605
## 529/603 [=====>....] - ETA: 1s - loss: 0.2513 - auc: 0.9605
## 533/603 [=====>....] - ETA: 1s - loss: 0.2515 - auc: 0.9603
## 537/603 [=====>....] - ETA: 1s - loss: 0.2505 - auc: 0.9607
## 541/603 [=====>....] - ETA: 0s - loss: 0.2501 - auc: 0.9607
## 545/603 [=====>...] - ETA: 0s - loss: 0.2498 - auc: 0.9608
## 549/603 [=====>...] - ETA: 0s - loss: 0.2492 - auc: 0.9609
## 553/603 [=====>...] - ETA: 0s - loss: 0.2487 - auc: 0.9610
## 557/603 [=====>...] - ETA: 0s - loss: 0.2478 - auc: 0.9613
## 561/603 [=====>...] - ETA: 0s - loss: 0.2472 - auc: 0.9615
## 565/603 [=====>..] - ETA: 0s - loss: 0.2461 - auc: 0.9618
## 569/603 [=====>..] - ETA: 0s - loss: 0.2453 - auc: 0.9621
## 573/603 [=====>..] - ETA: 0s - loss: 0.2447 - auc: 0.9622
## 577/603 [=====>..] - ETA: 0s - loss: 0.2437 - auc: 0.9625
## 581/603 [=====>..] - ETA: 0s - loss: 0.2434 - auc: 0.9626
## 585/603 [=====>.] - ETA: 0s - loss: 0.2425 - auc: 0.9628
## 589/603 [=====>.] - ETA: 0s - loss: 0.2425 - auc: 0.9628
## 593/603 [=====>.] - ETA: 0s - loss: 0.2427 - auc: 0.9628
## 597/603 [=====>.] - ETA: 0s - loss: 0.2418 - auc: 0.9630
## 601/603 [=====>.] - ETA: 0s - loss: 0.2419 - auc: 0.9630
## 603/603 [=====] - ETA: 0s - loss: 0.2416 - auc: 0.9631
## 603/603 [=====] - 15s 18ms/step - loss: 0.2416 - auc: 0.9631 - va
l_loss: 0.3098 - val_auc: 0.9824
## Epoch 2/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.1933 - auc: 0.9765
## 5/603 [.....] - ETA: 9s - loss: 0.1944 - auc: 0.9792
## 9/603 [.....] - ETA: 9s - loss: 0.1653 - auc: 0.9815
```

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## 13/603 [.....] - ETA: 9s - loss: 0.1429 - auc: 0.9853
## 17/603 [.....] - ETA: 9s - loss: 0.1461 - auc: 0.9835
## 21/603 [>.....] - ETA: 9s - loss: 0.1452 - auc: 0.9836
## 24/603 [>.....] - ETA: 9s - loss: 0.1463 - auc: 0.9834
## 28/603 [>.....] - ETA: 9s - loss: 0.1478 - auc: 0.9821
## 32/603 [>.....] - ETA: 9s - loss: 0.1481 - auc: 0.9816
## 36/603 [>.....] - ETA: 9s - loss: 0.1518 - auc: 0.9807
## 40/603 [>.....] - ETA: 9s - loss: 0.1541 - auc: 0.9807
## 44/603 [=>.....] - ETA: 9s - loss: 0.1502 - auc: 0.9813
## 48/603 [=>.....] - ETA: 8s - loss: 0.1439 - auc: 0.9830
## 52/603 [=>.....] - ETA: 8s - loss: 0.1477 - auc: 0.9820
## 56/603 [=>.....] - ETA: 8s - loss: 0.1493 - auc: 0.9818
## 59/603 [=>.....] - ETA: 8s - loss: 0.1524 - auc: 0.9809
## 63/603 [==>.....] - ETA: 8s - loss: 0.1516 - auc: 0.9814
## 67/603 [==>.....] - ETA: 8s - loss: 0.1494 - auc: 0.9818
## 71/603 [==>.....] - ETA: 8s - loss: 0.1536 - auc: 0.9810
## 75/603 [==>.....] - ETA: 8s - loss: 0.1520 - auc: 0.9816
## 79/603 [==>.....] - ETA: 8s - loss: 0.1525 - auc: 0.9810
## 83/603 [===>.....] - ETA: 8s - loss: 0.1487 - auc: 0.9820
## 87/603 [===>.....] - ETA: 8s - loss: 0.1452 - auc: 0.9829
## 91/603 [===>.....] - ETA: 8s - loss: 0.1475 - auc: 0.9822
## 95/603 [===>.....] - ETA: 8s - loss: 0.1472 - auc: 0.9822
## 99/603 [===>.....] - ETA: 8s - loss: 0.1471 - auc: 0.9826
## 103/603 [====>.....] - ETA: 8s - loss: 0.1487 - auc: 0.9821
## 107/603 [====>.....] - ETA: 7s - loss: 0.1497 - auc: 0.9819
## 110/603 [====>.....] - ETA: 7s - loss: 0.1522 - auc: 0.9814
## 113/603 [====>.....] - ETA: 7s - loss: 0.1533 - auc: 0.9811
## 117/603 [====>.....] - ETA: 7s - loss: 0.1519 - auc: 0.9815
## 121/603 [=====>.....] - ETA: 7s - loss: 0.1521 - auc: 0.9816
## 124/603 [=====>.....] - ETA: 7s - loss: 0.1529 - auc: 0.9816
## 128/603 [=====>.....] - ETA: 7s - loss: 0.1533 - auc: 0.9815
## 131/603 [=====>.....] - ETA: 7s - loss: 0.1552 - auc: 0.9811
## 135/603 [=====>.....] - ETA: 7s - loss: 0.1575 - auc: 0.9806
## 139/603 [=====>.....] - ETA: 7s - loss: 0.1588 - auc: 0.9805
## 143/603 [=====>.....] - ETA: 7s - loss: 0.1590 - auc: 0.9803
## 147/603 [=====>.....] - ETA: 7s - loss: 0.1582 - auc: 0.9807
## 151/603 [=====>.....] - ETA: 7s - loss: 0.1583 - auc: 0.9807
## 155/603 [=====>.....] - ETA: 7s - loss: 0.1607 - auc: 0.9802
## 159/603 [=====>.....] - ETA: 7s - loss: 0.1604 - auc: 0.9802
## 163/603 [=====>.....] - ETA: 7s - loss: 0.1612 - auc: 0.9801
## 167/603 [=====>.....] - ETA: 7s - loss: 0.1622 - auc: 0.9799
## 170/603 [=====>.....] - ETA: 7s - loss: 0.1617 - auc: 0.9800
## 174/603 [=====>.....] - ETA: 6s - loss: 0.1619 - auc: 0.9799
## 178/603 [=====>.....] - ETA: 6s - loss: 0.1638 - auc: 0.9795
## 182/603 [=====>.....] - ETA: 6s - loss: 0.1635 - auc: 0.9795
## 186/603 [=====>.....] - ETA: 6s - loss: 0.1646 - auc: 0.9792
## 190/603 [=====>.....] - ETA: 6s - loss: 0.1664 - auc: 0.9790
## 194/603 [=====>.....] - ETA: 6s - loss: 0.1675 - auc: 0.9787
## 198/603 [=====>.....] - ETA: 6s - loss: 0.1677 - auc: 0.9786
## 202/603 [=====>.....] - ETA: 6s - loss: 0.1673 - auc: 0.9787
## 206/603 [=====>.....] - ETA: 6s - loss: 0.1660 - auc: 0.9791
## 210/603 [=====>.....] - ETA: 6s - loss: 0.1657 - auc: 0.9792
## 214/603 [=====>.....] - ETA: 6s - loss: 0.1651 - auc: 0.9793
## 218/603 [=====>.....] - ETA: 6s - loss: 0.1644 - auc: 0.9796
## 221/603 [=====>.....] - ETA: 6s - loss: 0.1639 - auc: 0.9797
## 225/603 [=====>.....] - ETA: 6s - loss: 0.1643 - auc: 0.9797
```

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## 229/603 [=====>.....] - ETA: 6s - loss: 0.1628 - auc: 0.9800
## 233/603 [=====>.....] - ETA: 5s - loss: 0.1616 - auc: 0.9802
## 237/603 [=====>.....] - ETA: 5s - loss: 0.1631 - auc: 0.9800
## 241/603 [=====>.....] - ETA: 5s - loss: 0.1631 - auc: 0.9799
## 245/603 [=====>.....] - ETA: 5s - loss: 0.1629 - auc: 0.9799
## 249/603 [=====>.....] - ETA: 5s - loss: 0.1617 - auc: 0.9801
## 253/603 [=====>.....] - ETA: 5s - loss: 0.1616 - auc: 0.9801
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1612 - auc: 0.9802
## 261/603 [=====>.....] - ETA: 5s - loss: 0.1607 - auc: 0.9801
## 265/603 [=====>.....] - ETA: 5s - loss: 0.1600 - auc: 0.9803
## 269/603 [=====>.....] - ETA: 5s - loss: 0.1599 - auc: 0.9803
## 273/603 [=====>.....] - ETA: 5s - loss: 0.1607 - auc: 0.9801
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1609 - auc: 0.9800
## 280/603 [=====>.....] - ETA: 5s - loss: 0.1605 - auc: 0.9802
## 284/603 [=====>.....] - ETA: 5s - loss: 0.1608 - auc: 0.9801
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1599 - auc: 0.9803
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1598 - auc: 0.9803
## 296/603 [=====>.....] - ETA: 4s - loss: 0.1598 - auc: 0.9803
## 300/603 [=====>.....] - ETA: 4s - loss: 0.1602 - auc: 0.9802
## 303/603 [=====>.....] - ETA: 4s - loss: 0.1604 - auc: 0.9802
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1607 - auc: 0.9800
## 310/603 [=====>.....] - ETA: 4s - loss: 0.1621 - auc: 0.9797
## 313/603 [=====>.....] - ETA: 4s - loss: 0.1630 - auc: 0.9795
## 317/603 [=====>.....] - ETA: 4s - loss: 0.1650 - auc: 0.9791
## 321/603 [=====>.....] - ETA: 4s - loss: 0.1642 - auc: 0.9793
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1636 - auc: 0.9794
## 329/603 [=====>.....] - ETA: 4s - loss: 0.1633 - auc: 0.9795
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1631 - auc: 0.9795
## 336/603 [=====>.....] - ETA: 4s - loss: 0.1627 - auc: 0.9796
## 340/603 [=====>.....] - ETA: 4s - loss: 0.1623 - auc: 0.9797
## 343/603 [=====>.....] - ETA: 4s - loss: 0.1621 - auc: 0.9798
## 347/603 [=====>.....] - ETA: 4s - loss: 0.1617 - auc: 0.9799
## 351/603 [=====>.....] - ETA: 4s - loss: 0.1616 - auc: 0.9799
## 354/603 [=====>.....] - ETA: 4s - loss: 0.1614 - auc: 0.9800
## 358/603 [=====>.....] - ETA: 3s - loss: 0.1611 - auc: 0.9801
## 362/603 [=====>.....] - ETA: 3s - loss: 0.1610 - auc: 0.9801
## 365/603 [=====>.....] - ETA: 3s - loss: 0.1607 - auc: 0.9802
## 368/603 [=====>.....] - ETA: 3s - loss: 0.1602 - auc: 0.9803
## 372/603 [=====>.....] - ETA: 3s - loss: 0.1598 - auc: 0.9804
## 376/603 [=====>.....] - ETA: 3s - loss: 0.1596 - auc: 0.9805
## 380/603 [=====>.....] - ETA: 3s - loss: 0.1593 - auc: 0.9805
## 384/603 [=====>.....] - ETA: 3s - loss: 0.1586 - auc: 0.9806
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1591 - auc: 0.9806
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1588 - auc: 0.9806
## 394/603 [=====>.....] - ETA: 3s - loss: 0.1583 - auc: 0.9808
## 398/603 [=====>.....] - ETA: 3s - loss: 0.1589 - auc: 0.9806
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1587 - auc: 0.9807
## 406/603 [=====>.....] - ETA: 3s - loss: 0.1587 - auc: 0.9807
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1597 - auc: 0.9805
## 412/603 [=====>.....] - ETA: 3s - loss: 0.1593 - auc: 0.9805
## 416/603 [=====>.....] - ETA: 3s - loss: 0.1593 - auc: 0.9806
## 420/603 [=====>.....] - ETA: 2s - loss: 0.1590 - auc: 0.9807
## 424/603 [=====>.....] - ETA: 2s - loss: 0.1600 - auc: 0.9805
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1596 - auc: 0.9806
## 431/603 [=====>.....] - ETA: 2s - loss: 0.1591 - auc: 0.9807
## 435/603 [=====>.....] - ETA: 2s - loss: 0.1584 - auc: 0.9809
```

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## 439/603 [=====>.....] - ETA: 2s - loss: 0.1577 - auc: 0.9811
## 443/603 [=====>.....] - ETA: 2s - loss: 0.1586 - auc: 0.9808
## 447/603 [=====>.....] - ETA: 2s - loss: 0.1583 - auc: 0.9809
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1580 - auc: 0.9809
## 455/603 [=====>.....] - ETA: 2s - loss: 0.1576 - auc: 0.9810
## 459/603 [=====>.....] - ETA: 2s - loss: 0.1579 - auc: 0.9809
## 463/603 [=====>.....] - ETA: 2s - loss: 0.1580 - auc: 0.9809
## 466/603 [=====>.....] - ETA: 2s - loss: 0.1583 - auc: 0.9809
## 469/603 [=====>.....] - ETA: 2s - loss: 0.1594 - auc: 0.9806
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1594 - auc: 0.9806
## 475/603 [=====>.....] - ETA: 2s - loss: 0.1590 - auc: 0.9807
## 479/603 [=====>.....] - ETA: 2s - loss: 0.1586 - auc: 0.9808
## 482/603 [=====>.....] - ETA: 1s - loss: 0.1589 - auc: 0.9807
## 486/603 [=====>.....] - ETA: 1s - loss: 0.1583 - auc: 0.9809
## 490/603 [=====>.....] - ETA: 1s - loss: 0.1582 - auc: 0.9809
## 494/603 [=====>.....] - ETA: 1s - loss: 0.1584 - auc: 0.9809
## 498/603 [=====>.....] - ETA: 1s - loss: 0.1580 - auc: 0.9810
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1578 - auc: 0.9810
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1572 - auc: 0.9811
## 509/603 [=====>.....] - ETA: 1s - loss: 0.1566 - auc: 0.9813
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1567 - auc: 0.9812
## 517/603 [=====>.....] - ETA: 1s - loss: 0.1567 - auc: 0.9812
## 521/603 [=====>.....] - ETA: 1s - loss: 0.1568 - auc: 0.9812
## 525/603 [=====>....] - ETA: 1s - loss: 0.1567 - auc: 0.9811
## 528/603 [=====>....] - ETA: 1s - loss: 0.1562 - auc: 0.9812
## 531/603 [=====>....] - ETA: 1s - loss: 0.1559 - auc: 0.9813
## 534/603 [=====>....] - ETA: 1s - loss: 0.1558 - auc: 0.9813
## 537/603 [=====>....] - ETA: 1s - loss: 0.1554 - auc: 0.9814
## 541/603 [=====>....] - ETA: 1s - loss: 0.1550 - auc: 0.9815
## 544/603 [=====>...] - ETA: 0s - loss: 0.1553 - auc: 0.9814
## 547/603 [=====>...] - ETA: 0s - loss: 0.1553 - auc: 0.9814
## 551/603 [=====>...] - ETA: 0s - loss: 0.1552 - auc: 0.9814
## 555/603 [=====>...] - ETA: 0s - loss: 0.1554 - auc: 0.9813
## 558/603 [=====>...] - ETA: 0s - loss: 0.1552 - auc: 0.9814
## 562/603 [=====>...] - ETA: 0s - loss: 0.1550 - auc: 0.9814
## 566/603 [=====>..] - ETA: 0s - loss: 0.1547 - auc: 0.9814
## 569/603 [=====>..] - ETA: 0s - loss: 0.1552 - auc: 0.9813
## 573/603 [=====>..] - ETA: 0s - loss: 0.1559 - auc: 0.9813
## 577/603 [=====>..] - ETA: 0s - loss: 0.1567 - auc: 0.9811
## 581/603 [=====>..] - ETA: 0s - loss: 0.1564 - auc: 0.9811
## 585/603 [=====>.] - ETA: 0s - loss: 0.1566 - auc: 0.9811
## 589/603 [=====>.] - ETA: 0s - loss: 0.1565 - auc: 0.9811
## 592/603 [=====>.] - ETA: 0s - loss: 0.1567 - auc: 0.9811
## 596/603 [=====>.] - ETA: 0s - loss: 0.1564 - auc: 0.9811
## 600/603 [=====>.] - ETA: 0s - loss: 0.1564 - auc: 0.9811
## 603/603 [=====] - 10s 17ms/step - loss: 0.1568 - auc: 0.9811 - va
l_loss: 0.5769 - val_auc: 0.9525
## Epoch 3/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.1848 - auc: 0.9833
## 5/603 [.....] - ETA: 9s - loss: 0.1055 - auc: 0.9945
## 9/603 [.....] - ETA: 9s - loss: 0.1208 - auc: 0.9917
## 13/603 [.....] - ETA: 9s - loss: 0.1418 - auc: 0.9865
## 17/603 [.....] - ETA: 9s - loss: 0.1344 - auc: 0.9883
## 20/603 [.....] - ETA: 9s - loss: 0.1214 - auc: 0.9903
## 23/603 [>.....] - ETA: 9s - loss: 0.1228 - auc: 0.9896
```

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## 27/603 [>.....] - ETA: 9s - loss: 0.1228 - auc: 0.9899
## 30/603 [>.....] - ETA: 9s - loss: 0.1284 - auc: 0.9882
## 33/603 [>.....] - ETA: 9s - loss: 0.1281 - auc: 0.9879
## 36/603 [>.....] - ETA: 9s - loss: 0.1272 - auc: 0.9883
## 40/603 [>.....] - ETA: 9s - loss: 0.1304 - auc: 0.9877
## 43/603 [=>.....] - ETA: 9s - loss: 0.1274 - auc: 0.9884
## 47/603 [=>.....] - ETA: 9s - loss: 0.1245 - auc: 0.9890
## 51/603 [=>.....] - ETA: 9s - loss: 0.1248 - auc: 0.9889
## 54/603 [=>.....] - ETA: 9s - loss: 0.1284 - auc: 0.9882
## 58/603 [=>.....] - ETA: 9s - loss: 0.1257 - auc: 0.9886
## 62/603 [==>.....] - ETA: 9s - loss: 0.1242 - auc: 0.9887
## 65/603 [==>.....] - ETA: 9s - loss: 0.1279 - auc: 0.9881
## 68/603 [==>.....] - ETA: 8s - loss: 0.1308 - auc: 0.9874
## 72/603 [==>.....] - ETA: 8s - loss: 0.1277 - auc: 0.9880
## 75/603 [==>.....] - ETA: 8s - loss: 0.1296 - auc: 0.9875
## 79/603 [==>.....] - ETA: 8s - loss: 0.1311 - auc: 0.9871
## 82/603 [===>.....] - ETA: 8s - loss: 0.1325 - auc: 0.9865
## 85/603 [===>.....] - ETA: 8s - loss: 0.1319 - auc: 0.9867
## 88/603 [===>.....] - ETA: 8s - loss: 0.1321 - auc: 0.9864
## 91/603 [===>.....] - ETA: 8s - loss: 0.1363 - auc: 0.9856
## 94/603 [===>.....] - ETA: 8s - loss: 0.1374 - auc: 0.9853
## 98/603 [===>.....] - ETA: 8s - loss: 0.1400 - auc: 0.9845
## 102/603 [====>.....] - ETA: 8s - loss: 0.1392 - auc: 0.9848
## 106/603 [====>.....] - ETA: 8s - loss: 0.1443 - auc: 0.9834
## 110/603 [====>.....] - ETA: 8s - loss: 0.1464 - auc: 0.9832
## 113/603 [====>.....] - ETA: 8s - loss: 0.1459 - auc: 0.9833
## 116/603 [====>.....] - ETA: 8s - loss: 0.1438 - auc: 0.9836
## 119/603 [====>.....] - ETA: 8s - loss: 0.1451 - auc: 0.9834
## 123/603 [=====>.....] - ETA: 8s - loss: 0.1441 - auc: 0.9836
## 127/603 [=====>.....] - ETA: 8s - loss: 0.1441 - auc: 0.9835
## 130/603 [=====>.....] - ETA: 7s - loss: 0.1431 - auc: 0.9837
## 133/603 [=====>.....] - ETA: 7s - loss: 0.1431 - auc: 0.9838
## 137/603 [=====>.....] - ETA: 7s - loss: 0.1471 - auc: 0.9829
## 141/603 [=====>.....] - ETA: 7s - loss: 0.1473 - auc: 0.9831
## 145/603 [=====>.....] - ETA: 7s - loss: 0.1496 - auc: 0.9827
## 148/603 [=====>.....] - ETA: 7s - loss: 0.1539 - auc: 0.9819
## 152/603 [=====>.....] - ETA: 7s - loss: 0.1557 - auc: 0.9816
## 156/603 [=====>.....] - ETA: 7s - loss: 0.1545 - auc: 0.9819
## 160/603 [=====>.....] - ETA: 7s - loss: 0.1552 - auc: 0.9816
## 164/603 [=====>.....] - ETA: 7s - loss: 0.1570 - auc: 0.9814
## 168/603 [=====>.....] - ETA: 7s - loss: 0.1556 - auc: 0.9818
## 172/603 [=====>.....] - ETA: 7s - loss: 0.1538 - auc: 0.9822
## 176/603 [=====>.....] - ETA: 7s - loss: 0.1553 - auc: 0.9819
## 180/603 [=====>.....] - ETA: 7s - loss: 0.1557 - auc: 0.9819
## 184/603 [=====>.....] - ETA: 6s - loss: 0.1580 - auc: 0.9813
## 188/603 [=====>.....] - ETA: 6s - loss: 0.1574 - auc: 0.9814
## 192/603 [=====>.....] - ETA: 6s - loss: 0.1561 - auc: 0.9817
## 196/603 [=====>.....] - ETA: 6s - loss: 0.1541 - auc: 0.9821
## 200/603 [=====>.....] - ETA: 6s - loss: 0.1553 - auc: 0.9818
## 204/603 [=====>.....] - ETA: 6s - loss: 0.1547 - auc: 0.9820
## 208/603 [=====>.....] - ETA: 6s - loss: 0.1529 - auc: 0.9823
## 212/603 [=====>.....] - ETA: 6s - loss: 0.1534 - auc: 0.9822
## 216/603 [=====>.....] - ETA: 6s - loss: 0.1524 - auc: 0.9824
## 220/603 [=====>.....] - ETA: 6s - loss: 0.1534 - auc: 0.9822
## 224/603 [=====>.....] - ETA: 6s - loss: 0.1546 - auc: 0.9821
## 227/603 [=====>.....] - ETA: 6s - loss: 0.1553 - auc: 0.9820
```

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## 231/603 [=====>.....] - ETA: 6s - loss: 0.1550 - auc: 0.9820
## 235/603 [=====>.....] - ETA: 6s - loss: 0.1542 - auc: 0.9822
## 239/603 [=====>.....] - ETA: 6s - loss: 0.1550 - auc: 0.9820
## 243/603 [=====>.....] - ETA: 5s - loss: 0.1546 - auc: 0.9821
## 247/603 [=====>.....] - ETA: 5s - loss: 0.1543 - auc: 0.9822
## 251/603 [=====>.....] - ETA: 5s - loss: 0.1538 - auc: 0.9823
## 255/603 [=====>.....] - ETA: 5s - loss: 0.1533 - auc: 0.9824
## 259/603 [=====>.....] - ETA: 5s - loss: 0.1529 - auc: 0.9825
## 263/603 [=====>.....] - ETA: 5s - loss: 0.1533 - auc: 0.9823
## 267/603 [=====>.....] - ETA: 5s - loss: 0.1529 - auc: 0.9824
## 271/603 [=====>.....] - ETA: 5s - loss: 0.1521 - auc: 0.9826
## 274/603 [=====>.....] - ETA: 5s - loss: 0.1524 - auc: 0.9825
## 278/603 [=====>.....] - ETA: 5s - loss: 0.1519 - auc: 0.9827
## 282/603 [=====>.....] - ETA: 5s - loss: 0.1513 - auc: 0.9828
## 286/603 [=====>.....] - ETA: 5s - loss: 0.1504 - auc: 0.9830
## 290/603 [=====>.....] - ETA: 5s - loss: 0.1501 - auc: 0.9831
## 294/603 [=====>.....] - ETA: 5s - loss: 0.1514 - auc: 0.9829
## 298/603 [=====>.....] - ETA: 5s - loss: 0.1518 - auc: 0.9827
## 302/603 [=====>.....] - ETA: 4s - loss: 0.1518 - auc: 0.9827
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1517 - auc: 0.9828
## 310/603 [=====>.....] - ETA: 4s - loss: 0.1512 - auc: 0.9829
## 314/603 [=====>.....] - ETA: 4s - loss: 0.1509 - auc: 0.9829
## 318/603 [=====>.....] - ETA: 4s - loss: 0.1511 - auc: 0.9828
## 322/603 [=====>.....] - ETA: 4s - loss: 0.1510 - auc: 0.9828
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1504 - auc: 0.9830
## 329/603 [=====>.....] - ETA: 4s - loss: 0.1496 - auc: 0.9831
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1489 - auc: 0.9832
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1483 - auc: 0.9833
## 341/603 [=====>.....] - ETA: 4s - loss: 0.1487 - auc: 0.9833
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1483 - auc: 0.9833
## 349/603 [=====>.....] - ETA: 4s - loss: 0.1498 - auc: 0.9831
## 353/603 [=====>.....] - ETA: 4s - loss: 0.1505 - auc: 0.9829
## 357/603 [=====>.....] - ETA: 4s - loss: 0.1504 - auc: 0.9830
## 361/603 [=====>.....] - ETA: 3s - loss: 0.1498 - auc: 0.9831
## 365/603 [=====>.....] - ETA: 3s - loss: 0.1492 - auc: 0.9832
## 369/603 [=====>.....] - ETA: 3s - loss: 0.1483 - auc: 0.9834
## 373/603 [=====>.....] - ETA: 3s - loss: 0.1489 - auc: 0.9833
## 377/603 [=====>.....] - ETA: 3s - loss: 0.1493 - auc: 0.9833
## 381/603 [=====>.....] - ETA: 3s - loss: 0.1488 - auc: 0.9834
## 385/603 [=====>.....] - ETA: 3s - loss: 0.1487 - auc: 0.9834
## 388/603 [=====>.....] - ETA: 3s - loss: 0.1489 - auc: 0.9834
## 392/603 [=====>.....] - ETA: 3s - loss: 0.1483 - auc: 0.9834
## 396/603 [=====>.....] - ETA: 3s - loss: 0.1486 - auc: 0.9834
## 400/603 [=====>.....] - ETA: 3s - loss: 0.1490 - auc: 0.9833
## 404/603 [=====>.....] - ETA: 3s - loss: 0.1486 - auc: 0.9834
## 408/603 [=====>.....] - ETA: 3s - loss: 0.1483 - auc: 0.9834
## 412/603 [=====>.....] - ETA: 3s - loss: 0.1478 - auc: 0.9835
## 416/603 [=====>.....] - ETA: 3s - loss: 0.1477 - auc: 0.9836
## 420/603 [=====>.....] - ETA: 2s - loss: 0.1471 - auc: 0.9837
## 424/603 [=====>.....] - ETA: 2s - loss: 0.1469 - auc: 0.9838
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1469 - auc: 0.9837
## 432/603 [=====>.....] - ETA: 2s - loss: 0.1476 - auc: 0.9836
## 436/603 [=====>.....] - ETA: 2s - loss: 0.1473 - auc: 0.9837
## 440/603 [=====>.....] - ETA: 2s - loss: 0.1470 - auc: 0.9837
## 443/603 [=====>.....] - ETA: 2s - loss: 0.1472 - auc: 0.9837
## 446/603 [=====>.....] - ETA: 2s - loss: 0.1467 - auc: 0.9838
```

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## 450/603 [=====>.....] - ETA: 2s - loss: 0.1465 - auc: 0.9838
## 453/603 [=====>.....] - ETA: 2s - loss: 0.1470 - auc: 0.9837
## 457/603 [=====>.....] - ETA: 2s - loss: 0.1469 - auc: 0.9837
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1468 - auc: 0.9838
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1469 - auc: 0.9837
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1463 - auc: 0.9838
## 471/603 [=====>.....] - ETA: 2s - loss: 0.1459 - auc: 0.9839
## 474/603 [=====>.....] - ETA: 2s - loss: 0.1458 - auc: 0.9839
## 477/603 [=====>.....] - ETA: 2s - loss: 0.1463 - auc: 0.9838
## 480/603 [=====>.....] - ETA: 2s - loss: 0.1460 - auc: 0.9839
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1464 - auc: 0.9838
## 486/603 [=====>.....] - ETA: 1s - loss: 0.1460 - auc: 0.9839
## 489/603 [=====>.....] - ETA: 1s - loss: 0.1461 - auc: 0.9839
## 493/603 [=====>.....] - ETA: 1s - loss: 0.1468 - auc: 0.9837
## 496/603 [=====>.....] - ETA: 1s - loss: 0.1464 - auc: 0.9838
## 499/603 [=====>.....] - ETA: 1s - loss: 0.1461 - auc: 0.9839
## 503/603 [=====>.....] - ETA: 1s - loss: 0.1457 - auc: 0.9840
## 507/603 [=====>.....] - ETA: 1s - loss: 0.1459 - auc: 0.9840
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1458 - auc: 0.9840
## 514/603 [=====>.....] - ETA: 1s - loss: 0.1459 - auc: 0.9840
## 518/603 [=====>.....] - ETA: 1s - loss: 0.1453 - auc: 0.9841
## 522/603 [=====>.....] - ETA: 1s - loss: 0.1456 - auc: 0.9840
## 525/603 [=====>....] - ETA: 1s - loss: 0.1455 - auc: 0.9841
## 529/603 [=====>....] - ETA: 1s - loss: 0.1454 - auc: 0.9841
## 532/603 [=====>....] - ETA: 1s - loss: 0.1454 - auc: 0.9841
## 535/603 [=====>....] - ETA: 1s - loss: 0.1459 - auc: 0.9840
## 538/603 [=====>....] - ETA: 1s - loss: 0.1458 - auc: 0.9840
## 542/603 [=====>....] - ETA: 1s - loss: 0.1457 - auc: 0.9840
## 546/603 [=====>...] - ETA: 0s - loss: 0.1454 - auc: 0.9841
## 549/603 [=====>...] - ETA: 0s - loss: 0.1454 - auc: 0.9840
## 553/603 [=====>...] - ETA: 0s - loss: 0.1449 - auc: 0.9842
## 556/603 [=====>...] - ETA: 0s - loss: 0.1448 - auc: 0.9842
## 560/603 [=====>...] - ETA: 0s - loss: 0.1447 - auc: 0.9842
## 564/603 [=====>..] - ETA: 0s - loss: 0.1443 - auc: 0.9843
## 567/603 [=====>..] - ETA: 0s - loss: 0.1445 - auc: 0.9842
## 571/603 [=====>..] - ETA: 0s - loss: 0.1446 - auc: 0.9842
## 575/603 [=====>..] - ETA: 0s - loss: 0.1452 - auc: 0.9841
## 579/603 [=====>..] - ETA: 0s - loss: 0.1450 - auc: 0.9841
## 583/603 [=====>.] - ETA: 0s - loss: 0.1454 - auc: 0.9841
## 586/603 [=====>.] - ETA: 0s - loss: 0.1453 - auc: 0.9841
## 590/603 [=====>.] - ETA: 0s - loss: 0.1457 - auc: 0.9840
## 594/603 [=====>.] - ETA: 0s - loss: 0.1456 - auc: 0.9840
## 598/603 [=====>.] - ETA: 0s - loss: 0.1458 - auc: 0.9840
## 602/603 [=====>.] - ETA: 0s - loss: 0.1466 - auc: 0.9839
## 603/603 [=====] - 10s 17ms/step - loss: 0.1465 - auc: 0.9839 - va
l_loss: 0.2535 - val_auc: 0.9790
## Epoch 4/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0481 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.1214 - auc: 0.9886
## 8/603 [.....] - ETA: 9s - loss: 0.1342 - auc: 0.9813
## 11/603 [.....] - ETA: 9s - loss: 0.1470 - auc: 0.9817
## 15/603 [.....] - ETA: 9s - loss: 0.1591 - auc: 0.9805
## 18/603 [.....] - ETA: 9s - loss: 0.1528 - auc: 0.9807
## 22/603 [>.....] - ETA: 9s - loss: 0.1395 - auc: 0.9839
## 26/603 [>.....] - ETA: 9s - loss: 0.1395 - auc: 0.9841
```

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## 30/603 [>.....] - ETA: 9s - loss: 0.1433 - auc: 0.9840
## 33/603 [>.....] - ETA: 9s - loss: 0.1420 - auc: 0.9846
## 37/603 [>.....] - ETA: 9s - loss: 0.1354 - auc: 0.9860
## 41/603 [=>.....] - ETA: 9s - loss: 0.1330 - auc: 0.9868
## 45/603 [=>.....] - ETA: 9s - loss: 0.1393 - auc: 0.9854
## 49/603 [=>.....] - ETA: 9s - loss: 0.1516 - auc: 0.9829
## 53/603 [=>.....] - ETA: 8s - loss: 0.1547 - auc: 0.9822
## 57/603 [=>.....] - ETA: 8s - loss: 0.1520 - auc: 0.9826
## 60/603 [=>.....] - ETA: 8s - loss: 0.1527 - auc: 0.9826
## 64/603 [==>.....] - ETA: 8s - loss: 0.1511 - auc: 0.9826
## 68/603 [==>.....] - ETA: 8s - loss: 0.1462 - auc: 0.9834
## 71/603 [==>.....] - ETA: 8s - loss: 0.1532 - auc: 0.9832
## 74/603 [==>.....] - ETA: 8s - loss: 0.1500 - auc: 0.9838
## 78/603 [==>.....] - ETA: 8s - loss: 0.1504 - auc: 0.9836
## 81/603 [===>.....] - ETA: 8s - loss: 0.1482 - auc: 0.9841
## 85/603 [===>.....] - ETA: 8s - loss: 0.1485 - auc: 0.9836
## 89/603 [===>.....] - ETA: 8s - loss: 0.1483 - auc: 0.9837
## 92/603 [===>.....] - ETA: 8s - loss: 0.1498 - auc: 0.9834
## 95/603 [===>.....] - ETA: 8s - loss: 0.1513 - auc: 0.9832
## 99/603 [===>.....] - ETA: 8s - loss: 0.1495 - auc: 0.9836
## 103/603 [====>.....] - ETA: 8s - loss: 0.1518 - auc: 0.9828
## 107/603 [====>.....] - ETA: 8s - loss: 0.1513 - auc: 0.9830
## 110/603 [====>.....] - ETA: 8s - loss: 0.1502 - auc: 0.9832
## 113/603 [====>.....] - ETA: 8s - loss: 0.1479 - auc: 0.9838
## 116/603 [====>.....] - ETA: 8s - loss: 0.1477 - auc: 0.9837
## 120/603 [====>.....] - ETA: 7s - loss: 0.1466 - auc: 0.9838
## 123/603 [====>.....] - ETA: 7s - loss: 0.1466 - auc: 0.9840
## 127/603 [====>.....] - ETA: 7s - loss: 0.1458 - auc: 0.9840
## 131/603 [====>.....] - ETA: 7s - loss: 0.1466 - auc: 0.9836
## 135/603 [====>.....] - ETA: 7s - loss: 0.1454 - auc: 0.9839
## 139/603 [====>.....] - ETA: 7s - loss: 0.1464 - auc: 0.9836
## 143/603 [====>.....] - ETA: 7s - loss: 0.1460 - auc: 0.9838
## 146/603 [====>.....] - ETA: 7s - loss: 0.1451 - auc: 0.9840
## 150/603 [====>.....] - ETA: 7s - loss: 0.1447 - auc: 0.9842
## 153/603 [====>.....] - ETA: 7s - loss: 0.1446 - auc: 0.9843
## 156/603 [====>.....] - ETA: 7s - loss: 0.1450 - auc: 0.9843
## 160/603 [====>.....] - ETA: 7s - loss: 0.1450 - auc: 0.9843
## 164/603 [====>.....] - ETA: 7s - loss: 0.1477 - auc: 0.9838
## 168/603 [====>.....] - ETA: 7s - loss: 0.1461 - auc: 0.9842
## 172/603 [====>.....] - ETA: 7s - loss: 0.1456 - auc: 0.9843
## 175/603 [====>.....] - ETA: 7s - loss: 0.1456 - auc: 0.9842
## 179/603 [====>.....] - ETA: 6s - loss: 0.1464 - auc: 0.9838
## 183/603 [====>.....] - ETA: 6s - loss: 0.1453 - auc: 0.9841
## 187/603 [====>.....] - ETA: 6s - loss: 0.1442 - auc: 0.9843
## 191/603 [====>.....] - ETA: 6s - loss: 0.1447 - auc: 0.9842
## 195/603 [====>.....] - ETA: 6s - loss: 0.1452 - auc: 0.9839
## 199/603 [====>.....] - ETA: 6s - loss: 0.1447 - auc: 0.9840
## 202/603 [====>.....] - ETA: 6s - loss: 0.1441 - auc: 0.9841
## 206/603 [====>.....] - ETA: 6s - loss: 0.1441 - auc: 0.9842
## 209/603 [====>.....] - ETA: 6s - loss: 0.1435 - auc: 0.9843
## 213/603 [====>.....] - ETA: 6s - loss: 0.1427 - auc: 0.9844
## 217/603 [====>.....] - ETA: 6s - loss: 0.1431 - auc: 0.9844
## 221/603 [====>.....] - ETA: 6s - loss: 0.1435 - auc: 0.9842
## 225/603 [====>.....] - ETA: 6s - loss: 0.1447 - auc: 0.9840
## 229/603 [====>.....] - ETA: 6s - loss: 0.1446 - auc: 0.9840
## 233/603 [====>.....] - ETA: 6s - loss: 0.1433 - auc: 0.9842
```

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## 237/603 [=====>.....] - ETA: 5s - loss: 0.1443 - auc: 0.9839
## 241/603 [=====>.....] - ETA: 5s - loss: 0.1445 - auc: 0.9837
## 245/603 [=====>.....] - ETA: 5s - loss: 0.1435 - auc: 0.9839
## 249/603 [=====>.....] - ETA: 5s - loss: 0.1427 - auc: 0.9841
## 253/603 [=====>.....] - ETA: 5s - loss: 0.1418 - auc: 0.9843
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1427 - auc: 0.9841
## 261/603 [=====>.....] - ETA: 5s - loss: 0.1420 - auc: 0.9843
## 265/603 [=====>.....] - ETA: 5s - loss: 0.1414 - auc: 0.9845
## 269/603 [=====>.....] - ETA: 5s - loss: 0.1423 - auc: 0.9843
## 273/603 [=====>.....] - ETA: 5s - loss: 0.1428 - auc: 0.9842
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1431 - auc: 0.9842
## 281/603 [=====>.....] - ETA: 5s - loss: 0.1433 - auc: 0.9841
## 285/603 [=====>.....] - ETA: 5s - loss: 0.1429 - auc: 0.9842
## 289/603 [=====>.....] - ETA: 5s - loss: 0.1424 - auc: 0.9843
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1421 - auc: 0.9844
## 296/603 [=====>.....] - ETA: 5s - loss: 0.1415 - auc: 0.9845
## 300/603 [=====>.....] - ETA: 4s - loss: 0.1403 - auc: 0.9847
## 304/603 [=====>.....] - ETA: 4s - loss: 0.1398 - auc: 0.9847
## 308/603 [=====>.....] - ETA: 4s - loss: 0.1397 - auc: 0.9848
## 312/603 [=====>.....] - ETA: 4s - loss: 0.1398 - auc: 0.9848
## 316/603 [=====>.....] - ETA: 4s - loss: 0.1408 - auc: 0.9845
## 320/603 [=====>.....] - ETA: 4s - loss: 0.1423 - auc: 0.9842
## 323/603 [=====>.....] - ETA: 4s - loss: 0.1425 - auc: 0.9842
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1415 - auc: 0.9844
## 331/603 [=====>.....] - ETA: 4s - loss: 0.1418 - auc: 0.9843
## 335/603 [=====>.....] - ETA: 4s - loss: 0.1420 - auc: 0.9843
## 339/603 [=====>.....] - ETA: 4s - loss: 0.1421 - auc: 0.9844
## 343/603 [=====>.....] - ETA: 4s - loss: 0.1416 - auc: 0.9845
## 346/603 [=====>.....] - ETA: 4s - loss: 0.1420 - auc: 0.9845
## 350/603 [=====>.....] - ETA: 4s - loss: 0.1419 - auc: 0.9845
## 354/603 [=====>.....] - ETA: 4s - loss: 0.1422 - auc: 0.9845
## 358/603 [=====>.....] - ETA: 4s - loss: 0.1426 - auc: 0.9844
## 362/603 [=====>.....] - ETA: 3s - loss: 0.1426 - auc: 0.9845
## 366/603 [=====>.....] - ETA: 3s - loss: 0.1422 - auc: 0.9845
## 370/603 [=====>.....] - ETA: 3s - loss: 0.1425 - auc: 0.9845
## 374/603 [=====>.....] - ETA: 3s - loss: 0.1424 - auc: 0.9846
## 378/603 [=====>.....] - ETA: 3s - loss: 0.1424 - auc: 0.9846
## 381/603 [=====>.....] - ETA: 3s - loss: 0.1432 - auc: 0.9844
## 385/603 [=====>.....] - ETA: 3s - loss: 0.1425 - auc: 0.9845
## 389/603 [=====>.....] - ETA: 3s - loss: 0.1421 - auc: 0.9846
## 392/603 [=====>.....] - ETA: 3s - loss: 0.1420 - auc: 0.9847
## 396/603 [=====>.....] - ETA: 3s - loss: 0.1417 - auc: 0.9848
## 399/603 [=====>.....] - ETA: 3s - loss: 0.1425 - auc: 0.9846
## 403/603 [=====>.....] - ETA: 3s - loss: 0.1422 - auc: 0.9846
## 407/603 [=====>.....] - ETA: 3s - loss: 0.1419 - auc: 0.9847
## 410/603 [=====>.....] - ETA: 3s - loss: 0.1418 - auc: 0.9847
## 414/603 [=====>.....] - ETA: 3s - loss: 0.1420 - auc: 0.9847
## 418/603 [=====>.....] - ETA: 3s - loss: 0.1420 - auc: 0.9847
## 422/603 [=====>.....] - ETA: 2s - loss: 0.1416 - auc: 0.9847
## 426/603 [=====>.....] - ETA: 2s - loss: 0.1417 - auc: 0.9847
## 430/603 [=====>.....] - ETA: 2s - loss: 0.1415 - auc: 0.9847
## 433/603 [=====>.....] - ETA: 2s - loss: 0.1415 - auc: 0.9848
## 437/603 [=====>.....] - ETA: 2s - loss: 0.1415 - auc: 0.9848
## 441/603 [=====>.....] - ETA: 2s - loss: 0.1412 - auc: 0.9849
## 445/603 [=====>.....] - ETA: 2s - loss: 0.1412 - auc: 0.9849
## 448/603 [=====>.....] - ETA: 2s - loss: 0.1416 - auc: 0.9849
```

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## 452/603 [=====>.....] - ETA: 2s - loss: 0.1422 - auc: 0.9847
## 456/603 [=====>.....] - ETA: 2s - loss: 0.1419 - auc: 0.9848
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1419 - auc: 0.9848
## 463/603 [=====>.....] - ETA: 2s - loss: 0.1414 - auc: 0.9849
## 467/603 [=====>.....] - ETA: 2s - loss: 0.1413 - auc: 0.9850
## 470/603 [=====>.....] - ETA: 2s - loss: 0.1411 - auc: 0.9850
## 474/603 [=====>.....] - ETA: 2s - loss: 0.1407 - auc: 0.9851
## 477/603 [=====>.....] - ETA: 2s - loss: 0.1406 - auc: 0.9852
## 480/603 [=====>.....] - ETA: 2s - loss: 0.1407 - auc: 0.9852
## 484/603 [=====>.....] - ETA: 1s - loss: 0.1402 - auc: 0.9853
## 488/603 [=====>.....] - ETA: 1s - loss: 0.1399 - auc: 0.9854
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1407 - auc: 0.9852
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1418 - auc: 0.9850
## 499/603 [=====>.....] - ETA: 1s - loss: 0.1416 - auc: 0.9851
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1418 - auc: 0.9851
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1413 - auc: 0.9852
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1412 - auc: 0.9852
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1413 - auc: 0.9852
## 517/603 [=====>.....] - ETA: 1s - loss: 0.1410 - auc: 0.9852
## 521/603 [=====>.....] - ETA: 1s - loss: 0.1409 - auc: 0.9852
## 524/603 [=====>....] - ETA: 1s - loss: 0.1416 - auc: 0.9851
## 528/603 [=====>....] - ETA: 1s - loss: 0.1416 - auc: 0.9851
## 531/603 [=====>....] - ETA: 1s - loss: 0.1415 - auc: 0.9851
## 534/603 [=====>....] - ETA: 1s - loss: 0.1416 - auc: 0.9851
## 538/603 [=====>....] - ETA: 1s - loss: 0.1413 - auc: 0.9852
## 542/603 [=====>....] - ETA: 0s - loss: 0.1412 - auc: 0.9852
## 546/603 [=====>...] - ETA: 0s - loss: 0.1405 - auc: 0.9853
## 550/603 [=====>...] - ETA: 0s - loss: 0.1400 - auc: 0.9854
## 553/603 [=====>...] - ETA: 0s - loss: 0.1402 - auc: 0.9854
## 557/603 [=====>...] - ETA: 0s - loss: 0.1401 - auc: 0.9854
## 561/603 [=====>...] - ETA: 0s - loss: 0.1402 - auc: 0.9854
## 565/603 [=====>..] - ETA: 0s - loss: 0.1404 - auc: 0.9854
## 568/603 [=====>..] - ETA: 0s - loss: 0.1404 - auc: 0.9854
## 571/603 [=====>..] - ETA: 0s - loss: 0.1405 - auc: 0.9854
## 575/603 [=====>..] - ETA: 0s - loss: 0.1406 - auc: 0.9854
## 579/603 [=====>..] - ETA: 0s - loss: 0.1407 - auc: 0.9853
## 583/603 [=====>.] - ETA: 0s - loss: 0.1404 - auc: 0.9854
## 586/603 [=====>.] - ETA: 0s - loss: 0.1404 - auc: 0.9854
## 590/603 [=====>.] - ETA: 0s - loss: 0.1409 - auc: 0.9853
## 594/603 [=====>.] - ETA: 0s - loss: 0.1406 - auc: 0.9853
## 598/603 [=====>.] - ETA: 0s - loss: 0.1407 - auc: 0.9853
## 602/603 [=====>.] - ETA: 0s - loss: 0.1413 - auc: 0.9851
## 603/603 [=====] - 10s 17ms/step - loss: 0.1412 - auc: 0.9852 - va
l_loss: 0.3616 - val_auc: 0.9752
## Epoch 5/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0228 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.1148 - auc: 0.9892
## 9/603 [.....] - ETA: 9s - loss: 0.1056 - auc: 0.9916
## 13/603 [.....] - ETA: 9s - loss: 0.0962 - auc: 0.9933
## 17/603 [.....] - ETA: 9s - loss: 0.1248 - auc: 0.9868
## 20/603 [.....] - ETA: 9s - loss: 0.1291 - auc: 0.9865
## 24/603 [>.....] - ETA: 9s - loss: 0.1275 - auc: 0.9872
## 28/603 [>.....] - ETA: 9s - loss: 0.1240 - auc: 0.9880
## 32/603 [>.....] - ETA: 9s - loss: 0.1202 - auc: 0.9888
## 36/603 [>.....] - ETA: 9s - loss: 0.1218 - auc: 0.9884
```

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## 40/603 [>.....] - ETA: 9s - loss: 0.1246 - auc: 0.9880
## 44/603 [=>.....] - ETA: 9s - loss: 0.1183 - auc: 0.9891
## 47/603 [=>.....] - ETA: 9s - loss: 0.1187 - auc: 0.9892
## 51/603 [=>.....] - ETA: 9s - loss: 0.1138 - auc: 0.9900
## 55/603 [=>.....] - ETA: 8s - loss: 0.1162 - auc: 0.9892
## 59/603 [=>.....] - ETA: 8s - loss: 0.1157 - auc: 0.9893
## 63/603 [==>.....] - ETA: 8s - loss: 0.1150 - auc: 0.9891
## 66/603 [==>.....] - ETA: 8s - loss: 0.1177 - auc: 0.9890
## 70/603 [==>.....] - ETA: 8s - loss: 0.1154 - auc: 0.9895
## 73/603 [==>.....] - ETA: 8s - loss: 0.1152 - auc: 0.9897
## 77/603 [==>.....] - ETA: 8s - loss: 0.1137 - auc: 0.9900
## 81/603 [===>.....] - ETA: 8s - loss: 0.1202 - auc: 0.9892
## 85/603 [===>.....] - ETA: 8s - loss: 0.1219 - auc: 0.9885
## 88/603 [===>.....] - ETA: 8s - loss: 0.1221 - auc: 0.9883
## 92/603 [===>.....] - ETA: 8s - loss: 0.1217 - auc: 0.9885
## 95/603 [===>.....] - ETA: 8s - loss: 0.1245 - auc: 0.9880
## 99/603 [===>.....] - ETA: 8s - loss: 0.1263 - auc: 0.9873
## 103/603 [====>.....] - ETA: 8s - loss: 0.1253 - auc: 0.9874
## 106/603 [====>.....] - ETA: 8s - loss: 0.1249 - auc: 0.9875
## 109/603 [====>.....] - ETA: 8s - loss: 0.1244 - auc: 0.9877
## 112/603 [====>.....] - ETA: 8s - loss: 0.1267 - auc: 0.9871
## 116/603 [====>.....] - ETA: 7s - loss: 0.1282 - auc: 0.9870
## 120/603 [====>.....] - ETA: 7s - loss: 0.1275 - auc: 0.9872
## 124/603 [====>.....] - ETA: 7s - loss: 0.1275 - auc: 0.9872
## 128/603 [====>.....] - ETA: 7s - loss: 0.1302 - auc: 0.9868
## 132/603 [====>.....] - ETA: 7s - loss: 0.1322 - auc: 0.9862
## 136/603 [====>.....] - ETA: 7s - loss: 0.1305 - auc: 0.9866
## 140/603 [====>.....] - ETA: 7s - loss: 0.1283 - auc: 0.9870
## 144/603 [====>.....] - ETA: 7s - loss: 0.1287 - auc: 0.9869
## 148/603 [====>.....] - ETA: 7s - loss: 0.1294 - auc: 0.9868
## 152/603 [====>.....] - ETA: 7s - loss: 0.1297 - auc: 0.9868
## 156/603 [====>.....] - ETA: 7s - loss: 0.1293 - auc: 0.9869
## 160/603 [====>.....] - ETA: 7s - loss: 0.1291 - auc: 0.9870
## 164/603 [====>.....] - ETA: 7s - loss: 0.1298 - auc: 0.9869
## 167/603 [====>.....] - ETA: 7s - loss: 0.1293 - auc: 0.9870
## 171/603 [====>.....] - ETA: 7s - loss: 0.1283 - auc: 0.9873
## 175/603 [====>.....] - ETA: 7s - loss: 0.1286 - auc: 0.9873
## 179/603 [====>.....] - ETA: 6s - loss: 0.1294 - auc: 0.9871
## 183/603 [====>.....] - ETA: 6s - loss: 0.1332 - auc: 0.9867
## 187/603 [====>.....] - ETA: 6s - loss: 0.1313 - auc: 0.9870
## 191/603 [====>.....] - ETA: 6s - loss: 0.1312 - auc: 0.9870
## 195/603 [====>.....] - ETA: 6s - loss: 0.1321 - auc: 0.9869
## 199/603 [====>.....] - ETA: 6s - loss: 0.1311 - auc: 0.9872
## 203/603 [====>.....] - ETA: 6s - loss: 0.1294 - auc: 0.9874
## 207/603 [====>.....] - ETA: 6s - loss: 0.1290 - auc: 0.9874
## 211/603 [====>.....] - ETA: 6s - loss: 0.1304 - auc: 0.9871
## 215/603 [====>.....] - ETA: 6s - loss: 0.1311 - auc: 0.9869
## 219/603 [====>.....] - ETA: 6s - loss: 0.1309 - auc: 0.9869
## 223/603 [====>.....] - ETA: 6s - loss: 0.1314 - auc: 0.9869
## 227/603 [====>.....] - ETA: 6s - loss: 0.1322 - auc: 0.9867
## 231/603 [====>.....] - ETA: 6s - loss: 0.1335 - auc: 0.9866
## 234/603 [====>.....] - ETA: 6s - loss: 0.1330 - auc: 0.9866
## 238/603 [====>.....] - ETA: 5s - loss: 0.1315 - auc: 0.9868
## 242/603 [====>.....] - ETA: 5s - loss: 0.1329 - auc: 0.9867
## 246/603 [====>.....] - ETA: 5s - loss: 0.1334 - auc: 0.9865
## 249/603 [====>.....] - ETA: 5s - loss: 0.1334 - auc: 0.9865
```

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## 253/603 [=====>.....] - ETA: 5s - loss: 0.1351 - auc: 0.9862
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1351 - auc: 0.9861
## 260/603 [=====>.....] - ETA: 5s - loss: 0.1356 - auc: 0.9861
## 264/603 [=====>.....] - ETA: 5s - loss: 0.1364 - auc: 0.9859
## 267/603 [=====>.....] - ETA: 5s - loss: 0.1366 - auc: 0.9859
## 270/603 [=====>.....] - ETA: 5s - loss: 0.1371 - auc: 0.9858
## 273/603 [=====>.....] - ETA: 5s - loss: 0.1366 - auc: 0.9859
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1362 - auc: 0.9860
## 281/603 [=====>.....] - ETA: 5s - loss: 0.1359 - auc: 0.9861
## 285/603 [=====>.....] - ETA: 5s - loss: 0.1360 - auc: 0.9861
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1364 - auc: 0.9860
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1366 - auc: 0.9860
## 296/603 [=====>.....] - ETA: 5s - loss: 0.1373 - auc: 0.9859
## 300/603 [=====>.....] - ETA: 4s - loss: 0.1385 - auc: 0.9856
## 304/603 [=====>.....] - ETA: 4s - loss: 0.1378 - auc: 0.9858
## 308/603 [=====>.....] - ETA: 4s - loss: 0.1380 - auc: 0.9857
## 311/603 [=====>.....] - ETA: 4s - loss: 0.1377 - auc: 0.9858
## 314/603 [=====>.....] - ETA: 4s - loss: 0.1376 - auc: 0.9858
## 318/603 [=====>.....] - ETA: 4s - loss: 0.1373 - auc: 0.9858
## 321/603 [=====>.....] - ETA: 4s - loss: 0.1372 - auc: 0.9858
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1363 - auc: 0.9860
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1362 - auc: 0.9860
## 329/603 [=====>.....] - ETA: 4s - loss: 0.1352 - auc: 0.9862
## 332/603 [=====>.....] - ETA: 4s - loss: 0.1352 - auc: 0.9862
## 336/603 [=====>.....] - ETA: 4s - loss: 0.1346 - auc: 0.9863
## 340/603 [=====>.....] - ETA: 4s - loss: 0.1347 - auc: 0.9863
## 344/603 [=====>.....] - ETA: 4s - loss: 0.1349 - auc: 0.9862
## 348/603 [=====>.....] - ETA: 4s - loss: 0.1346 - auc: 0.9863
## 352/603 [=====>.....] - ETA: 4s - loss: 0.1345 - auc: 0.9864
## 355/603 [=====>.....] - ETA: 4s - loss: 0.1345 - auc: 0.9864
## 359/603 [=====>.....] - ETA: 4s - loss: 0.1337 - auc: 0.9865
## 362/603 [=====>.....] - ETA: 4s - loss: 0.1340 - auc: 0.9864
## 366/603 [=====>.....] - ETA: 4s - loss: 0.1348 - auc: 0.9863
## 370/603 [=====>.....] - ETA: 3s - loss: 0.1345 - auc: 0.9863
## 374/603 [=====>.....] - ETA: 3s - loss: 0.1338 - auc: 0.9865
## 378/603 [=====>.....] - ETA: 3s - loss: 0.1335 - auc: 0.9865
## 382/603 [=====>.....] - ETA: 3s - loss: 0.1333 - auc: 0.9865
## 385/603 [=====>.....] - ETA: 3s - loss: 0.1328 - auc: 0.9866
## 389/603 [=====>.....] - ETA: 3s - loss: 0.1330 - auc: 0.9865
## 393/603 [=====>.....] - ETA: 3s - loss: 0.1333 - auc: 0.9865
## 397/603 [=====>.....] - ETA: 3s - loss: 0.1334 - auc: 0.9864
## 401/603 [=====>.....] - ETA: 3s - loss: 0.1336 - auc: 0.9864
## 405/603 [=====>.....] - ETA: 3s - loss: 0.1336 - auc: 0.9864
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1329 - auc: 0.9866
## 413/603 [=====>.....] - ETA: 3s - loss: 0.1332 - auc: 0.9865
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1332 - auc: 0.9865
## 421/603 [=====>.....] - ETA: 3s - loss: 0.1332 - auc: 0.9865
## 424/603 [=====>.....] - ETA: 3s - loss: 0.1325 - auc: 0.9866
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1325 - auc: 0.9866
## 432/603 [=====>.....] - ETA: 2s - loss: 0.1327 - auc: 0.9866
## 435/603 [=====>.....] - ETA: 2s - loss: 0.1338 - auc: 0.9864
## 438/603 [=====>.....] - ETA: 2s - loss: 0.1344 - auc: 0.9863
## 442/603 [=====>.....] - ETA: 2s - loss: 0.1346 - auc: 0.9863
## 445/603 [=====>.....] - ETA: 2s - loss: 0.1345 - auc: 0.9863
## 449/603 [=====>.....] - ETA: 2s - loss: 0.1339 - auc: 0.9864
## 453/603 [=====>.....] - ETA: 2s - loss: 0.1343 - auc: 0.9864
```

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## 456/603 [=====>.....] - ETA: 2s - loss: 0.1338 - auc: 0.9865
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1338 - auc: 0.9864
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1336 - auc: 0.9865
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1333 - auc: 0.9866
## 471/603 [=====>.....] - ETA: 2s - loss: 0.1333 - auc: 0.9866
## 475/603 [=====>.....] - ETA: 2s - loss: 0.1330 - auc: 0.9867
## 479/603 [=====>.....] - ETA: 2s - loss: 0.1331 - auc: 0.9867
## 483/603 [=====>.....] - ETA: 2s - loss: 0.1324 - auc: 0.9868
## 487/603 [=====>.....] - ETA: 1s - loss: 0.1330 - auc: 0.9867
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1332 - auc: 0.9866
## 494/603 [=====>.....] - ETA: 1s - loss: 0.1326 - auc: 0.9867
## 498/603 [=====>.....] - ETA: 1s - loss: 0.1324 - auc: 0.9868
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1323 - auc: 0.9868
## 505/603 [=====>.....] - ETA: 1s - loss: 0.1321 - auc: 0.9869
## 508/603 [=====>.....] - ETA: 1s - loss: 0.1317 - auc: 0.9870
## 512/603 [=====>.....] - ETA: 1s - loss: 0.1318 - auc: 0.9870
## 516/603 [=====>.....] - ETA: 1s - loss: 0.1315 - auc: 0.9870
## 520/603 [=====>.....] - ETA: 1s - loss: 0.1321 - auc: 0.9869
## 524/603 [=====>....] - ETA: 1s - loss: 0.1320 - auc: 0.9869
## 528/603 [=====>....] - ETA: 1s - loss: 0.1319 - auc: 0.9870
## 532/603 [=====>....] - ETA: 1s - loss: 0.1316 - auc: 0.9870
## 535/603 [=====>....] - ETA: 1s - loss: 0.1321 - auc: 0.9869
## 539/603 [=====>....] - ETA: 1s - loss: 0.1325 - auc: 0.9869
## 543/603 [=====>...] - ETA: 1s - loss: 0.1326 - auc: 0.9869
## 547/603 [=====>...] - ETA: 0s - loss: 0.1324 - auc: 0.9869
## 551/603 [=====>...] - ETA: 0s - loss: 0.1322 - auc: 0.9870
## 555/603 [=====>...] - ETA: 0s - loss: 0.1316 - auc: 0.9871
## 558/603 [=====>...] - ETA: 0s - loss: 0.1315 - auc: 0.9871
## 562/603 [=====>...] - ETA: 0s - loss: 0.1317 - auc: 0.9871
## 566/603 [=====>..] - ETA: 0s - loss: 0.1322 - auc: 0.9870
## 570/603 [=====>..] - ETA: 0s - loss: 0.1322 - auc: 0.9870
## 574/603 [=====>..] - ETA: 0s - loss: 0.1330 - auc: 0.9869
## 578/603 [=====>..] - ETA: 0s - loss: 0.1330 - auc: 0.9869
## 582/603 [=====>..] - ETA: 0s - loss: 0.1330 - auc: 0.9869
## 585/603 [=====>.] - ETA: 0s - loss: 0.1329 - auc: 0.9869
## 589/603 [=====>.] - ETA: 0s - loss: 0.1328 - auc: 0.9869
## 593/603 [=====>.] - ETA: 0s - loss: 0.1333 - auc: 0.9868
## 597/603 [=====>.] - ETA: 0s - loss: 0.1342 - auc: 0.9866
## 600/603 [=====>.] - ETA: 0s - loss: 0.1345 - auc: 0.9866
## 603/603 [=====] - 11s 17ms/step - loss: 0.1342 - auc: 0.9866 - va
l_loss: 0.2225 - val_auc: 0.9813
## Epoch 6/20
##
## 1/603 [.....] - ETA: 11s - loss: 0.1654 - auc: 0.9960
## 5/603 [.....] - ETA: 9s - loss: 0.0819 - auc: 0.9959
## 8/603 [.....] - ETA: 9s - loss: 0.0746 - auc: 0.9970
## 12/603 [.....] - ETA: 9s - loss: 0.0672 - auc: 0.9975
## 16/603 [.....] - ETA: 9s - loss: 0.0679 - auc: 0.9974
## 20/603 [.....] - ETA: 9s - loss: 0.0920 - auc: 0.9937
## 24/603 [>.....] - ETA: 9s - loss: 0.0978 - auc: 0.9919
## 27/603 [>.....] - ETA: 9s - loss: 0.0986 - auc: 0.9913
## 31/603 [>.....] - ETA: 9s - loss: 0.0985 - auc: 0.9917
## 35/603 [>.....] - ETA: 9s - loss: 0.1082 - auc: 0.9895
## 39/603 [>.....] - ETA: 9s - loss: 0.1218 - auc: 0.9878
## 43/603 [=>.....] - ETA: 9s - loss: 0.1184 - auc: 0.9887
## 47/603 [=>.....] - ETA: 9s - loss: 0.1145 - auc: 0.9896
```

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## 51/603 [=>.....] - ETA: 8s - loss: 0.1129 - auc: 0.9901
## 54/603 [=>.....] - ETA: 8s - loss: 0.1205 - auc: 0.9883
## 57/603 [=>.....] - ETA: 8s - loss: 0.1199 - auc: 0.9886
## 60/603 [=>.....] - ETA: 8s - loss: 0.1176 - auc: 0.9890
## 63/603 [==>.....] - ETA: 8s - loss: 0.1198 - auc: 0.9883
## 66/603 [==>.....] - ETA: 8s - loss: 0.1189 - auc: 0.9884
## 70/603 [==>.....] - ETA: 8s - loss: 0.1216 - auc: 0.9884
## 74/603 [==>.....] - ETA: 8s - loss: 0.1245 - auc: 0.9877
## 78/603 [==>.....] - ETA: 8s - loss: 0.1255 - auc: 0.9877
## 81/603 [===>.....] - ETA: 8s - loss: 0.1289 - auc: 0.9870
## 85/603 [===>.....] - ETA: 8s - loss: 0.1291 - auc: 0.9870
## 89/603 [===>.....] - ETA: 8s - loss: 0.1266 - auc: 0.9875
## 93/603 [===>.....] - ETA: 8s - loss: 0.1247 - auc: 0.9878
## 97/603 [===>.....] - ETA: 8s - loss: 0.1273 - auc: 0.9873
## 101/603 [====>.....] - ETA: 8s - loss: 0.1255 - auc: 0.9876
## 104/603 [====>.....] - ETA: 8s - loss: 0.1265 - auc: 0.9874
## 108/603 [====>.....] - ETA: 8s - loss: 0.1262 - auc: 0.9875
## 111/603 [====>.....] - ETA: 8s - loss: 0.1259 - auc: 0.9874
## 115/603 [====>.....] - ETA: 7s - loss: 0.1306 - auc: 0.9869
## 118/603 [====>.....] - ETA: 7s - loss: 0.1296 - auc: 0.9871
## 122/603 [====>.....] - ETA: 7s - loss: 0.1271 - auc: 0.9875
## 126/603 [====>.....] - ETA: 7s - loss: 0.1272 - auc: 0.9875
## 130/603 [====>.....] - ETA: 7s - loss: 0.1287 - auc: 0.9871
## 134/603 [====>.....] - ETA: 7s - loss: 0.1300 - auc: 0.9869
## 138/603 [====>.....] - ETA: 7s - loss: 0.1293 - auc: 0.9870
## 142/603 [====>.....] - ETA: 7s - loss: 0.1304 - auc: 0.9868
## 146/603 [====>.....] - ETA: 7s - loss: 0.1296 - auc: 0.9871
## 150/603 [====>.....] - ETA: 7s - loss: 0.1296 - auc: 0.9872
## 153/603 [====>.....] - ETA: 7s - loss: 0.1301 - auc: 0.9871
## 157/603 [====>.....] - ETA: 7s - loss: 0.1314 - auc: 0.9869
## 161/603 [====>.....] - ETA: 7s - loss: 0.1309 - auc: 0.9871
## 165/603 [====>.....] - ETA: 7s - loss: 0.1331 - auc: 0.9866
## 169/603 [====>.....] - ETA: 7s - loss: 0.1319 - auc: 0.9868
## 172/603 [====>.....] - ETA: 7s - loss: 0.1311 - auc: 0.9870
## 176/603 [====>.....] - ETA: 6s - loss: 0.1310 - auc: 0.9871
## 180/603 [====>.....] - ETA: 6s - loss: 0.1303 - auc: 0.9872
## 184/603 [====>.....] - ETA: 6s - loss: 0.1296 - auc: 0.9874
## 187/603 [====>.....] - ETA: 6s - loss: 0.1310 - auc: 0.9870
## 190/603 [====>.....] - ETA: 6s - loss: 0.1300 - auc: 0.9872
## 194/603 [====>.....] - ETA: 6s - loss: 0.1310 - auc: 0.9870
## 198/603 [====>.....] - ETA: 6s - loss: 0.1305 - auc: 0.9872
## 202/603 [====>.....] - ETA: 6s - loss: 0.1297 - auc: 0.9873
## 206/603 [====>.....] - ETA: 6s - loss: 0.1327 - auc: 0.9868
## 210/603 [====>.....] - ETA: 6s - loss: 0.1321 - auc: 0.9869
## 213/603 [====>.....] - ETA: 6s - loss: 0.1329 - auc: 0.9867
## 217/603 [====>.....] - ETA: 6s - loss: 0.1323 - auc: 0.9867
## 221/603 [====>.....] - ETA: 6s - loss: 0.1321 - auc: 0.9868
## 225/603 [====>.....] - ETA: 6s - loss: 0.1316 - auc: 0.9869
## 228/603 [====>.....] - ETA: 6s - loss: 0.1322 - auc: 0.9868
## 231/603 [====>.....] - ETA: 6s - loss: 0.1320 - auc: 0.9868
## 234/603 [====>.....] - ETA: 6s - loss: 0.1321 - auc: 0.9869
## 237/603 [====>.....] - ETA: 6s - loss: 0.1316 - auc: 0.9870
## 241/603 [====>.....] - ETA: 5s - loss: 0.1320 - auc: 0.9869
## 245/603 [====>.....] - ETA: 5s - loss: 0.1310 - auc: 0.9871
## 249/603 [====>.....] - ETA: 5s - loss: 0.1310 - auc: 0.9871
## 252/603 [====>.....] - ETA: 5s - loss: 0.1306 - auc: 0.9872
```

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## 256/603 [=====>.....] - ETA: 5s - loss: 0.1294 - auc: 0.9874
## 260/603 [=====>.....] - ETA: 5s - loss: 0.1293 - auc: 0.9874
## 264/603 [=====>.....] - ETA: 5s - loss: 0.1295 - auc: 0.9873
## 268/603 [=====>.....] - ETA: 5s - loss: 0.1287 - auc: 0.9875
## 272/603 [=====>.....] - ETA: 5s - loss: 0.1280 - auc: 0.9876
## 276/603 [=====>.....] - ETA: 5s - loss: 0.1276 - auc: 0.9877
## 280/603 [=====>.....] - ETA: 5s - loss: 0.1276 - auc: 0.9877
## 284/603 [=====>.....] - ETA: 5s - loss: 0.1276 - auc: 0.9878
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1281 - auc: 0.9876
## 291/603 [=====>.....] - ETA: 5s - loss: 0.1276 - auc: 0.9877
## 294/603 [=====>.....] - ETA: 5s - loss: 0.1273 - auc: 0.9877
## 298/603 [=====>.....] - ETA: 5s - loss: 0.1279 - auc: 0.9876
## 302/603 [=====>.....] - ETA: 4s - loss: 0.1281 - auc: 0.9876
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1276 - auc: 0.9877
## 309/603 [=====>.....] - ETA: 4s - loss: 0.1271 - auc: 0.9878
## 312/603 [=====>.....] - ETA: 4s - loss: 0.1269 - auc: 0.9878
## 315/603 [=====>.....] - ETA: 4s - loss: 0.1265 - auc: 0.9879
## 319/603 [=====>.....] - ETA: 4s - loss: 0.1264 - auc: 0.9880
## 323/603 [=====>.....] - ETA: 4s - loss: 0.1256 - auc: 0.9881
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1272 - auc: 0.9878
## 331/603 [=====>.....] - ETA: 4s - loss: 0.1264 - auc: 0.9880
## 335/603 [=====>.....] - ETA: 4s - loss: 0.1261 - auc: 0.9881
## 338/603 [=====>.....] - ETA: 4s - loss: 0.1262 - auc: 0.9881
## 341/603 [=====>.....] - ETA: 4s - loss: 0.1263 - auc: 0.9880
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1273 - auc: 0.9879
## 349/603 [=====>.....] - ETA: 4s - loss: 0.1275 - auc: 0.9878
## 352/603 [=====>.....] - ETA: 4s - loss: 0.1272 - auc: 0.9879
## 356/603 [=====>.....] - ETA: 4s - loss: 0.1280 - auc: 0.9877
## 360/603 [=====>.....] - ETA: 3s - loss: 0.1278 - auc: 0.9878
## 364/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9878
## 367/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9878
## 371/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9878
## 375/603 [=====>.....] - ETA: 3s - loss: 0.1283 - auc: 0.9877
## 378/603 [=====>.....] - ETA: 3s - loss: 0.1279 - auc: 0.9878
## 382/603 [=====>.....] - ETA: 3s - loss: 0.1278 - auc: 0.9878
## 386/603 [=====>.....] - ETA: 3s - loss: 0.1279 - auc: 0.9878
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1280 - auc: 0.9878
## 394/603 [=====>.....] - ETA: 3s - loss: 0.1280 - auc: 0.9878
## 398/603 [=====>.....] - ETA: 3s - loss: 0.1286 - auc: 0.9877
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1280 - auc: 0.9878
## 406/603 [=====>.....] - ETA: 3s - loss: 0.1285 - auc: 0.9877
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1283 - auc: 0.9878
## 413/603 [=====>.....] - ETA: 3s - loss: 0.1283 - auc: 0.9877
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1287 - auc: 0.9877
## 421/603 [=====>.....] - ETA: 2s - loss: 0.1291 - auc: 0.9876
## 425/603 [=====>.....] - ETA: 2s - loss: 0.1296 - auc: 0.9875
## 429/603 [=====>.....] - ETA: 2s - loss: 0.1306 - auc: 0.9874
## 433/603 [=====>.....] - ETA: 2s - loss: 0.1303 - auc: 0.9874
## 437/603 [=====>.....] - ETA: 2s - loss: 0.1304 - auc: 0.9874
## 441/603 [=====>.....] - ETA: 2s - loss: 0.1303 - auc: 0.9874
## 444/603 [=====>.....] - ETA: 2s - loss: 0.1310 - auc: 0.9872
## 448/603 [=====>.....] - ETA: 2s - loss: 0.1305 - auc: 0.9873
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1305 - auc: 0.9873
## 455/603 [=====>.....] - ETA: 2s - loss: 0.1307 - auc: 0.9873
## 458/603 [=====>.....] - ETA: 2s - loss: 0.1305 - auc: 0.9873
## 461/603 [=====>.....] - ETA: 2s - loss: 0.1306 - auc: 0.9873
```

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## 465/603 [=====>.....] - ETA: 2s - loss: 0.1302 - auc: 0.9874
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1297 - auc: 0.9874
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1295 - auc: 0.9875
## 475/603 [=====>.....] - ETA: 2s - loss: 0.1291 - auc: 0.9876
## 478/603 [=====>.....] - ETA: 2s - loss: 0.1291 - auc: 0.9876
## 482/603 [=====>.....] - ETA: 1s - loss: 0.1288 - auc: 0.9876
## 485/603 [=====>.....] - ETA: 1s - loss: 0.1289 - auc: 0.9876
## 488/603 [=====>.....] - ETA: 1s - loss: 0.1288 - auc: 0.9876
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1286 - auc: 0.9877
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1283 - auc: 0.9878
## 499/603 [=====>.....] - ETA: 1s - loss: 0.1280 - auc: 0.9878
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1279 - auc: 0.9878
## 505/603 [=====>.....] - ETA: 1s - loss: 0.1282 - auc: 0.9877
## 509/603 [=====>.....] - ETA: 1s - loss: 0.1280 - auc: 0.9878
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1279 - auc: 0.9878
## 517/603 [=====>.....] - ETA: 1s - loss: 0.1284 - auc: 0.9877
## 521/603 [=====>.....] - ETA: 1s - loss: 0.1282 - auc: 0.9877
## 524/603 [=====>....] - ETA: 1s - loss: 0.1278 - auc: 0.9878
## 528/603 [=====>....] - ETA: 1s - loss: 0.1276 - auc: 0.9878
## 532/603 [=====>....] - ETA: 1s - loss: 0.1271 - auc: 0.9879
## 535/603 [=====>....] - ETA: 1s - loss: 0.1276 - auc: 0.9878
## 538/603 [=====>....] - ETA: 1s - loss: 0.1272 - auc: 0.9879
## 542/603 [=====>....] - ETA: 1s - loss: 0.1274 - auc: 0.9879
## 545/603 [=====>...] - ETA: 0s - loss: 0.1276 - auc: 0.9878
## 549/603 [=====>...] - ETA: 0s - loss: 0.1281 - auc: 0.9877
## 553/603 [=====>...] - ETA: 0s - loss: 0.1277 - auc: 0.9878
## 557/603 [=====>...] - ETA: 0s - loss: 0.1278 - auc: 0.9878
## 561/603 [=====>...] - ETA: 0s - loss: 0.1277 - auc: 0.9878
## 565/603 [=====>..] - ETA: 0s - loss: 0.1279 - auc: 0.9877
## 569/603 [=====>..] - ETA: 0s - loss: 0.1275 - auc: 0.9878
## 572/603 [=====>..] - ETA: 0s - loss: 0.1272 - auc: 0.9879
## 576/603 [=====>..] - ETA: 0s - loss: 0.1272 - auc: 0.9879
## 580/603 [=====>..] - ETA: 0s - loss: 0.1275 - auc: 0.9878
## 584/603 [=====>.] - ETA: 0s - loss: 0.1273 - auc: 0.9879
## 588/603 [=====>.] - ETA: 0s - loss: 0.1270 - auc: 0.9879
## 591/603 [=====>.] - ETA: 0s - loss: 0.1275 - auc: 0.9879
## 595/603 [=====>.] - ETA: 0s - loss: 0.1274 - auc: 0.9879
## 599/603 [=====>.] - ETA: 0s - loss: 0.1271 - auc: 0.9879
## 602/603 [=====>.] - ETA: 0s - loss: 0.1267 - auc: 0.9880
## 603/603 [=====] - 10s 17ms/step - loss: 0.1267 - auc: 0.9880 - va
l_loss: 0.1586 - val_auc: 0.9867
## Epoch 7/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0425 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.1090 - auc: 0.9896
## 8/603 [.....] - ETA: 9s - loss: 0.1028 - auc: 0.9933
## 12/603 [.....] - ETA: 9s - loss: 0.1101 - auc: 0.9909
## 16/603 [.....] - ETA: 9s - loss: 0.1253 - auc: 0.9899
## 20/603 [.....] - ETA: 9s - loss: 0.1332 - auc: 0.9857
## 24/603 [>.....] - ETA: 9s - loss: 0.1242 - auc: 0.9868
## 27/603 [>.....] - ETA: 9s - loss: 0.1275 - auc: 0.9862
## 30/603 [>.....] - ETA: 9s - loss: 0.1249 - auc: 0.9870
## 34/603 [>.....] - ETA: 9s - loss: 0.1221 - auc: 0.9872
## 38/603 [>.....] - ETA: 9s - loss: 0.1184 - auc: 0.9881
## 41/603 [=>.....] - ETA: 9s - loss: 0.1134 - auc: 0.9891
## 45/603 [=>.....] - ETA: 9s - loss: 0.1231 - auc: 0.9874
```

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## 49/603 [=>.....] - ETA: 9s - loss: 0.1224 - auc: 0.9877
## 53/603 [=>.....] - ETA: 9s - loss: 0.1292 - auc: 0.9868
## 56/603 [=>.....] - ETA: 9s - loss: 0.1289 - auc: 0.9871
## 59/603 [=>.....] - ETA: 8s - loss: 0.1300 - auc: 0.9871
## 63/603 [==>.....] - ETA: 8s - loss: 0.1303 - auc: 0.9870
## 67/603 [==>.....] - ETA: 8s - loss: 0.1285 - auc: 0.9874
## 71/603 [==>.....] - ETA: 8s - loss: 0.1310 - auc: 0.9871
## 75/603 [==>.....] - ETA: 8s - loss: 0.1321 - auc: 0.9870
## 79/603 [==>.....] - ETA: 8s - loss: 0.1304 - auc: 0.9874
## 82/603 [===>.....] - ETA: 8s - loss: 0.1292 - auc: 0.9876
## 86/603 [===>.....] - ETA: 8s - loss: 0.1308 - auc: 0.9874
## 90/603 [===>.....] - ETA: 8s - loss: 0.1293 - auc: 0.9877
## 94/603 [===>.....] - ETA: 8s - loss: 0.1283 - auc: 0.9880
## 98/603 [===>.....] - ETA: 8s - loss: 0.1311 - auc: 0.9872
## 102/603 [====>.....] - ETA: 8s - loss: 0.1294 - auc: 0.9875
## 106/603 [====>.....] - ETA: 8s - loss: 0.1296 - auc: 0.9875
## 110/603 [====>.....] - ETA: 8s - loss: 0.1292 - auc: 0.9877
## 114/603 [====>.....] - ETA: 8s - loss: 0.1304 - auc: 0.9874
## 117/603 [====>.....] - ETA: 8s - loss: 0.1299 - auc: 0.9875
## 121/603 [=====>.....] - ETA: 7s - loss: 0.1286 - auc: 0.9877
## 125/603 [=====>.....] - ETA: 7s - loss: 0.1263 - auc: 0.9880
## 128/603 [=====>.....] - ETA: 7s - loss: 0.1261 - auc: 0.9881
## 132/603 [=====>.....] - ETA: 7s - loss: 0.1256 - auc: 0.9883
## 136/603 [=====>.....] - ETA: 7s - loss: 0.1250 - auc: 0.9884
## 140/603 [=====>.....] - ETA: 7s - loss: 0.1242 - auc: 0.9886
## 143/603 [=====>.....] - ETA: 7s - loss: 0.1242 - auc: 0.9886
## 147/603 [=====>.....] - ETA: 7s - loss: 0.1246 - auc: 0.9886
## 150/603 [=====>.....] - ETA: 7s - loss: 0.1242 - auc: 0.9886
## 154/603 [=====>.....] - ETA: 7s - loss: 0.1235 - auc: 0.9888
## 158/603 [=====>.....] - ETA: 7s - loss: 0.1241 - auc: 0.9888
## 162/603 [=====>.....] - ETA: 7s - loss: 0.1255 - auc: 0.9886
## 166/603 [=====>.....] - ETA: 7s - loss: 0.1249 - auc: 0.9887
## 169/603 [=====>.....] - ETA: 7s - loss: 0.1245 - auc: 0.9888
## 172/603 [=====>.....] - ETA: 7s - loss: 0.1267 - auc: 0.9884
## 176/603 [=====>.....] - ETA: 7s - loss: 0.1267 - auc: 0.9884
## 179/603 [=====>.....] - ETA: 6s - loss: 0.1265 - auc: 0.9885
## 183/603 [=====>.....] - ETA: 6s - loss: 0.1267 - auc: 0.9885
## 186/603 [=====>.....] - ETA: 6s - loss: 0.1267 - auc: 0.9885
## 190/603 [=====>.....] - ETA: 6s - loss: 0.1287 - auc: 0.9882
## 194/603 [=====>.....] - ETA: 6s - loss: 0.1291 - auc: 0.9882
## 198/603 [=====>.....] - ETA: 6s - loss: 0.1278 - auc: 0.9883
## 202/603 [=====>.....] - ETA: 6s - loss: 0.1285 - auc: 0.9883
## 205/603 [=====>.....] - ETA: 6s - loss: 0.1283 - auc: 0.9882
## 209/603 [=====>.....] - ETA: 6s - loss: 0.1287 - auc: 0.9881
## 213/603 [=====>.....] - ETA: 6s - loss: 0.1280 - auc: 0.9883
## 217/603 [=====>.....] - ETA: 6s - loss: 0.1278 - auc: 0.9883
## 221/603 [=====>.....] - ETA: 6s - loss: 0.1280 - auc: 0.9882
## 225/603 [=====>.....] - ETA: 6s - loss: 0.1276 - auc: 0.9883
## 229/603 [=====>.....] - ETA: 6s - loss: 0.1296 - auc: 0.9879
## 233/603 [=====>.....] - ETA: 6s - loss: 0.1294 - auc: 0.9879
## 236/603 [=====>.....] - ETA: 6s - loss: 0.1292 - auc: 0.9879
## 239/603 [=====>.....] - ETA: 5s - loss: 0.1295 - auc: 0.9878
## 242/603 [=====>.....] - ETA: 5s - loss: 0.1300 - auc: 0.9877
## 246/603 [=====>.....] - ETA: 5s - loss: 0.1312 - auc: 0.9874
## 250/603 [=====>.....] - ETA: 5s - loss: 0.1320 - auc: 0.9873
## 254/603 [=====>.....] - ETA: 5s - loss: 0.1319 - auc: 0.9873
```

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## 258/603 [=====>.....] - ETA: 5s - loss: 0.1309 - auc: 0.9874
## 261/603 [=====>.....] - ETA: 5s - loss: 0.1313 - auc: 0.9874
## 265/603 [=====>.....] - ETA: 5s - loss: 0.1324 - auc: 0.9873
## 269/603 [=====>.....] - ETA: 5s - loss: 0.1328 - auc: 0.9872
## 272/603 [=====>.....] - ETA: 5s - loss: 0.1323 - auc: 0.9873
## 276/603 [=====>.....] - ETA: 5s - loss: 0.1320 - auc: 0.9873
## 280/603 [=====>.....] - ETA: 5s - loss: 0.1318 - auc: 0.9874
## 284/603 [=====>.....] - ETA: 5s - loss: 0.1320 - auc: 0.9874
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1314 - auc: 0.9875
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1313 - auc: 0.9875
## 295/603 [=====>.....] - ETA: 5s - loss: 0.1318 - auc: 0.9874
## 299/603 [=====>.....] - ETA: 4s - loss: 0.1312 - auc: 0.9875
## 303/603 [=====>.....] - ETA: 4s - loss: 0.1313 - auc: 0.9875
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1314 - auc: 0.9875
## 310/603 [=====>.....] - ETA: 4s - loss: 0.1312 - auc: 0.9875
## 314/603 [=====>.....] - ETA: 4s - loss: 0.1307 - auc: 0.9876
## 317/603 [=====>.....] - ETA: 4s - loss: 0.1308 - auc: 0.9876
## 320/603 [=====>.....] - ETA: 4s - loss: 0.1310 - auc: 0.9875
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1305 - auc: 0.9876
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1304 - auc: 0.9876
## 331/603 [=====>.....] - ETA: 4s - loss: 0.1296 - auc: 0.9877
## 335/603 [=====>.....] - ETA: 4s - loss: 0.1292 - auc: 0.9878
## 339/603 [=====>.....] - ETA: 4s - loss: 0.1283 - auc: 0.9880
## 342/603 [=====>.....] - ETA: 4s - loss: 0.1279 - auc: 0.9880
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1278 - auc: 0.9881
## 349/603 [=====>.....] - ETA: 4s - loss: 0.1282 - auc: 0.9880
## 352/603 [=====>.....] - ETA: 4s - loss: 0.1282 - auc: 0.9879
## 356/603 [=====>.....] - ETA: 4s - loss: 0.1279 - auc: 0.9880
## 360/603 [=====>.....] - ETA: 3s - loss: 0.1269 - auc: 0.9882
## 364/603 [=====>.....] - ETA: 3s - loss: 0.1276 - auc: 0.9880
## 368/603 [=====>.....] - ETA: 3s - loss: 0.1282 - auc: 0.9879
## 372/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9880
## 376/603 [=====>.....] - ETA: 3s - loss: 0.1269 - auc: 0.9881
## 380/603 [=====>.....] - ETA: 3s - loss: 0.1279 - auc: 0.9879
## 384/603 [=====>.....] - ETA: 3s - loss: 0.1275 - auc: 0.9880
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1272 - auc: 0.9880
## 391/603 [=====>.....] - ETA: 3s - loss: 0.1275 - auc: 0.9880
## 394/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9879
## 398/603 [=====>.....] - ETA: 3s - loss: 0.1273 - auc: 0.9880
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1277 - auc: 0.9880
## 406/603 [=====>.....] - ETA: 3s - loss: 0.1276 - auc: 0.9880
## 410/603 [=====>.....] - ETA: 3s - loss: 0.1289 - auc: 0.9878
## 414/603 [=====>.....] - ETA: 3s - loss: 0.1286 - auc: 0.9878
## 418/603 [=====>.....] - ETA: 3s - loss: 0.1288 - auc: 0.9877
## 421/603 [=====>.....] - ETA: 2s - loss: 0.1291 - auc: 0.9877
## 425/603 [=====>.....] - ETA: 2s - loss: 0.1297 - auc: 0.9876
## 429/603 [=====>.....] - ETA: 2s - loss: 0.1295 - auc: 0.9876
## 432/603 [=====>.....] - ETA: 2s - loss: 0.1295 - auc: 0.9876
## 436/603 [=====>.....] - ETA: 2s - loss: 0.1298 - auc: 0.9876
## 439/603 [=====>.....] - ETA: 2s - loss: 0.1295 - auc: 0.9876
## 443/603 [=====>.....] - ETA: 2s - loss: 0.1292 - auc: 0.9877
## 447/603 [=====>.....] - ETA: 2s - loss: 0.1293 - auc: 0.9876
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1288 - auc: 0.9877
## 455/603 [=====>.....] - ETA: 2s - loss: 0.1293 - auc: 0.9876
## 459/603 [=====>.....] - ETA: 2s - loss: 0.1290 - auc: 0.9877
## 462/603 [=====>.....] - ETA: 2s - loss: 0.1294 - auc: 0.9877
```

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## 466/603 [=====>.....] - ETA: 2s - loss: 0.1303 - auc: 0.9875
## 470/603 [=====>.....] - ETA: 2s - loss: 0.1299 - auc: 0.9876
## 474/603 [=====>.....] - ETA: 2s - loss: 0.1298 - auc: 0.9876
## 478/603 [=====>.....] - ETA: 2s - loss: 0.1297 - auc: 0.9876
## 482/603 [=====>.....] - ETA: 1s - loss: 0.1293 - auc: 0.9877
## 486/603 [=====>.....] - ETA: 1s - loss: 0.1296 - auc: 0.9877
## 490/603 [=====>.....] - ETA: 1s - loss: 0.1293 - auc: 0.9877
## 493/603 [=====>.....] - ETA: 1s - loss: 0.1298 - auc: 0.9876
## 497/603 [=====>.....] - ETA: 1s - loss: 0.1294 - auc: 0.9877
## 500/603 [=====>.....] - ETA: 1s - loss: 0.1293 - auc: 0.9877
## 504/603 [=====>.....] - ETA: 1s - loss: 0.1295 - auc: 0.9877
## 508/603 [=====>.....] - ETA: 1s - loss: 0.1300 - auc: 0.9877
## 511/603 [=====>.....] - ETA: 1s - loss: 0.1297 - auc: 0.9877
## 515/603 [=====>.....] - ETA: 1s - loss: 0.1301 - auc: 0.9876
## 519/603 [=====>.....] - ETA: 1s - loss: 0.1300 - auc: 0.9877
## 523/603 [=====>....] - ETA: 1s - loss: 0.1298 - auc: 0.9877
## 526/603 [=====>....] - ETA: 1s - loss: 0.1293 - auc: 0.9878
## 530/603 [=====>....] - ETA: 1s - loss: 0.1292 - auc: 0.9878
## 534/603 [=====>....] - ETA: 1s - loss: 0.1290 - auc: 0.9879
## 538/603 [=====>....] - ETA: 1s - loss: 0.1284 - auc: 0.9880
## 542/603 [=====>....] - ETA: 0s - loss: 0.1284 - auc: 0.9880
## 546/603 [=====>...] - ETA: 0s - loss: 0.1285 - auc: 0.9879
## 550/603 [=====>...] - ETA: 0s - loss: 0.1282 - auc: 0.9880
## 554/603 [=====>...] - ETA: 0s - loss: 0.1282 - auc: 0.9879
## 557/603 [=====>...] - ETA: 0s - loss: 0.1285 - auc: 0.9879
## 561/603 [=====>...] - ETA: 0s - loss: 0.1281 - auc: 0.9880
## 565/603 [=====>..] - ETA: 0s - loss: 0.1276 - auc: 0.9881
## 569/603 [=====>..] - ETA: 0s - loss: 0.1273 - auc: 0.9881
## 573/603 [=====>..] - ETA: 0s - loss: 0.1274 - auc: 0.9881
## 577/603 [=====>..] - ETA: 0s - loss: 0.1278 - auc: 0.9879
## 581/603 [=====>..] - ETA: 0s - loss: 0.1281 - auc: 0.9879
## 585/603 [=====>.] - ETA: 0s - loss: 0.1280 - auc: 0.9879
## 589/603 [=====>.] - ETA: 0s - loss: 0.1281 - auc: 0.9878
## 593/603 [=====>.] - ETA: 0s - loss: 0.1281 - auc: 0.9878
## 597/603 [=====>.] - ETA: 0s - loss: 0.1278 - auc: 0.9879
## 601/603 [=====>.] - ETA: 0s - loss: 0.1276 - auc: 0.9879
## 603/603 [=====] - 10s 17ms/step - loss: 0.1275 - auc: 0.9879 - va
l_loss: 0.2368 - val_auc: 0.9833
## Epoch 8/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.1753 - auc: 0.9917
## 5/603 [.....] - ETA: 9s - loss: 0.1185 - auc: 0.9907
## 9/603 [.....] - ETA: 9s - loss: 0.0953 - auc: 0.9952
## 13/603 [.....] - ETA: 9s - loss: 0.0896 - auc: 0.9947
## 17/603 [.....] - ETA: 9s - loss: 0.0860 - auc: 0.9957
## 20/603 [.....] - ETA: 9s - loss: 0.1039 - auc: 0.9918
## 23/603 [>.....] - ETA: 9s - loss: 0.1064 - auc: 0.9915
## 26/603 [>.....] - ETA: 9s - loss: 0.1191 - auc: 0.9897
## 30/603 [>.....] - ETA: 9s - loss: 0.1105 - auc: 0.9911
## 34/603 [>.....] - ETA: 9s - loss: 0.1088 - auc: 0.9916
## 38/603 [>.....] - ETA: 9s - loss: 0.1149 - auc: 0.9901
## 42/603 [=>.....] - ETA: 9s - loss: 0.1123 - auc: 0.9907
## 46/603 [=>.....] - ETA: 9s - loss: 0.1099 - auc: 0.9911
## 50/603 [=>.....] - ETA: 8s - loss: 0.1268 - auc: 0.9883
## 54/603 [=>.....] - ETA: 8s - loss: 0.1251 - auc: 0.9885
## 58/603 [=>.....] - ETA: 8s - loss: 0.1303 - auc: 0.9877
```

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## 62/603 [==>.....] - ETA: 8s - loss: 0.1319 - auc: 0.9876
## 66/603 [==>.....] - ETA: 8s - loss: 0.1334 - auc: 0.9873
## 70/603 [==>.....] - ETA: 8s - loss: 0.1373 - auc: 0.9868
## 73/603 [==>.....] - ETA: 8s - loss: 0.1348 - auc: 0.9873
## 77/603 [==>.....] - ETA: 8s - loss: 0.1344 - auc: 0.9873
## 81/603 [===>.....] - ETA: 8s - loss: 0.1342 - auc: 0.9874
## 85/603 [===>.....] - ETA: 8s - loss: 0.1312 - auc: 0.9880
## 89/603 [===>.....] - ETA: 8s - loss: 0.1316 - auc: 0.9880
## 93/603 [===>.....] - ETA: 8s - loss: 0.1300 - auc: 0.9883
## 97/603 [===>.....] - ETA: 8s - loss: 0.1293 - auc: 0.9883
## 100/603 [===>.....] - ETA: 8s - loss: 0.1298 - auc: 0.9881
## 103/603 [====>.....] - ETA: 8s - loss: 0.1298 - auc: 0.9882
## 107/603 [====>.....] - ETA: 8s - loss: 0.1279 - auc: 0.9886
## 111/603 [====>.....] - ETA: 8s - loss: 0.1260 - auc: 0.9888
## 115/603 [====>.....] - ETA: 7s - loss: 0.1249 - auc: 0.9891
## 118/603 [====>.....] - ETA: 7s - loss: 0.1259 - auc: 0.9890
## 121/603 [=====>.....] - ETA: 7s - loss: 0.1252 - auc: 0.9891
## 125/603 [=====>.....] - ETA: 7s - loss: 0.1261 - auc: 0.9891
## 129/603 [=====>.....] - ETA: 7s - loss: 0.1239 - auc: 0.9894
## 133/603 [=====>.....] - ETA: 7s - loss: 0.1266 - auc: 0.9890
## 136/603 [=====>.....] - ETA: 7s - loss: 0.1253 - auc: 0.9892
## 140/603 [=====>.....] - ETA: 7s - loss: 0.1258 - auc: 0.9891
## 144/603 [=====>.....] - ETA: 7s - loss: 0.1258 - auc: 0.9891
## 148/603 [=====>.....] - ETA: 7s - loss: 0.1282 - auc: 0.9886
## 152/603 [=====>.....] - ETA: 7s - loss: 0.1291 - auc: 0.9886
## 155/603 [=====>.....] - ETA: 7s - loss: 0.1290 - auc: 0.9886
## 159/603 [=====>.....] - ETA: 7s - loss: 0.1300 - auc: 0.9885
## 163/603 [=====>.....] - ETA: 7s - loss: 0.1290 - auc: 0.9887
## 167/603 [=====>.....] - ETA: 7s - loss: 0.1286 - auc: 0.9888
## 171/603 [=====>.....] - ETA: 7s - loss: 0.1282 - auc: 0.9889
## 175/603 [=====>.....] - ETA: 6s - loss: 0.1282 - auc: 0.9889
## 179/603 [=====>.....] - ETA: 6s - loss: 0.1281 - auc: 0.9890
## 182/603 [=====>.....] - ETA: 6s - loss: 0.1276 - auc: 0.9890
## 186/603 [=====>.....] - ETA: 6s - loss: 0.1279 - auc: 0.9890
## 190/603 [=====>.....] - ETA: 6s - loss: 0.1284 - auc: 0.9889
## 194/603 [=====>.....] - ETA: 6s - loss: 0.1300 - auc: 0.9886
## 197/603 [=====>.....] - ETA: 6s - loss: 0.1302 - auc: 0.9885
## 201/603 [=====>.....] - ETA: 6s - loss: 0.1292 - auc: 0.9887
## 205/603 [=====>.....] - ETA: 6s - loss: 0.1298 - auc: 0.9886
## 209/603 [=====>.....] - ETA: 6s - loss: 0.1292 - auc: 0.9887
## 212/603 [=====>.....] - ETA: 6s - loss: 0.1280 - auc: 0.9888
## 216/603 [=====>.....] - ETA: 6s - loss: 0.1282 - auc: 0.9887
## 219/603 [=====>.....] - ETA: 6s - loss: 0.1285 - auc: 0.9887
## 223/603 [=====>.....] - ETA: 6s - loss: 0.1284 - auc: 0.9888
## 226/603 [=====>.....] - ETA: 6s - loss: 0.1280 - auc: 0.9888
## 230/603 [=====>.....] - ETA: 6s - loss: 0.1275 - auc: 0.9889
## 234/603 [=====>.....] - ETA: 6s - loss: 0.1264 - auc: 0.9891
## 238/603 [=====>.....] - ETA: 5s - loss: 0.1257 - auc: 0.9892
## 242/603 [=====>.....] - ETA: 5s - loss: 0.1246 - auc: 0.9894
## 245/603 [=====>.....] - ETA: 5s - loss: 0.1257 - auc: 0.9891
## 249/603 [=====>.....] - ETA: 5s - loss: 0.1274 - auc: 0.9889
## 253/603 [=====>.....] - ETA: 5s - loss: 0.1275 - auc: 0.9889
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1277 - auc: 0.9888
## 261/603 [=====>.....] - ETA: 5s - loss: 0.1270 - auc: 0.9889
## 265/603 [=====>.....] - ETA: 5s - loss: 0.1267 - auc: 0.9890
## 269/603 [=====>.....] - ETA: 5s - loss: 0.1274 - auc: 0.9888
```

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## 273/603 [=====>.....] - ETA: 5s - loss: 0.1272 - auc: 0.9888
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1281 - auc: 0.9886
## 281/603 [=====>.....] - ETA: 5s - loss: 0.1272 - auc: 0.9888
## 285/603 [=====>.....] - ETA: 5s - loss: 0.1273 - auc: 0.9888
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1287 - auc: 0.9886
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1281 - auc: 0.9887
## 296/603 [=====>.....] - ETA: 5s - loss: 0.1276 - auc: 0.9888
## 300/603 [=====>.....] - ETA: 4s - loss: 0.1276 - auc: 0.9888
## 304/603 [=====>.....] - ETA: 4s - loss: 0.1272 - auc: 0.9888
## 307/603 [=====>.....] - ETA: 4s - loss: 0.1268 - auc: 0.9889
## 311/603 [=====>.....] - ETA: 4s - loss: 0.1276 - auc: 0.9887
## 315/603 [=====>.....] - ETA: 4s - loss: 0.1281 - auc: 0.9886
## 319/603 [=====>.....] - ETA: 4s - loss: 0.1287 - auc: 0.9884
## 323/603 [=====>.....] - ETA: 4s - loss: 0.1286 - auc: 0.9884
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1289 - auc: 0.9883
## 330/603 [=====>.....] - ETA: 4s - loss: 0.1288 - auc: 0.9883
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1294 - auc: 0.9882
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1296 - auc: 0.9881
## 341/603 [=====>.....] - ETA: 4s - loss: 0.1292 - auc: 0.9882
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1286 - auc: 0.9883
## 349/603 [=====>.....] - ETA: 4s - loss: 0.1290 - auc: 0.9882
## 353/603 [=====>.....] - ETA: 4s - loss: 0.1284 - auc: 0.9883
## 357/603 [=====>.....] - ETA: 4s - loss: 0.1285 - auc: 0.9883
## 361/603 [=====>.....] - ETA: 3s - loss: 0.1279 - auc: 0.9884
## 365/603 [=====>.....] - ETA: 3s - loss: 0.1275 - auc: 0.9885
## 369/603 [=====>.....] - ETA: 3s - loss: 0.1270 - auc: 0.9886
## 373/603 [=====>.....] - ETA: 3s - loss: 0.1269 - auc: 0.9886
## 377/603 [=====>.....] - ETA: 3s - loss: 0.1270 - auc: 0.9885
## 381/603 [=====>.....] - ETA: 3s - loss: 0.1275 - auc: 0.9885
## 385/603 [=====>.....] - ETA: 3s - loss: 0.1268 - auc: 0.9886
## 389/603 [=====>.....] - ETA: 3s - loss: 0.1266 - auc: 0.9886
## 393/603 [=====>.....] - ETA: 3s - loss: 0.1273 - auc: 0.9885
## 397/603 [=====>.....] - ETA: 3s - loss: 0.1268 - auc: 0.9885
## 400/603 [=====>.....] - ETA: 3s - loss: 0.1268 - auc: 0.9885
## 404/603 [=====>.....] - ETA: 3s - loss: 0.1266 - auc: 0.9886
## 408/603 [=====>.....] - ETA: 3s - loss: 0.1274 - auc: 0.9884
## 412/603 [=====>.....] - ETA: 3s - loss: 0.1275 - auc: 0.9884
## 416/603 [=====>.....] - ETA: 3s - loss: 0.1274 - auc: 0.9883
## 419/603 [=====>.....] - ETA: 3s - loss: 0.1269 - auc: 0.9884
## 423/603 [=====>.....] - ETA: 2s - loss: 0.1268 - auc: 0.9884
## 427/603 [=====>.....] - ETA: 2s - loss: 0.1262 - auc: 0.9886
## 431/603 [=====>.....] - ETA: 2s - loss: 0.1257 - auc: 0.9886
## 434/603 [=====>.....] - ETA: 2s - loss: 0.1255 - auc: 0.9887
## 438/603 [=====>.....] - ETA: 2s - loss: 0.1266 - auc: 0.9884
## 442/603 [=====>.....] - ETA: 2s - loss: 0.1267 - auc: 0.9884
## 445/603 [=====>.....] - ETA: 2s - loss: 0.1265 - auc: 0.9884
## 449/603 [=====>.....] - ETA: 2s - loss: 0.1263 - auc: 0.9885
## 453/603 [=====>.....] - ETA: 2s - loss: 0.1267 - auc: 0.9884
## 456/603 [=====>.....] - ETA: 2s - loss: 0.1269 - auc: 0.9884
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1270 - auc: 0.9884
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1265 - auc: 0.9885
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1265 - auc: 0.9885
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1263 - auc: 0.9885
## 476/603 [=====>.....] - ETA: 2s - loss: 0.1259 - auc: 0.9886
## 480/603 [=====>.....] - ETA: 2s - loss: 0.1256 - auc: 0.9886
## 484/603 [=====>.....] - ETA: 1s - loss: 0.1252 - auc: 0.9887
```

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## 488/603 [=====>.....] - ETA: 1s - loss: 0.1248 - auc: 0.9888
## 492/603 [=====>.....] - ETA: 1s - loss: 0.1245 - auc: 0.9888
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1243 - auc: 0.9888
## 498/603 [=====>.....] - ETA: 1s - loss: 0.1241 - auc: 0.9889
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1242 - auc: 0.9888
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1238 - auc: 0.9889
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1236 - auc: 0.9889
## 514/603 [=====>.....] - ETA: 1s - loss: 0.1240 - auc: 0.9888
## 517/603 [=====>.....] - ETA: 1s - loss: 0.1235 - auc: 0.9889
## 521/603 [=====>.....] - ETA: 1s - loss: 0.1237 - auc: 0.9889
## 525/603 [=====>....] - ETA: 1s - loss: 0.1237 - auc: 0.9889
## 529/603 [=====>....] - ETA: 1s - loss: 0.1239 - auc: 0.9889
## 533/603 [=====>....] - ETA: 1s - loss: 0.1237 - auc: 0.9889
## 537/603 [=====>....] - ETA: 1s - loss: 0.1234 - auc: 0.9890
## 541/603 [=====>....] - ETA: 1s - loss: 0.1232 - auc: 0.9890
## 545/603 [=====>...] - ETA: 0s - loss: 0.1230 - auc: 0.9890
## 549/603 [=====>...] - ETA: 0s - loss: 0.1231 - auc: 0.9890
## 553/603 [=====>...] - ETA: 0s - loss: 0.1227 - auc: 0.9891
## 557/603 [=====>...] - ETA: 0s - loss: 0.1224 - auc: 0.9891
## 561/603 [=====>...] - ETA: 0s - loss: 0.1226 - auc: 0.9891
## 565/603 [=====>..] - ETA: 0s - loss: 0.1225 - auc: 0.9891
## 568/603 [=====>..] - ETA: 0s - loss: 0.1229 - auc: 0.9890
## 572/603 [=====>..] - ETA: 0s - loss: 0.1228 - auc: 0.9890
## 575/603 [=====>..] - ETA: 0s - loss: 0.1231 - auc: 0.9890
## 579/603 [=====>..] - ETA: 0s - loss: 0.1233 - auc: 0.9889
## 582/603 [=====>..] - ETA: 0s - loss: 0.1236 - auc: 0.9889
## 586/603 [=====>.] - ETA: 0s - loss: 0.1235 - auc: 0.9889
## 590/603 [=====>.] - ETA: 0s - loss: 0.1238 - auc: 0.9888
## 594/603 [=====>.] - ETA: 0s - loss: 0.1237 - auc: 0.9889
## 598/603 [=====>.] - ETA: 0s - loss: 0.1235 - auc: 0.9889
## 602/603 [=====>.] - ETA: 0s - loss: 0.1242 - auc: 0.9887
## 603/603 [=====] - 10s 17ms/step - loss: 0.1242 - auc: 0.9887 - va
l_loss: 0.2119 - val_auc: 0.9818
## Epoch 9/20
##
## 1/603 [.....] - ETA: 10s - loss: 0.2715 - auc: 0.9708
## 5/603 [.....] - ETA: 9s - loss: 0.2071 - auc: 0.9828
## 9/603 [.....] - ETA: 9s - loss: 0.1551 - auc: 0.9885
## 13/603 [.....] - ETA: 9s - loss: 0.1423 - auc: 0.9897
## 17/603 [.....] - ETA: 9s - loss: 0.1270 - auc: 0.9915
## 21/603 [>.....] - ETA: 9s - loss: 0.1226 - auc: 0.9919
## 24/603 [>.....] - ETA: 9s - loss: 0.1190 - auc: 0.9922
## 28/603 [>.....] - ETA: 9s - loss: 0.1116 - auc: 0.9927
## 32/603 [>.....] - ETA: 9s - loss: 0.1196 - auc: 0.9903
## 36/603 [>.....] - ETA: 9s - loss: 0.1154 - auc: 0.9911
## 40/603 [>.....] - ETA: 9s - loss: 0.1175 - auc: 0.9905
## 44/603 [=>.....] - ETA: 9s - loss: 0.1121 - auc: 0.9914
## 48/603 [=>.....] - ETA: 8s - loss: 0.1112 - auc: 0.9912
## 52/603 [=>.....] - ETA: 8s - loss: 0.1124 - auc: 0.9914
## 55/603 [=>.....] - ETA: 8s - loss: 0.1164 - auc: 0.9907
## 58/603 [=>.....] - ETA: 8s - loss: 0.1161 - auc: 0.9908
## 62/603 [==>.....] - ETA: 8s - loss: 0.1150 - auc: 0.9911
## 66/603 [==>.....] - ETA: 8s - loss: 0.1116 - auc: 0.9915
## 70/603 [==>.....] - ETA: 8s - loss: 0.1111 - auc: 0.9916
## 74/603 [==>.....] - ETA: 8s - loss: 0.1148 - auc: 0.9909
## 78/603 [==>.....] - ETA: 8s - loss: 0.1159 - auc: 0.9904
```

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## 81/603 [===>.....] - ETA: 8s - loss: 0.1147 - auc: 0.9906
## 85/603 [===>.....] - ETA: 8s - loss: 0.1135 - auc: 0.9908
## 88/603 [===>.....] - ETA: 8s - loss: 0.1124 - auc: 0.9909
## 92/603 [===>.....] - ETA: 8s - loss: 0.1126 - auc: 0.9910
## 96/603 [===>.....] - ETA: 8s - loss: 0.1128 - auc: 0.9911
## 100/603 [===>.....] - ETA: 8s - loss: 0.1107 - auc: 0.9914
## 104/603 [====>.....] - ETA: 8s - loss: 0.1118 - auc: 0.9913
## 108/603 [====>.....] - ETA: 8s - loss: 0.1111 - auc: 0.9914
## 112/603 [====>.....] - ETA: 7s - loss: 0.1111 - auc: 0.9915
## 116/603 [====>.....] - ETA: 7s - loss: 0.1098 - auc: 0.9917
## 119/603 [====>.....] - ETA: 7s - loss: 0.1089 - auc: 0.9919
## 123/603 [=====>.....] - ETA: 7s - loss: 0.1095 - auc: 0.9918
## 127/603 [=====>.....] - ETA: 7s - loss: 0.1102 - auc: 0.9918
## 131/603 [=====>.....] - ETA: 7s - loss: 0.1109 - auc: 0.9917
## 135/603 [=====>.....] - ETA: 7s - loss: 0.1095 - auc: 0.9919
## 138/603 [=====>.....] - ETA: 7s - loss: 0.1118 - auc: 0.9917
## 142/603 [=====>.....] - ETA: 7s - loss: 0.1120 - auc: 0.9916
## 145/603 [=====>.....] - ETA: 7s - loss: 0.1115 - auc: 0.9916
## 149/603 [=====>.....] - ETA: 7s - loss: 0.1120 - auc: 0.9915
## 152/603 [=====>.....] - ETA: 7s - loss: 0.1106 - auc: 0.9917
## 156/603 [=====>.....] - ETA: 7s - loss: 0.1102 - auc: 0.9917
## 160/603 [=====>.....] - ETA: 7s - loss: 0.1117 - auc: 0.9916
## 164/603 [=====>.....] - ETA: 7s - loss: 0.1140 - auc: 0.9910
## 168/603 [=====>.....] - ETA: 7s - loss: 0.1160 - auc: 0.9907
## 172/603 [=====>.....] - ETA: 7s - loss: 0.1160 - auc: 0.9907
## 175/603 [=====>.....] - ETA: 6s - loss: 0.1154 - auc: 0.9908
## 179/603 [=====>.....] - ETA: 6s - loss: 0.1169 - auc: 0.9905
## 183/603 [=====>.....] - ETA: 6s - loss: 0.1182 - auc: 0.9903
## 187/603 [=====>.....] - ETA: 6s - loss: 0.1176 - auc: 0.9905
## 190/603 [=====>.....] - ETA: 6s - loss: 0.1180 - auc: 0.9903
## 194/603 [=====>.....] - ETA: 6s - loss: 0.1181 - auc: 0.9903
## 198/603 [=====>.....] - ETA: 6s - loss: 0.1192 - auc: 0.9900
## 202/603 [=====>.....] - ETA: 6s - loss: 0.1195 - auc: 0.9900
## 206/603 [=====>.....] - ETA: 6s - loss: 0.1194 - auc: 0.9900
## 209/603 [=====>.....] - ETA: 6s - loss: 0.1193 - auc: 0.9900
## 213/603 [=====>.....] - ETA: 6s - loss: 0.1208 - auc: 0.9898
## 217/603 [=====>.....] - ETA: 6s - loss: 0.1206 - auc: 0.9898
## 221/603 [=====>.....] - ETA: 6s - loss: 0.1201 - auc: 0.9899
## 225/603 [=====>.....] - ETA: 6s - loss: 0.1197 - auc: 0.9900
## 229/603 [=====>.....] - ETA: 6s - loss: 0.1196 - auc: 0.9899
## 233/603 [=====>.....] - ETA: 6s - loss: 0.1207 - auc: 0.9896
## 237/603 [=====>.....] - ETA: 5s - loss: 0.1217 - auc: 0.9895
## 241/603 [=====>.....] - ETA: 5s - loss: 0.1218 - auc: 0.9895
## 244/603 [=====>.....] - ETA: 5s - loss: 0.1215 - auc: 0.9895
## 248/603 [=====>.....] - ETA: 5s - loss: 0.1208 - auc: 0.9897
## 252/603 [=====>.....] - ETA: 5s - loss: 0.1213 - auc: 0.9895
## 256/603 [=====>.....] - ETA: 5s - loss: 0.1209 - auc: 0.9895
## 260/603 [=====>.....] - ETA: 5s - loss: 0.1206 - auc: 0.9896
## 264/603 [=====>.....] - ETA: 5s - loss: 0.1204 - auc: 0.9895
## 268/603 [=====>.....] - ETA: 5s - loss: 0.1210 - auc: 0.9894
## 271/603 [=====>.....] - ETA: 5s - loss: 0.1203 - auc: 0.9895
## 275/603 [=====>.....] - ETA: 5s - loss: 0.1210 - auc: 0.9893
## 279/603 [=====>.....] - ETA: 5s - loss: 0.1218 - auc: 0.9891
## 283/603 [=====>.....] - ETA: 5s - loss: 0.1214 - auc: 0.9892
## 287/603 [=====>.....] - ETA: 5s - loss: 0.1211 - auc: 0.9892
## 291/603 [=====>.....] - ETA: 5s - loss: 0.1209 - auc: 0.9893
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## 295/603 [=====>.....] - ETA: 5s - loss: 0.1208 - auc: 0.9893
## 299/603 [=====>.....] - ETA: 4s - loss: 0.1220 - auc: 0.9891
## 303/603 [=====>.....] - ETA: 4s - loss: 0.1213 - auc: 0.9892
## 307/603 [=====>.....] - ETA: 4s - loss: 0.1219 - auc: 0.9891
## 311/603 [=====>.....] - ETA: 4s - loss: 0.1212 - auc: 0.9892
## 315/603 [=====>.....] - ETA: 4s - loss: 0.1209 - auc: 0.9892
## 318/603 [=====>.....] - ETA: 4s - loss: 0.1200 - auc: 0.9894
## 322/603 [=====>.....] - ETA: 4s - loss: 0.1193 - auc: 0.9895
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1185 - auc: 0.9896
## 329/603 [=====>.....] - ETA: 4s - loss: 0.1178 - auc: 0.9897
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9898
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1173 - auc: 0.9897
## 341/603 [=====>.....] - ETA: 4s - loss: 0.1173 - auc: 0.9897
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9897
## 349/603 [=====>.....] - ETA: 4s - loss: 0.1167 - auc: 0.9898
## 353/603 [=====>.....] - ETA: 4s - loss: 0.1159 - auc: 0.9899
## 357/603 [=====>.....] - ETA: 4s - loss: 0.1156 - auc: 0.9899
## 361/603 [=====>.....] - ETA: 3s - loss: 0.1158 - auc: 0.9899
## 364/603 [=====>.....] - ETA: 3s - loss: 0.1162 - auc: 0.9899
## 368/603 [=====>.....] - ETA: 3s - loss: 0.1165 - auc: 0.9898
## 371/603 [=====>.....] - ETA: 3s - loss: 0.1162 - auc: 0.9899
## 375/603 [=====>.....] - ETA: 3s - loss: 0.1166 - auc: 0.9897
## 379/603 [=====>.....] - ETA: 3s - loss: 0.1172 - auc: 0.9896
## 383/603 [=====>.....] - ETA: 3s - loss: 0.1168 - auc: 0.9897
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1169 - auc: 0.9897
## 391/603 [=====>.....] - ETA: 3s - loss: 0.1163 - auc: 0.9898
## 395/603 [=====>.....] - ETA: 3s - loss: 0.1164 - auc: 0.9898
## 399/603 [=====>.....] - ETA: 3s - loss: 0.1159 - auc: 0.9899
## 403/603 [=====>.....] - ETA: 3s - loss: 0.1162 - auc: 0.9899
## 407/603 [=====>.....] - ETA: 3s - loss: 0.1160 - auc: 0.9899
## 411/603 [=====>.....] - ETA: 3s - loss: 0.1164 - auc: 0.9898
## 414/603 [=====>.....] - ETA: 3s - loss: 0.1159 - auc: 0.9899
## 418/603 [=====>.....] - ETA: 3s - loss: 0.1160 - auc: 0.9899
## 421/603 [=====>.....] - ETA: 2s - loss: 0.1154 - auc: 0.9900
## 425/603 [=====>.....] - ETA: 2s - loss: 0.1151 - auc: 0.9900
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1149 - auc: 0.9901
## 431/603 [=====>.....] - ETA: 2s - loss: 0.1150 - auc: 0.9901
## 435/603 [=====>.....] - ETA: 2s - loss: 0.1149 - auc: 0.9901
## 439/603 [=====>.....] - ETA: 2s - loss: 0.1146 - auc: 0.9901
## 443/603 [=====>.....] - ETA: 2s - loss: 0.1144 - auc: 0.9902
## 447/603 [=====>.....] - ETA: 2s - loss: 0.1148 - auc: 0.9901
## 450/603 [=====>.....] - ETA: 2s - loss: 0.1148 - auc: 0.9901
## 453/603 [=====>.....] - ETA: 2s - loss: 0.1150 - auc: 0.9901
## 457/603 [=====>.....] - ETA: 2s - loss: 0.1145 - auc: 0.9901
## 461/603 [=====>.....] - ETA: 2s - loss: 0.1143 - auc: 0.9902
## 465/603 [=====>.....] - ETA: 2s - loss: 0.1147 - auc: 0.9901
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1151 - auc: 0.9901
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1154 - auc: 0.9900
## 476/603 [=====>.....] - ETA: 2s - loss: 0.1165 - auc: 0.9898
## 480/603 [=====>.....] - ETA: 2s - loss: 0.1167 - auc: 0.9897
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1166 - auc: 0.9897
## 487/603 [=====>.....] - ETA: 1s - loss: 0.1164 - auc: 0.9898
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1160 - auc: 0.9898
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1159 - auc: 0.9899
## 498/603 [=====>.....] - ETA: 1s - loss: 0.1160 - auc: 0.9898
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1162 - auc: 0.9898
```

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## 506/603 [=====>.....] - ETA: 1s - loss: 0.1167 - auc: 0.9898
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1164 - auc: 0.9898
## 514/603 [=====>.....] - ETA: 1s - loss: 0.1164 - auc: 0.9898
## 518/603 [=====>.....] - ETA: 1s - loss: 0.1165 - auc: 0.9898
## 522/603 [=====>.....] - ETA: 1s - loss: 0.1171 - auc: 0.9897
## 526/603 [=====>.....] - ETA: 1s - loss: 0.1172 - auc: 0.9897
## 530/603 [=====>....] - ETA: 1s - loss: 0.1172 - auc: 0.9897
## 534/603 [=====>....] - ETA: 1s - loss: 0.1172 - auc: 0.9897
## 537/603 [=====>....] - ETA: 1s - loss: 0.1171 - auc: 0.9897
## 541/603 [=====>....] - ETA: 1s - loss: 0.1167 - auc: 0.9898
## 544/603 [=====>...] - ETA: 0s - loss: 0.1164 - auc: 0.9899
## 548/603 [=====>...] - ETA: 0s - loss: 0.1161 - auc: 0.9899
## 552/603 [=====>...] - ETA: 0s - loss: 0.1164 - auc: 0.9899
## 555/603 [=====>...] - ETA: 0s - loss: 0.1167 - auc: 0.9899
## 559/603 [=====>...] - ETA: 0s - loss: 0.1165 - auc: 0.9899
## 562/603 [=====>...] - ETA: 0s - loss: 0.1165 - auc: 0.9899
## 565/603 [=====>..] - ETA: 0s - loss: 0.1164 - auc: 0.9899
## 568/603 [=====>..] - ETA: 0s - loss: 0.1162 - auc: 0.9899
## 572/603 [=====>..] - ETA: 0s - loss: 0.1161 - auc: 0.9899
## 576/603 [=====>..] - ETA: 0s - loss: 0.1162 - auc: 0.9899
## 580/603 [=====>..] - ETA: 0s - loss: 0.1157 - auc: 0.9900
## 584/603 [=====>.] - ETA: 0s - loss: 0.1156 - auc: 0.9900
## 587/603 [=====>.] - ETA: 0s - loss: 0.1155 - auc: 0.9900
## 590/603 [=====>.] - ETA: 0s - loss: 0.1155 - auc: 0.9900
## 593/603 [=====>.] - ETA: 0s - loss: 0.1163 - auc: 0.9899
## 597/603 [=====>.] - ETA: 0s - loss: 0.1166 - auc: 0.9898
## 601/603 [=====>.] - ETA: 0s - loss: 0.1168 - auc: 0.9898
## 603/603 [=====] - 10s 17ms/step - loss: 0.1169 - auc: 0.9898 - va
l_loss: 0.2047 - val_auc: 0.9824
## Epoch 10/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0986 - auc: 0.9957
## 5/603 [.....] - ETA: 9s - loss: 0.0785 - auc: 0.9979
## 9/603 [.....] - ETA: 9s - loss: 0.0671 - auc: 0.9979
## 12/603 [.....] - ETA: 9s - loss: 0.0877 - auc: 0.9927
## 15/603 [.....] - ETA: 9s - loss: 0.0940 - auc: 0.9920
## 18/603 [.....] - ETA: 9s - loss: 0.0886 - auc: 0.9928
## 22/603 [>.....] - ETA: 9s - loss: 0.0962 - auc: 0.9926
## 25/603 [>.....] - ETA: 9s - loss: 0.0953 - auc: 0.9931
## 29/603 [>.....] - ETA: 9s - loss: 0.0973 - auc: 0.9931
## 32/603 [>.....] - ETA: 9s - loss: 0.1019 - auc: 0.9916
## 35/603 [>.....] - ETA: 9s - loss: 0.1014 - auc: 0.9921
## 38/603 [>.....] - ETA: 9s - loss: 0.0988 - auc: 0.9924
## 42/603 [=>.....] - ETA: 9s - loss: 0.0999 - auc: 0.9925
## 46/603 [=>.....] - ETA: 9s - loss: 0.1008 - auc: 0.9923
## 49/603 [=>.....] - ETA: 9s - loss: 0.1004 - auc: 0.9925
## 53/603 [=>.....] - ETA: 9s - loss: 0.1050 - auc: 0.9920
## 57/603 [=>.....] - ETA: 9s - loss: 0.1038 - auc: 0.9923
## 61/603 [==>.....] - ETA: 8s - loss: 0.1026 - auc: 0.9926
## 64/603 [==>.....] - ETA: 8s - loss: 0.1043 - auc: 0.9920
## 68/603 [==>.....] - ETA: 8s - loss: 0.1061 - auc: 0.9916
## 72/603 [==>.....] - ETA: 8s - loss: 0.1111 - auc: 0.9912
## 75/603 [==>.....] - ETA: 8s - loss: 0.1102 - auc: 0.9914
## 78/603 [==>.....] - ETA: 8s - loss: 0.1126 - auc: 0.9908
## 81/603 [===>.....] - ETA: 8s - loss: 0.1110 - auc: 0.9910
## 85/603 [===>.....] - ETA: 8s - loss: 0.1076 - auc: 0.9914
```

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## 88/603 [===>.....] - ETA: 8s - loss: 0.1080 - auc: 0.9912
## 91/603 [===>.....] - ETA: 8s - loss: 0.1061 - auc: 0.9915
## 94/603 [===>.....] - ETA: 8s - loss: 0.1067 - auc: 0.9914
## 98/603 [===>.....] - ETA: 8s - loss: 0.1061 - auc: 0.9915
## 101/603 [====>.....] - ETA: 8s - loss: 0.1050 - auc: 0.9916
## 105/603 [====>.....] - ETA: 8s - loss: 0.1046 - auc: 0.9918
## 109/603 [====>.....] - ETA: 8s - loss: 0.1072 - auc: 0.9912
## 113/603 [====>.....] - ETA: 8s - loss: 0.1054 - auc: 0.9915
## 117/603 [====>.....] - ETA: 8s - loss: 0.1049 - auc: 0.9916
## 121/603 [=====>.....] - ETA: 7s - loss: 0.1062 - auc: 0.9914
## 124/603 [=====>.....] - ETA: 7s - loss: 0.1061 - auc: 0.9915
## 127/603 [=====>.....] - ETA: 7s - loss: 0.1063 - auc: 0.9915
## 131/603 [=====>.....] - ETA: 7s - loss: 0.1059 - auc: 0.9914
## 135/603 [=====>.....] - ETA: 7s - loss: 0.1069 - auc: 0.9913
## 139/603 [=====>.....] - ETA: 7s - loss: 0.1064 - auc: 0.9913
## 143/603 [=====>.....] - ETA: 7s - loss: 0.1070 - auc: 0.9911
## 147/603 [=====>.....] - ETA: 7s - loss: 0.1067 - auc: 0.9912
## 151/603 [=====>.....] - ETA: 7s - loss: 0.1076 - auc: 0.9911
## 155/603 [=====>.....] - ETA: 7s - loss: 0.1069 - auc: 0.9912
## 158/603 [=====>.....] - ETA: 7s - loss: 0.1062 - auc: 0.9913
## 162/603 [=====>.....] - ETA: 7s - loss: 0.1080 - auc: 0.9910
## 166/603 [=====>.....] - ETA: 7s - loss: 0.1079 - auc: 0.9909
## 169/603 [=====>.....] - ETA: 7s - loss: 0.1085 - auc: 0.9908
## 172/603 [=====>.....] - ETA: 7s - loss: 0.1096 - auc: 0.9905
## 175/603 [=====>.....] - ETA: 7s - loss: 0.1102 - auc: 0.9904
## 179/603 [=====>.....] - ETA: 6s - loss: 0.1118 - auc: 0.9902
## 183/603 [=====>.....] - ETA: 6s - loss: 0.1123 - auc: 0.9901
## 187/603 [=====>.....] - ETA: 6s - loss: 0.1155 - auc: 0.9895
## 190/603 [=====>.....] - ETA: 6s - loss: 0.1148 - auc: 0.9896
## 194/603 [=====>.....] - ETA: 6s - loss: 0.1147 - auc: 0.9897
## 198/603 [=====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9897
## 202/603 [=====>.....] - ETA: 6s - loss: 0.1144 - auc: 0.9898
## 206/603 [=====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9897
## 210/603 [=====>.....] - ETA: 6s - loss: 0.1137 - auc: 0.9898
## 214/603 [=====>.....] - ETA: 6s - loss: 0.1136 - auc: 0.9898
## 218/603 [=====>.....] - ETA: 6s - loss: 0.1150 - auc: 0.9895
## 222/603 [=====>.....] - ETA: 6s - loss: 0.1147 - auc: 0.9895
## 226/603 [=====>.....] - ETA: 6s - loss: 0.1134 - auc: 0.9897
## 230/603 [=====>.....] - ETA: 6s - loss: 0.1142 - auc: 0.9895
## 234/603 [=====>.....] - ETA: 6s - loss: 0.1145 - auc: 0.9895
## 238/603 [=====>.....] - ETA: 5s - loss: 0.1140 - auc: 0.9896
## 242/603 [=====>.....] - ETA: 5s - loss: 0.1136 - auc: 0.9897
## 246/603 [=====>.....] - ETA: 5s - loss: 0.1133 - auc: 0.9898
## 250/603 [=====>.....] - ETA: 5s - loss: 0.1130 - auc: 0.9898
## 253/603 [=====>.....] - ETA: 5s - loss: 0.1122 - auc: 0.9899
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1119 - auc: 0.9900
## 260/603 [=====>.....] - ETA: 5s - loss: 0.1129 - auc: 0.9898
## 263/603 [=====>.....] - ETA: 5s - loss: 0.1138 - auc: 0.9897
## 267/603 [=====>.....] - ETA: 5s - loss: 0.1143 - auc: 0.9896
## 271/603 [=====>.....] - ETA: 5s - loss: 0.1132 - auc: 0.9898
## 275/603 [=====>.....] - ETA: 5s - loss: 0.1131 - auc: 0.9899
## 279/603 [=====>.....] - ETA: 5s - loss: 0.1125 - auc: 0.9900
## 283/603 [=====>.....] - ETA: 5s - loss: 0.1122 - auc: 0.9900
## 287/603 [=====>.....] - ETA: 5s - loss: 0.1126 - auc: 0.9899
## 291/603 [=====>.....] - ETA: 5s - loss: 0.1130 - auc: 0.9898
## 295/603 [=====>.....] - ETA: 5s - loss: 0.1126 - auc: 0.9899
```

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## 299/603 [=====>.....] - ETA: 4s - loss: 0.1140 - auc: 0.9897
## 302/603 [=====>.....] - ETA: 4s - loss: 0.1136 - auc: 0.9897
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1131 - auc: 0.9898
## 310/603 [=====>.....] - ETA: 4s - loss: 0.1135 - auc: 0.9898
## 313/603 [=====>.....] - ETA: 4s - loss: 0.1132 - auc: 0.9898
## 316/603 [=====>.....] - ETA: 4s - loss: 0.1137 - auc: 0.9898
## 320/603 [=====>.....] - ETA: 4s - loss: 0.1140 - auc: 0.9897
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1137 - auc: 0.9898
## 328/603 [=====>.....] - ETA: 4s - loss: 0.1138 - auc: 0.9898
## 332/603 [=====>.....] - ETA: 4s - loss: 0.1155 - auc: 0.9895
## 336/603 [=====>.....] - ETA: 4s - loss: 0.1152 - auc: 0.9896
## 340/603 [=====>.....] - ETA: 4s - loss: 0.1155 - auc: 0.9895
## 344/603 [=====>.....] - ETA: 4s - loss: 0.1162 - auc: 0.9895
## 348/603 [=====>.....] - ETA: 4s - loss: 0.1155 - auc: 0.9896
## 352/603 [=====>.....] - ETA: 4s - loss: 0.1162 - auc: 0.9895
## 356/603 [=====>.....] - ETA: 4s - loss: 0.1170 - auc: 0.9893
## 360/603 [=====>.....] - ETA: 3s - loss: 0.1172 - auc: 0.9893
## 364/603 [=====>.....] - ETA: 3s - loss: 0.1174 - auc: 0.9893
## 368/603 [=====>.....] - ETA: 3s - loss: 0.1189 - auc: 0.9890
## 372/603 [=====>.....] - ETA: 3s - loss: 0.1191 - auc: 0.9890
## 376/603 [=====>.....] - ETA: 3s - loss: 0.1192 - auc: 0.9890
## 379/603 [=====>.....] - ETA: 3s - loss: 0.1189 - auc: 0.9891
## 383/603 [=====>.....] - ETA: 3s - loss: 0.1188 - auc: 0.9891
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1189 - auc: 0.9890
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1191 - auc: 0.9890
## 394/603 [=====>.....] - ETA: 3s - loss: 0.1182 - auc: 0.9892
## 398/603 [=====>.....] - ETA: 3s - loss: 0.1180 - auc: 0.9892
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1187 - auc: 0.9891
## 405/603 [=====>.....] - ETA: 3s - loss: 0.1183 - auc: 0.9891
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1181 - auc: 0.9892
## 412/603 [=====>.....] - ETA: 3s - loss: 0.1181 - auc: 0.9892
## 416/603 [=====>.....] - ETA: 3s - loss: 0.1178 - auc: 0.9892
## 420/603 [=====>.....] - ETA: 2s - loss: 0.1179 - auc: 0.9892
## 424/603 [=====>.....] - ETA: 2s - loss: 0.1173 - auc: 0.9893
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1169 - auc: 0.9894
## 432/603 [=====>.....] - ETA: 2s - loss: 0.1167 - auc: 0.9895
## 436/603 [=====>.....] - ETA: 2s - loss: 0.1168 - auc: 0.9895
## 440/603 [=====>.....] - ETA: 2s - loss: 0.1164 - auc: 0.9895
## 444/603 [=====>.....] - ETA: 2s - loss: 0.1168 - auc: 0.9895
## 448/603 [=====>.....] - ETA: 2s - loss: 0.1171 - auc: 0.9895
## 452/603 [=====>.....] - ETA: 2s - loss: 0.1180 - auc: 0.9893
## 456/603 [=====>.....] - ETA: 2s - loss: 0.1183 - auc: 0.9893
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1187 - auc: 0.9892
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1184 - auc: 0.9893
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1183 - auc: 0.9893
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1179 - auc: 0.9894
## 475/603 [=====>.....] - ETA: 2s - loss: 0.1178 - auc: 0.9894
## 479/603 [=====>.....] - ETA: 2s - loss: 0.1176 - auc: 0.9894
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1175 - auc: 0.9894
## 487/603 [=====>.....] - ETA: 1s - loss: 0.1178 - auc: 0.9893
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1176 - auc: 0.9894
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1173 - auc: 0.9894
## 499/603 [=====>.....] - ETA: 1s - loss: 0.1170 - auc: 0.9895
## 503/603 [=====>.....] - ETA: 1s - loss: 0.1171 - auc: 0.9895
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1171 - auc: 0.9895
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1179 - auc: 0.9893
```

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## 514/603 [=====>.....] - ETA: 1s - loss: 0.1174 - auc: 0.9894
## 518/603 [=====>.....] - ETA: 1s - loss: 0.1178 - auc: 0.9893
## 522/603 [=====>.....] - ETA: 1s - loss: 0.1176 - auc: 0.9893
## 526/603 [=====>....] - ETA: 1s - loss: 0.1177 - auc: 0.9894
## 530/603 [=====>....] - ETA: 1s - loss: 0.1178 - auc: 0.9894
## 534/603 [=====>....] - ETA: 1s - loss: 0.1177 - auc: 0.9894
## 538/603 [=====>....] - ETA: 1s - loss: 0.1177 - auc: 0.9894
## 542/603 [=====>....] - ETA: 0s - loss: 0.1173 - auc: 0.9895
## 546/603 [=====>...] - ETA: 0s - loss: 0.1171 - auc: 0.9895
## 550/603 [=====>...] - ETA: 0s - loss: 0.1170 - auc: 0.9895
## 554/603 [=====>...] - ETA: 0s - loss: 0.1169 - auc: 0.9895
## 558/603 [=====>...] - ETA: 0s - loss: 0.1174 - auc: 0.9895
## 562/603 [=====>...] - ETA: 0s - loss: 0.1180 - auc: 0.9894
## 566/603 [=====>..] - ETA: 0s - loss: 0.1182 - auc: 0.9894
## 569/603 [=====>..] - ETA: 0s - loss: 0.1179 - auc: 0.9895
## 573/603 [=====>..] - ETA: 0s - loss: 0.1178 - auc: 0.9895
## 576/603 [=====>..] - ETA: 0s - loss: 0.1176 - auc: 0.9895
## 580/603 [=====>..] - ETA: 0s - loss: 0.1176 - auc: 0.9896
## 584/603 [=====>.] - ETA: 0s - loss: 0.1172 - auc: 0.9896
## 588/603 [=====>.] - ETA: 0s - loss: 0.1176 - auc: 0.9895
## 592/603 [=====>.] - ETA: 0s - loss: 0.1183 - auc: 0.9894
## 596/603 [=====>.] - ETA: 0s - loss: 0.1181 - auc: 0.9894
## 600/603 [=====>.] - ETA: 0s - loss: 0.1184 - auc: 0.9893
## 603/603 [=====] - 10s 17ms/step - loss: 0.1185 - auc: 0.9893 - va
l_loss: 0.2321 - val_auc: 0.9819
## Epoch 11/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.2390 - auc: 0.9686
## 5/603 [.....] - ETA: 9s - loss: 0.1708 - auc: 0.9784
## 9/603 [.....] - ETA: 9s - loss: 0.1435 - auc: 0.9850
## 13/603 [.....] - ETA: 9s - loss: 0.1294 - auc: 0.9875
## 17/603 [.....] - ETA: 9s - loss: 0.1164 - auc: 0.9902
## 21/603 [>.....] - ETA: 9s - loss: 0.1166 - auc: 0.9907
## 25/603 [>.....] - ETA: 9s - loss: 0.1226 - auc: 0.9900
## 29/603 [>.....] - ETA: 9s - loss: 0.1256 - auc: 0.9893
## 33/603 [>.....] - ETA: 9s - loss: 0.1226 - auc: 0.9894
## 37/603 [>.....] - ETA: 9s - loss: 0.1160 - auc: 0.9905
## 41/603 [=>.....] - ETA: 9s - loss: 0.1158 - auc: 0.9906
## 45/603 [=>.....] - ETA: 8s - loss: 0.1120 - auc: 0.9912
## 49/603 [=>.....] - ETA: 8s - loss: 0.1085 - auc: 0.9917
## 53/603 [=>.....] - ETA: 8s - loss: 0.1048 - auc: 0.9923
## 57/603 [=>.....] - ETA: 8s - loss: 0.1109 - auc: 0.9911
## 61/603 [==>.....] - ETA: 8s - loss: 0.1093 - auc: 0.9915
## 65/603 [==>.....] - ETA: 8s - loss: 0.1084 - auc: 0.9917
## 69/603 [==>.....] - ETA: 8s - loss: 0.1118 - auc: 0.9911
## 73/603 [==>.....] - ETA: 8s - loss: 0.1091 - auc: 0.9915
## 77/603 [==>.....] - ETA: 8s - loss: 0.1101 - auc: 0.9913
## 81/603 [===>.....] - ETA: 8s - loss: 0.1087 - auc: 0.9916
## 85/603 [===>.....] - ETA: 8s - loss: 0.1116 - auc: 0.9908
## 89/603 [===>.....] - ETA: 8s - loss: 0.1092 - auc: 0.9912
## 93/603 [===>.....] - ETA: 8s - loss: 0.1100 - auc: 0.9911
## 97/603 [===>.....] - ETA: 8s - loss: 0.1084 - auc: 0.9913
## 101/603 [===>.....] - ETA: 8s - loss: 0.1103 - auc: 0.9908
## 105/603 [===>.....] - ETA: 8s - loss: 0.1094 - auc: 0.9908
## 109/603 [===>.....] - ETA: 7s - loss: 0.1070 - auc: 0.9912
## 113/603 [===>.....] - ETA: 7s - loss: 0.1060 - auc: 0.9914
```

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## 117/603 [====>.....] - ETA: 7s - loss: 0.1064 - auc: 0.9914
## 121/603 [====>.....] - ETA: 7s - loss: 0.1054 - auc: 0.9915
## 125/603 [====>.....] - ETA: 7s - loss: 0.1047 - auc: 0.9917
## 129/603 [====>.....] - ETA: 7s - loss: 0.1031 - auc: 0.9919
## 132/603 [====>.....] - ETA: 7s - loss: 0.1026 - auc: 0.9921
## 135/603 [====>.....] - ETA: 7s - loss: 0.1034 - auc: 0.9918
## 139/603 [====>.....] - ETA: 7s - loss: 0.1034 - auc: 0.9919
## 143/603 [====>.....] - ETA: 7s - loss: 0.1038 - auc: 0.9919
## 147/603 [====>.....] - ETA: 7s - loss: 0.1028 - auc: 0.9920
## 151/603 [====>.....] - ETA: 7s - loss: 0.1020 - auc: 0.9922
## 155/603 [====>.....] - ETA: 7s - loss: 0.1044 - auc: 0.9918
## 159/603 [====>.....] - ETA: 7s - loss: 0.1066 - auc: 0.9914
## 163/603 [====>.....] - ETA: 7s - loss: 0.1071 - auc: 0.9914
## 167/603 [====>.....] - ETA: 7s - loss: 0.1094 - auc: 0.9911
## 171/603 [====>.....] - ETA: 6s - loss: 0.1107 - auc: 0.9909
## 175/603 [====>.....] - ETA: 6s - loss: 0.1117 - auc: 0.9908
## 179/603 [====>.....] - ETA: 6s - loss: 0.1105 - auc: 0.9910
## 182/603 [====>.....] - ETA: 6s - loss: 0.1101 - auc: 0.9911
## 186/603 [====>.....] - ETA: 6s - loss: 0.1111 - auc: 0.9910
## 190/603 [====>.....] - ETA: 6s - loss: 0.1097 - auc: 0.9912
## 194/603 [====>.....] - ETA: 6s - loss: 0.1100 - auc: 0.9911
## 198/603 [====>.....] - ETA: 6s - loss: 0.1127 - auc: 0.9906
## 202/603 [====>.....] - ETA: 6s - loss: 0.1128 - auc: 0.9906
## 206/603 [====>.....] - ETA: 6s - loss: 0.1122 - auc: 0.9907
## 210/603 [====>.....] - ETA: 6s - loss: 0.1127 - auc: 0.9907
## 214/603 [====>.....] - ETA: 6s - loss: 0.1144 - auc: 0.9903
## 218/603 [====>.....] - ETA: 6s - loss: 0.1144 - auc: 0.9903
## 222/603 [====>.....] - ETA: 6s - loss: 0.1144 - auc: 0.9903
## 226/603 [====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9903
## 230/603 [====>.....] - ETA: 6s - loss: 0.1160 - auc: 0.9900
## 234/603 [====>.....] - ETA: 5s - loss: 0.1159 - auc: 0.9900
## 238/603 [====>.....] - ETA: 5s - loss: 0.1154 - auc: 0.9901
## 242/603 [====>.....] - ETA: 5s - loss: 0.1161 - auc: 0.9901
## 246/603 [====>.....] - ETA: 5s - loss: 0.1160 - auc: 0.9901
## 250/603 [====>.....] - ETA: 5s - loss: 0.1176 - auc: 0.9899
## 254/603 [====>.....] - ETA: 5s - loss: 0.1173 - auc: 0.9899
## 258/603 [====>.....] - ETA: 5s - loss: 0.1170 - auc: 0.9899
## 261/603 [====>.....] - ETA: 5s - loss: 0.1164 - auc: 0.9900
## 265/603 [====>.....] - ETA: 5s - loss: 0.1159 - auc: 0.9901
## 269/603 [====>.....] - ETA: 5s - loss: 0.1160 - auc: 0.9901
## 273/603 [====>.....] - ETA: 5s - loss: 0.1156 - auc: 0.9902
## 276/603 [====>.....] - ETA: 5s - loss: 0.1170 - auc: 0.9898
## 280/603 [====>.....] - ETA: 5s - loss: 0.1170 - auc: 0.9899
## 284/603 [====>.....] - ETA: 5s - loss: 0.1175 - auc: 0.9898
## 288/603 [====>.....] - ETA: 5s - loss: 0.1169 - auc: 0.9899
## 292/603 [====>.....] - ETA: 5s - loss: 0.1164 - auc: 0.9900
## 296/603 [====>.....] - ETA: 4s - loss: 0.1169 - auc: 0.9900
## 300/603 [====>.....] - ETA: 4s - loss: 0.1177 - auc: 0.9898
## 304/603 [====>.....] - ETA: 4s - loss: 0.1182 - auc: 0.9897
## 308/603 [====>.....] - ETA: 4s - loss: 0.1185 - auc: 0.9896
## 312/603 [====>.....] - ETA: 4s - loss: 0.1182 - auc: 0.9897
## 316/603 [====>.....] - ETA: 4s - loss: 0.1177 - auc: 0.9898
## 320/603 [====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9899
## 323/603 [====>.....] - ETA: 4s - loss: 0.1173 - auc: 0.9899
## 327/603 [====>.....] - ETA: 4s - loss: 0.1178 - auc: 0.9898
## 331/603 [====>.....] - ETA: 4s - loss: 0.1174 - auc: 0.9899
```

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## 335/603 [=====>.....] - ETA: 4s - loss: 0.1174 - auc: 0.9899
## 339/603 [=====>.....] - ETA: 4s - loss: 0.1171 - auc: 0.9899
## 343/603 [=====>.....] - ETA: 4s - loss: 0.1169 - auc: 0.9900
## 347/603 [=====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9899
## 351/603 [=====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9899
## 355/603 [=====>.....] - ETA: 4s - loss: 0.1172 - auc: 0.9899
## 359/603 [=====>.....] - ETA: 3s - loss: 0.1177 - auc: 0.9898
## 363/603 [=====>.....] - ETA: 3s - loss: 0.1167 - auc: 0.9899
## 367/603 [=====>.....] - ETA: 3s - loss: 0.1164 - auc: 0.9900
## 371/603 [=====>.....] - ETA: 3s - loss: 0.1157 - auc: 0.9901
## 375/603 [=====>.....] - ETA: 3s - loss: 0.1164 - auc: 0.9900
## 379/603 [=====>.....] - ETA: 3s - loss: 0.1157 - auc: 0.9901
## 383/603 [=====>.....] - ETA: 3s - loss: 0.1155 - auc: 0.9902
## 386/603 [=====>.....] - ETA: 3s - loss: 0.1155 - auc: 0.9902
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1159 - auc: 0.9901
## 394/603 [=====>.....] - ETA: 3s - loss: 0.1153 - auc: 0.9902
## 398/603 [=====>.....] - ETA: 3s - loss: 0.1155 - auc: 0.9902
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1151 - auc: 0.9902
## 406/603 [=====>.....] - ETA: 3s - loss: 0.1147 - auc: 0.9903
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1144 - auc: 0.9903
## 413/603 [=====>.....] - ETA: 3s - loss: 0.1145 - auc: 0.9904
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1143 - auc: 0.9904
## 421/603 [=====>.....] - ETA: 2s - loss: 0.1147 - auc: 0.9903
## 425/603 [=====>.....] - ETA: 2s - loss: 0.1147 - auc: 0.9903
## 429/603 [=====>.....] - ETA: 2s - loss: 0.1146 - auc: 0.9903
## 433/603 [=====>.....] - ETA: 2s - loss: 0.1146 - auc: 0.9903
## 437/603 [=====>.....] - ETA: 2s - loss: 0.1142 - auc: 0.9904
## 441/603 [=====>.....] - ETA: 2s - loss: 0.1145 - auc: 0.9904
## 445/603 [=====>.....] - ETA: 2s - loss: 0.1137 - auc: 0.9905
## 448/603 [=====>.....] - ETA: 2s - loss: 0.1141 - auc: 0.9904
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1140 - auc: 0.9904
## 454/603 [=====>.....] - ETA: 2s - loss: 0.1140 - auc: 0.9904
## 457/603 [=====>.....] - ETA: 2s - loss: 0.1135 - auc: 0.9905
## 461/603 [=====>.....] - ETA: 2s - loss: 0.1132 - auc: 0.9905
## 465/603 [=====>.....] - ETA: 2s - loss: 0.1137 - auc: 0.9905
## 469/603 [=====>.....] - ETA: 2s - loss: 0.1138 - auc: 0.9905
## 473/603 [=====>.....] - ETA: 2s - loss: 0.1142 - auc: 0.9903
## 477/603 [=====>.....] - ETA: 2s - loss: 0.1140 - auc: 0.9904
## 481/603 [=====>.....] - ETA: 1s - loss: 0.1136 - auc: 0.9905
## 485/603 [=====>.....] - ETA: 1s - loss: 0.1142 - auc: 0.9903
## 489/603 [=====>.....] - ETA: 1s - loss: 0.1143 - auc: 0.9903
## 493/603 [=====>.....] - ETA: 1s - loss: 0.1141 - auc: 0.9903
## 497/603 [=====>.....] - ETA: 1s - loss: 0.1144 - auc: 0.9903
## 501/603 [=====>.....] - ETA: 1s - loss: 0.1142 - auc: 0.9904
## 505/603 [=====>.....] - ETA: 1s - loss: 0.1147 - auc: 0.9903
## 509/603 [=====>.....] - ETA: 1s - loss: 0.1147 - auc: 0.9902
## 512/603 [=====>.....] - ETA: 1s - loss: 0.1149 - auc: 0.9902
## 516/603 [=====>.....] - ETA: 1s - loss: 0.1150 - auc: 0.9902
## 520/603 [=====>.....] - ETA: 1s - loss: 0.1148 - auc: 0.9902
## 524/603 [=====>....] - ETA: 1s - loss: 0.1154 - auc: 0.9901
## 528/603 [=====>....] - ETA: 1s - loss: 0.1149 - auc: 0.9902
## 532/603 [=====>....] - ETA: 1s - loss: 0.1157 - auc: 0.9900
## 536/603 [=====>....] - ETA: 1s - loss: 0.1156 - auc: 0.9900
## 540/603 [=====>....] - ETA: 1s - loss: 0.1158 - auc: 0.9900
## 544/603 [=====>...] - ETA: 0s - loss: 0.1157 - auc: 0.9900
## 548/603 [=====>...] - ETA: 0s - loss: 0.1160 - auc: 0.9899
```

```
## 552/603 [=====>...] - ETA: 0s - loss: 0.1159 - auc: 0.9899
## 556/603 [=====>...] - ETA: 0s - loss: 0.1159 - auc: 0.9899
## 560/603 [=====>...] - ETA: 0s - loss: 0.1162 - auc: 0.9899
## 564/603 [=====>..] - ETA: 0s - loss: 0.1162 - auc: 0.9899
## 567/603 [=====>..] - ETA: 0s - loss: 0.1164 - auc: 0.9898
## 571/603 [=====>..] - ETA: 0s - loss: 0.1163 - auc: 0.9899
## 574/603 [=====>..] - ETA: 0s - loss: 0.1159 - auc: 0.9899
## 577/603 [=====>..] - ETA: 0s - loss: 0.1158 - auc: 0.9900
## 581/603 [=====>..] - ETA: 0s - loss: 0.1156 - auc: 0.9900
## 585/603 [=====>.] - ETA: 0s - loss: 0.1154 - auc: 0.9900
## 589/603 [=====>.] - ETA: 0s - loss: 0.1152 - auc: 0.9901
## 593/603 [=====>.] - ETA: 0s - loss: 0.1153 - auc: 0.9900
## 597/603 [=====>.] - ETA: 0s - loss: 0.1157 - auc: 0.9900
## 601/603 [=====>.] - ETA: 0s - loss: 0.1157 - auc: 0.9900
## 603/603 [=====] - 10s 17ms/step - loss: 0.1158 - auc: 0.9900 - va
l_loss: 0.2191 - val_auc: 0.9811
## Epoch 12/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0923 - auc: 0.9882
## 5/603 [.....] - ETA: 9s - loss: 0.1159 - auc: 0.9869
## 9/603 [.....] - ETA: 9s - loss: 0.1159 - auc: 0.9884
## 13/603 [.....] - ETA: 9s - loss: 0.1421 - auc: 0.9856
## 16/603 [.....] - ETA: 9s - loss: 0.1270 - auc: 0.9878
## 20/603 [.....] - ETA: 9s - loss: 0.1322 - auc: 0.9873
## 24/603 [>.....] - ETA: 9s - loss: 0.1297 - auc: 0.9880
## 28/603 [>.....] - ETA: 9s - loss: 0.1314 - auc: 0.9882
## 32/603 [>.....] - ETA: 9s - loss: 0.1279 - auc: 0.9887
## 36/603 [>.....] - ETA: 9s - loss: 0.1234 - auc: 0.9896
## 40/603 [>.....] - ETA: 9s - loss: 0.1293 - auc: 0.9889
## 44/603 [=>.....] - ETA: 9s - loss: 0.1266 - auc: 0.9893
## 48/603 [=>.....] - ETA: 9s - loss: 0.1245 - auc: 0.9896
## 52/603 [=>.....] - ETA: 8s - loss: 0.1262 - auc: 0.9887
## 56/603 [=>.....] - ETA: 8s - loss: 0.1225 - auc: 0.9893
## 60/603 [=>.....] - ETA: 8s - loss: 0.1169 - auc: 0.9901
## 64/603 [==>.....] - ETA: 8s - loss: 0.1148 - auc: 0.9906
## 67/603 [==>.....] - ETA: 8s - loss: 0.1169 - auc: 0.9902
## 70/603 [==>.....] - ETA: 8s - loss: 0.1173 - auc: 0.9902
## 73/603 [==>.....] - ETA: 8s - loss: 0.1188 - auc: 0.9901
## 77/603 [==>.....] - ETA: 8s - loss: 0.1167 - auc: 0.9904
## 81/603 [==>.....] - ETA: 8s - loss: 0.1150 - auc: 0.9908
## 85/603 [==>.....] - ETA: 8s - loss: 0.1150 - auc: 0.9907
## 89/603 [==>.....] - ETA: 8s - loss: 0.1141 - auc: 0.9910
## 93/603 [==>.....] - ETA: 8s - loss: 0.1132 - auc: 0.9912
## 97/603 [==>.....] - ETA: 8s - loss: 0.1161 - auc: 0.9904
## 101/603 [===>.....] - ETA: 8s - loss: 0.1187 - auc: 0.9902
## 105/603 [===>.....] - ETA: 8s - loss: 0.1183 - auc: 0.9902
## 109/603 [===>.....] - ETA: 8s - loss: 0.1161 - auc: 0.9905
## 113/603 [===>.....] - ETA: 7s - loss: 0.1155 - auc: 0.9905
## 117/603 [===>.....] - ETA: 7s - loss: 0.1139 - auc: 0.9907
## 121/603 [===>.....] - ETA: 7s - loss: 0.1144 - auc: 0.9906
## 125/603 [===>.....] - ETA: 7s - loss: 0.1149 - auc: 0.9906
## 129/603 [===>.....] - ETA: 7s - loss: 0.1144 - auc: 0.9907
## 133/603 [===>.....] - ETA: 7s - loss: 0.1125 - auc: 0.9909
## 136/603 [===>.....] - ETA: 7s - loss: 0.1129 - auc: 0.9909
## 140/603 [===>.....] - ETA: 7s - loss: 0.1125 - auc: 0.9910
## 143/603 [===>.....] - ETA: 7s - loss: 0.1117 - auc: 0.9911
```

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## 147/603 [=====>.....] - ETA: 7s - loss: 0.1117 - auc: 0.9911
## 151/603 [=====>.....] - ETA: 7s - loss: 0.1122 - auc: 0.9910
## 155/603 [=====>.....] - ETA: 7s - loss: 0.1119 - auc: 0.9911
## 159/603 [=====>.....] - ETA: 7s - loss: 0.1123 - auc: 0.9910
## 163/603 [=====>.....] - ETA: 7s - loss: 0.1135 - auc: 0.9909
## 167/603 [=====>.....] - ETA: 7s - loss: 0.1118 - auc: 0.9911
## 171/603 [=====>.....] - ETA: 7s - loss: 0.1125 - auc: 0.9911
## 175/603 [=====>.....] - ETA: 6s - loss: 0.1127 - auc: 0.9911
## 179/603 [=====>.....] - ETA: 6s - loss: 0.1137 - auc: 0.9909
## 183/603 [=====>.....] - ETA: 6s - loss: 0.1151 - auc: 0.9908
## 187/603 [=====>.....] - ETA: 6s - loss: 0.1139 - auc: 0.9909
## 191/603 [=====>.....] - ETA: 6s - loss: 0.1150 - auc: 0.9908
## 195/603 [=====>.....] - ETA: 6s - loss: 0.1141 - auc: 0.9909
## 198/603 [=====>.....] - ETA: 6s - loss: 0.1152 - auc: 0.9908
## 202/603 [=====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9909
## 206/603 [=====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9909
## 210/603 [=====>.....] - ETA: 6s - loss: 0.1139 - auc: 0.9910
## 214/603 [=====>.....] - ETA: 6s - loss: 0.1136 - auc: 0.9910
## 218/603 [=====>.....] - ETA: 6s - loss: 0.1135 - auc: 0.9911
## 222/603 [=====>.....] - ETA: 6s - loss: 0.1129 - auc: 0.9912
## 226/603 [=====>.....] - ETA: 6s - loss: 0.1131 - auc: 0.9911
## 230/603 [=====>.....] - ETA: 6s - loss: 0.1127 - auc: 0.9911
## 234/603 [=====>.....] - ETA: 5s - loss: 0.1123 - auc: 0.9912
## 238/603 [=====>.....] - ETA: 5s - loss: 0.1115 - auc: 0.9913
## 242/603 [=====>.....] - ETA: 5s - loss: 0.1119 - auc: 0.9912
## 246/603 [=====>.....] - ETA: 5s - loss: 0.1124 - auc: 0.9911
## 249/603 [=====>.....] - ETA: 5s - loss: 0.1125 - auc: 0.9910
## 253/603 [=====>.....] - ETA: 5s - loss: 0.1118 - auc: 0.9912
## 257/603 [=====>.....] - ETA: 5s - loss: 0.1119 - auc: 0.9911
## 261/603 [=====>.....] - ETA: 5s - loss: 0.1114 - auc: 0.9912
## 264/603 [=====>.....] - ETA: 5s - loss: 0.1116 - auc: 0.9912
## 268/603 [=====>.....] - ETA: 5s - loss: 0.1107 - auc: 0.9913
## 272/603 [=====>.....] - ETA: 5s - loss: 0.1104 - auc: 0.9914
## 276/603 [=====>.....] - ETA: 5s - loss: 0.1100 - auc: 0.9914
## 280/603 [=====>.....] - ETA: 5s - loss: 0.1104 - auc: 0.9914
## 284/603 [=====>.....] - ETA: 5s - loss: 0.1107 - auc: 0.9914
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1109 - auc: 0.9913
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1106 - auc: 0.9914
## 296/603 [=====>.....] - ETA: 4s - loss: 0.1108 - auc: 0.9913
## 300/603 [=====>.....] - ETA: 4s - loss: 0.1109 - auc: 0.9912
## 304/603 [=====>.....] - ETA: 4s - loss: 0.1140 - auc: 0.9907
## 308/603 [=====>.....] - ETA: 4s - loss: 0.1138 - auc: 0.9907
## 312/603 [=====>.....] - ETA: 4s - loss: 0.1134 - auc: 0.9907
## 316/603 [=====>.....] - ETA: 4s - loss: 0.1132 - auc: 0.9907
## 320/603 [=====>.....] - ETA: 4s - loss: 0.1140 - auc: 0.9907
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1137 - auc: 0.9906
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1140 - auc: 0.9906
## 331/603 [=====>.....] - ETA: 4s - loss: 0.1132 - auc: 0.9907
## 335/603 [=====>.....] - ETA: 4s - loss: 0.1133 - auc: 0.9907
## 339/603 [=====>.....] - ETA: 4s - loss: 0.1134 - auc: 0.9907
## 343/603 [=====>.....] - ETA: 4s - loss: 0.1129 - auc: 0.9908
## 347/603 [=====>.....] - ETA: 4s - loss: 0.1127 - auc: 0.9908
## 351/603 [=====>.....] - ETA: 4s - loss: 0.1125 - auc: 0.9908
## 355/603 [=====>.....] - ETA: 4s - loss: 0.1124 - auc: 0.9908
## 359/603 [=====>.....] - ETA: 3s - loss: 0.1123 - auc: 0.9908
## 363/603 [=====>.....] - ETA: 3s - loss: 0.1121 - auc: 0.9908
```

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## 367/603 [=====>.....] - ETA: 3s - loss: 0.1115 - auc: 0.9909
## 371/603 [=====>.....] - ETA: 3s - loss: 0.1112 - auc: 0.9909
## 375/603 [=====>.....] - ETA: 3s - loss: 0.1108 - auc: 0.9910
## 379/603 [=====>.....] - ETA: 3s - loss: 0.1110 - auc: 0.9909
## 383/603 [=====>.....] - ETA: 3s - loss: 0.1104 - auc: 0.9910
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1102 - auc: 0.9911
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1097 - auc: 0.9911
## 393/603 [=====>.....] - ETA: 3s - loss: 0.1098 - auc: 0.9911
## 397/603 [=====>.....] - ETA: 3s - loss: 0.1096 - auc: 0.9912
## 401/603 [=====>.....] - ETA: 3s - loss: 0.1105 - auc: 0.9910
## 405/603 [=====>.....] - ETA: 3s - loss: 0.1100 - auc: 0.9911
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1101 - auc: 0.9911
## 413/603 [=====>.....] - ETA: 3s - loss: 0.1102 - auc: 0.9911
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1096 - auc: 0.9912
## 421/603 [=====>.....] - ETA: 2s - loss: 0.1101 - auc: 0.9910
## 425/603 [=====>.....] - ETA: 2s - loss: 0.1102 - auc: 0.9910
## 429/603 [=====>.....] - ETA: 2s - loss: 0.1107 - auc: 0.9909
## 433/603 [=====>.....] - ETA: 2s - loss: 0.1102 - auc: 0.9910
## 437/603 [=====>.....] - ETA: 2s - loss: 0.1098 - auc: 0.9910
## 441/603 [=====>.....] - ETA: 2s - loss: 0.1102 - auc: 0.9910
## 445/603 [=====>.....] - ETA: 2s - loss: 0.1107 - auc: 0.9910
## 449/603 [=====>.....] - ETA: 2s - loss: 0.1108 - auc: 0.9909
## 453/603 [=====>.....] - ETA: 2s - loss: 0.1112 - auc: 0.9909
## 456/603 [=====>.....] - ETA: 2s - loss: 0.1117 - auc: 0.9907
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1112 - auc: 0.9908
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1113 - auc: 0.9908
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1115 - auc: 0.9908
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1110 - auc: 0.9909
## 476/603 [=====>.....] - ETA: 2s - loss: 0.1109 - auc: 0.9909
## 480/603 [=====>.....] - ETA: 1s - loss: 0.1110 - auc: 0.9909
## 484/603 [=====>.....] - ETA: 1s - loss: 0.1111 - auc: 0.9908
## 488/603 [=====>.....] - ETA: 1s - loss: 0.1112 - auc: 0.9908
## 492/603 [=====>.....] - ETA: 1s - loss: 0.1117 - auc: 0.9907
## 496/603 [=====>.....] - ETA: 1s - loss: 0.1118 - auc: 0.9906
## 500/603 [=====>.....] - ETA: 1s - loss: 0.1119 - auc: 0.9906
## 504/603 [=====>.....] - ETA: 1s - loss: 0.1119 - auc: 0.9906
## 507/603 [=====>.....] - ETA: 1s - loss: 0.1116 - auc: 0.9907
## 510/603 [=====>.....] - ETA: 1s - loss: 0.1115 - auc: 0.9907
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1114 - auc: 0.9907
## 516/603 [=====>.....] - ETA: 1s - loss: 0.1117 - auc: 0.9907
## 519/603 [=====>.....] - ETA: 1s - loss: 0.1120 - auc: 0.9907
## 523/603 [=====>....] - ETA: 1s - loss: 0.1118 - auc: 0.9907
## 526/603 [=====>....] - ETA: 1s - loss: 0.1119 - auc: 0.9907
## 529/603 [=====>....] - ETA: 1s - loss: 0.1117 - auc: 0.9907
## 532/603 [=====>....] - ETA: 1s - loss: 0.1116 - auc: 0.9908
## 536/603 [=====>....] - ETA: 1s - loss: 0.1113 - auc: 0.9908
## 540/603 [=====>....] - ETA: 1s - loss: 0.1110 - auc: 0.9908
## 543/603 [=====>...] - ETA: 0s - loss: 0.1120 - auc: 0.9907
## 547/603 [=====>...] - ETA: 0s - loss: 0.1128 - auc: 0.9906
## 551/603 [=====>...] - ETA: 0s - loss: 0.1129 - auc: 0.9906
## 555/603 [=====>...] - ETA: 0s - loss: 0.1131 - auc: 0.9906
## 559/603 [=====>...] - ETA: 0s - loss: 0.1126 - auc: 0.9906
## 563/603 [=====>..] - ETA: 0s - loss: 0.1124 - auc: 0.9907
## 567/603 [=====>..] - ETA: 0s - loss: 0.1126 - auc: 0.9906
## 571/603 [=====>..] - ETA: 0s - loss: 0.1130 - auc: 0.9906
## 575/603 [=====>..] - ETA: 0s - loss: 0.1127 - auc: 0.9906
```

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## 579/603 [=====>..] - ETA: 0s - loss: 0.1132 - auc: 0.9905
## 583/603 [=====>.] - ETA: 0s - loss: 0.1135 - auc: 0.9905
## 586/603 [=====>.] - ETA: 0s - loss: 0.1131 - auc: 0.9905
## 590/603 [=====>.] - ETA: 0s - loss: 0.1131 - auc: 0.9905
## 594/603 [=====>.] - ETA: 0s - loss: 0.1130 - auc: 0.9906
## 598/603 [=====>.] - ETA: 0s - loss: 0.1134 - auc: 0.9905
## 602/603 [=====>.] - ETA: 0s - loss: 0.1131 - auc: 0.9906
## 603/603 [=====] - 10s 17ms/step - loss: 0.1130 - auc: 0.9906 - va
l_loss: 0.1590 - val_auc: 0.9876
## Epoch 13/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0350 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.0525 - auc: 0.9982
## 8/603 [.....] - ETA: 9s - loss: 0.0601 - auc: 0.9982
## 12/603 [.....] - ETA: 9s - loss: 0.0787 - auc: 0.9953
## 16/603 [.....] - ETA: 9s - loss: 0.0932 - auc: 0.9944
## 19/603 [.....] - ETA: 9s - loss: 0.0918 - auc: 0.9933
## 22/603 [>.....] - ETA: 9s - loss: 0.0950 - auc: 0.9932
## 26/603 [>.....] - ETA: 9s - loss: 0.1026 - auc: 0.9914
## 29/603 [>.....] - ETA: 9s - loss: 0.1008 - auc: 0.9918
## 33/603 [>.....] - ETA: 9s - loss: 0.1140 - auc: 0.9897
## 37/603 [>.....] - ETA: 9s - loss: 0.1096 - auc: 0.9907
## 41/603 [=>.....] - ETA: 9s - loss: 0.1086 - auc: 0.9911
## 45/603 [=>.....] - ETA: 9s - loss: 0.1047 - auc: 0.9916
## 49/603 [=>.....] - ETA: 9s - loss: 0.1020 - auc: 0.9921
## 53/603 [=>.....] - ETA: 9s - loss: 0.1012 - auc: 0.9921
## 57/603 [=>.....] - ETA: 9s - loss: 0.1026 - auc: 0.9920
## 61/603 [==>.....] - ETA: 8s - loss: 0.1010 - auc: 0.9922
## 65/603 [==>.....] - ETA: 8s - loss: 0.0984 - auc: 0.9926
## 69/603 [==>.....] - ETA: 8s - loss: 0.1031 - auc: 0.9916
## 73/603 [==>.....] - ETA: 8s - loss: 0.1007 - auc: 0.9920
## 77/603 [==>.....] - ETA: 8s - loss: 0.1021 - auc: 0.9916
## 81/603 [===>.....] - ETA: 8s - loss: 0.1013 - auc: 0.9918
## 85/603 [===>.....] - ETA: 8s - loss: 0.0979 - auc: 0.9922
## 89/603 [===>.....] - ETA: 8s - loss: 0.1025 - auc: 0.9914
## 92/603 [===>.....] - ETA: 8s - loss: 0.1041 - auc: 0.9911
## 95/603 [===>.....] - ETA: 8s - loss: 0.1032 - auc: 0.9913
## 99/603 [===>.....] - ETA: 8s - loss: 0.1025 - auc: 0.9915
## 102/603 [====>.....] - ETA: 8s - loss: 0.1005 - auc: 0.9918
## 106/603 [====>.....] - ETA: 8s - loss: 0.0998 - auc: 0.9918
## 110/603 [====>.....] - ETA: 8s - loss: 0.0996 - auc: 0.9920
## 113/603 [====>.....] - ETA: 8s - loss: 0.0993 - auc: 0.9920
## 117/603 [====>.....] - ETA: 7s - loss: 0.0993 - auc: 0.9922
## 121/603 [====>.....] - ETA: 7s - loss: 0.1001 - auc: 0.9921
## 124/603 [====>.....] - ETA: 7s - loss: 0.1012 - auc: 0.9920
## 127/603 [====>.....] - ETA: 7s - loss: 0.1005 - auc: 0.9921
## 130/603 [====>.....] - ETA: 7s - loss: 0.1015 - auc: 0.9921
## 134/603 [====>.....] - ETA: 7s - loss: 0.1037 - auc: 0.9918
## 138/603 [====>.....] - ETA: 7s - loss: 0.1031 - auc: 0.9919
## 142/603 [====>.....] - ETA: 7s - loss: 0.1045 - auc: 0.9917
## 146/603 [====>.....] - ETA: 7s - loss: 0.1042 - auc: 0.9918
## 150/603 [====>.....] - ETA: 7s - loss: 0.1044 - auc: 0.9917
## 153/603 [====>.....] - ETA: 7s - loss: 0.1045 - auc: 0.9916
## 157/603 [====>.....] - ETA: 7s - loss: 0.1046 - auc: 0.9915
## 161/603 [====>.....] - ETA: 7s - loss: 0.1047 - auc: 0.9915
## 165/603 [====>.....] - ETA: 7s - loss: 0.1039 - auc: 0.9915
```

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## 169/603 [=====>.....] - ETA: 7s - loss: 0.1039 - auc: 0.9914
## 173/603 [=====>.....] - ETA: 7s - loss: 0.1032 - auc: 0.9915
## 177/603 [=====>.....] - ETA: 6s - loss: 0.1034 - auc: 0.9914
## 181/603 [=====>.....] - ETA: 6s - loss: 0.1034 - auc: 0.9914
## 185/603 [=====>.....] - ETA: 6s - loss: 0.1037 - auc: 0.9914
## 189/603 [=====>.....] - ETA: 6s - loss: 0.1038 - auc: 0.9914
## 193/603 [=====>.....] - ETA: 6s - loss: 0.1029 - auc: 0.9915
## 197/603 [=====>.....] - ETA: 6s - loss: 0.1041 - auc: 0.9914
## 201/603 [=====>.....] - ETA: 6s - loss: 0.1035 - auc: 0.9915
## 204/603 [=====>.....] - ETA: 6s - loss: 0.1033 - auc: 0.9915
## 208/603 [=====>.....] - ETA: 6s - loss: 0.1036 - auc: 0.9915
## 212/603 [=====>.....] - ETA: 6s - loss: 0.1033 - auc: 0.9915
## 216/603 [=====>.....] - ETA: 6s - loss: 0.1031 - auc: 0.9916
## 220/603 [=====>.....] - ETA: 6s - loss: 0.1030 - auc: 0.9916
## 224/603 [=====>.....] - ETA: 6s - loss: 0.1019 - auc: 0.9918
## 228/603 [=====>.....] - ETA: 6s - loss: 0.1017 - auc: 0.9918
## 232/603 [=====>.....] - ETA: 6s - loss: 0.1026 - auc: 0.9917
## 235/603 [=====>.....] - ETA: 6s - loss: 0.1037 - auc: 0.9915
## 238/603 [=====>.....] - ETA: 6s - loss: 0.1037 - auc: 0.9915
## 241/603 [=====>.....] - ETA: 5s - loss: 0.1032 - auc: 0.9916
## 244/603 [=====>.....] - ETA: 5s - loss: 0.1025 - auc: 0.9917
## 248/603 [=====>.....] - ETA: 5s - loss: 0.1023 - auc: 0.9917
## 252/603 [=====>.....] - ETA: 5s - loss: 0.1018 - auc: 0.9918
## 255/603 [=====>.....] - ETA: 5s - loss: 0.1018 - auc: 0.9918
## 259/603 [=====>.....] - ETA: 5s - loss: 0.1021 - auc: 0.9918
## 263/603 [=====>.....] - ETA: 5s - loss: 0.1015 - auc: 0.9919
## 266/603 [=====>.....] - ETA: 5s - loss: 0.1009 - auc: 0.9920
## 270/603 [=====>.....] - ETA: 5s - loss: 0.1014 - auc: 0.9918
## 273/603 [=====>.....] - ETA: 5s - loss: 0.1009 - auc: 0.9919
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1007 - auc: 0.9920
## 281/603 [=====>.....] - ETA: 5s - loss: 0.1013 - auc: 0.9919
## 285/603 [=====>.....] - ETA: 5s - loss: 0.1017 - auc: 0.9919
## 288/603 [=====>.....] - ETA: 5s - loss: 0.1016 - auc: 0.9919
## 292/603 [=====>.....] - ETA: 5s - loss: 0.1015 - auc: 0.9920
## 295/603 [=====>.....] - ETA: 5s - loss: 0.1017 - auc: 0.9920
## 298/603 [=====>.....] - ETA: 5s - loss: 0.1022 - auc: 0.9919
## 301/603 [=====>.....] - ETA: 4s - loss: 0.1024 - auc: 0.9919
## 305/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9919
## 308/603 [=====>.....] - ETA: 4s - loss: 0.1024 - auc: 0.9920
## 312/603 [=====>.....] - ETA: 4s - loss: 0.1026 - auc: 0.9920
## 315/603 [=====>.....] - ETA: 4s - loss: 0.1027 - auc: 0.9919
## 319/603 [=====>.....] - ETA: 4s - loss: 0.1028 - auc: 0.9919
## 322/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9919
## 326/603 [=====>.....] - ETA: 4s - loss: 0.1025 - auc: 0.9919
## 330/603 [=====>.....] - ETA: 4s - loss: 0.1028 - auc: 0.9919
## 334/603 [=====>.....] - ETA: 4s - loss: 0.1032 - auc: 0.9919
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1032 - auc: 0.9919
## 341/603 [=====>.....] - ETA: 4s - loss: 0.1030 - auc: 0.9919
## 344/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9920
## 348/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9920
## 352/603 [=====>.....] - ETA: 4s - loss: 0.1027 - auc: 0.9921
## 355/603 [=====>.....] - ETA: 4s - loss: 0.1025 - auc: 0.9921
## 359/603 [=====>.....] - ETA: 4s - loss: 0.1022 - auc: 0.9922
## 362/603 [=====>.....] - ETA: 3s - loss: 0.1020 - auc: 0.9922
## 366/603 [=====>.....] - ETA: 3s - loss: 0.1018 - auc: 0.9922
## 369/603 [=====>.....] - ETA: 3s - loss: 0.1014 - auc: 0.9922
```

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## 373/603 [=====>.....] - ETA: 3s - loss: 0.1012 - auc: 0.9923
## 376/603 [=====>.....] - ETA: 3s - loss: 0.1007 - auc: 0.9923
## 379/603 [=====>.....] - ETA: 3s - loss: 0.1009 - auc: 0.9923
## 383/603 [=====>.....] - ETA: 3s - loss: 0.1006 - auc: 0.9923
## 387/603 [=====>.....] - ETA: 3s - loss: 0.1005 - auc: 0.9924
## 390/603 [=====>.....] - ETA: 3s - loss: 0.1000 - auc: 0.9924
## 394/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9924
## 397/603 [=====>.....] - ETA: 3s - loss: 0.0996 - auc: 0.9925
## 401/603 [=====>.....] - ETA: 3s - loss: 0.0996 - auc: 0.9925
## 404/603 [=====>.....] - ETA: 3s - loss: 0.0996 - auc: 0.9925
## 408/603 [=====>.....] - ETA: 3s - loss: 0.0993 - auc: 0.9925
## 411/603 [=====>.....] - ETA: 3s - loss: 0.0998 - auc: 0.9925
## 415/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9925
## 419/603 [=====>.....] - ETA: 3s - loss: 0.0997 - auc: 0.9925
## 423/603 [=====>.....] - ETA: 2s - loss: 0.0997 - auc: 0.9925
## 427/603 [=====>.....] - ETA: 2s - loss: 0.0996 - auc: 0.9926
## 431/603 [=====>.....] - ETA: 2s - loss: 0.0999 - auc: 0.9925
## 435/603 [=====>.....] - ETA: 2s - loss: 0.0999 - auc: 0.9925
## 438/603 [=====>.....] - ETA: 2s - loss: 0.0996 - auc: 0.9926
## 442/603 [=====>.....] - ETA: 2s - loss: 0.0994 - auc: 0.9926
## 446/603 [=====>.....] - ETA: 2s - loss: 0.0995 - auc: 0.9926
## 449/603 [=====>.....] - ETA: 2s - loss: 0.0997 - auc: 0.9926
## 453/603 [=====>.....] - ETA: 2s - loss: 0.0992 - auc: 0.9927
## 457/603 [=====>.....] - ETA: 2s - loss: 0.0999 - auc: 0.9926
## 460/603 [=====>.....] - ETA: 2s - loss: 0.1000 - auc: 0.9925
## 464/603 [=====>.....] - ETA: 2s - loss: 0.1003 - auc: 0.9925
## 468/603 [=====>.....] - ETA: 2s - loss: 0.1006 - auc: 0.9924
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1004 - auc: 0.9925
## 476/603 [=====>.....] - ETA: 2s - loss: 0.1002 - auc: 0.9925
## 479/603 [=====>.....] - ETA: 2s - loss: 0.1004 - auc: 0.9925
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1011 - auc: 0.9924
## 486/603 [=====>.....] - ETA: 1s - loss: 0.1017 - auc: 0.9923
## 490/603 [=====>.....] - ETA: 1s - loss: 0.1014 - auc: 0.9924
## 493/603 [=====>.....] - ETA: 1s - loss: 0.1013 - auc: 0.9924
## 497/603 [=====>.....] - ETA: 1s - loss: 0.1016 - auc: 0.9923
## 501/603 [=====>.....] - ETA: 1s - loss: 0.1023 - auc: 0.9922
## 505/603 [=====>.....] - ETA: 1s - loss: 0.1018 - auc: 0.9923
## 508/603 [=====>.....] - ETA: 1s - loss: 0.1019 - auc: 0.9922
## 511/603 [=====>.....] - ETA: 1s - loss: 0.1018 - auc: 0.9923
## 515/603 [=====>.....] - ETA: 1s - loss: 0.1016 - auc: 0.9923
## 519/603 [=====>.....] - ETA: 1s - loss: 0.1017 - auc: 0.9922
## 523/603 [=====>....] - ETA: 1s - loss: 0.1021 - auc: 0.9922
## 527/603 [=====>....] - ETA: 1s - loss: 0.1027 - auc: 0.9921
## 531/603 [=====>....] - ETA: 1s - loss: 0.1029 - auc: 0.9920
## 534/603 [=====>....] - ETA: 1s - loss: 0.1030 - auc: 0.9920
## 538/603 [=====>....] - ETA: 1s - loss: 0.1035 - auc: 0.9918
## 542/603 [=====>....] - ETA: 1s - loss: 0.1033 - auc: 0.9918
## 546/603 [=====>...] - ETA: 0s - loss: 0.1035 - auc: 0.9918
## 549/603 [=====>...] - ETA: 0s - loss: 0.1033 - auc: 0.9918
## 552/603 [=====>...] - ETA: 0s - loss: 0.1031 - auc: 0.9919
## 555/603 [=====>...] - ETA: 0s - loss: 0.1028 - auc: 0.9919
## 558/603 [=====>...] - ETA: 0s - loss: 0.1032 - auc: 0.9918
## 561/603 [=====>...] - ETA: 0s - loss: 0.1029 - auc: 0.9918
## 564/603 [=====>..] - ETA: 0s - loss: 0.1030 - auc: 0.9918
## 567/603 [=====>..] - ETA: 0s - loss: 0.1033 - auc: 0.9918
## 570/603 [=====>..] - ETA: 0s - loss: 0.1039 - auc: 0.9917
```

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## 573/603 [=====>..] - ETA: 0s - loss: 0.1038 - auc: 0.9917
## 577/603 [=====>..] - ETA: 0s - loss: 0.1039 - auc: 0.9917
## 580/603 [=====>..] - ETA: 0s - loss: 0.1045 - auc: 0.9916
## 583/603 [=====>.] - ETA: 0s - loss: 0.1044 - auc: 0.9916
## 587/603 [=====>.] - ETA: 0s - loss: 0.1043 - auc: 0.9916
## 591/603 [=====>.] - ETA: 0s - loss: 0.1046 - auc: 0.9916
## 595/603 [=====>.] - ETA: 0s - loss: 0.1047 - auc: 0.9916
## 598/603 [=====>.] - ETA: 0s - loss: 0.1044 - auc: 0.9916
## 602/603 [=====>.] - ETA: 0s - loss: 0.1043 - auc: 0.9916
## 603/603 [=====] - 10s 17ms/step - loss: 0.1042 - auc: 0.9916 - va
l_loss: 0.1608 - val_auc: 0.9860
## Epoch 14/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0507 - auc: 1.0000
## 5/603 [.....] - ETA: 10s - loss: 0.0804 - auc: 0.9976
## 9/603 [.....] - ETA: 10s - loss: 0.0843 - auc: 0.9970
## 13/603 [.....] - ETA: 9s - loss: 0.0983 - auc: 0.9943
## 16/603 [.....] - ETA: 9s - loss: 0.1014 - auc: 0.9935
## 20/603 [.....] - ETA: 9s - loss: 0.0930 - auc: 0.9945
## 23/603 [>.....] - ETA: 9s - loss: 0.0973 - auc: 0.9937
## 27/603 [>.....] - ETA: 9s - loss: 0.0973 - auc: 0.9935
## 31/603 [>.....] - ETA: 9s - loss: 0.1051 - auc: 0.9924
## 34/603 [>.....] - ETA: 9s - loss: 0.1126 - auc: 0.9914
## 37/603 [>.....] - ETA: 9s - loss: 0.1096 - auc: 0.9918
## 41/603 [=>.....] - ETA: 9s - loss: 0.1077 - auc: 0.9921
## 44/603 [=>.....] - ETA: 9s - loss: 0.1038 - auc: 0.9927
## 48/603 [=>.....] - ETA: 9s - loss: 0.1001 - auc: 0.9933
## 52/603 [=>.....] - ETA: 9s - loss: 0.1011 - auc: 0.9930
## 55/603 [=>.....] - ETA: 9s - loss: 0.1011 - auc: 0.9931
## 58/603 [=>.....] - ETA: 9s - loss: 0.1007 - auc: 0.9932
## 61/603 [==>.....] - ETA: 9s - loss: 0.1055 - auc: 0.9926
## 65/603 [==>.....] - ETA: 8s - loss: 0.1069 - auc: 0.9922
## 68/603 [==>.....] - ETA: 8s - loss: 0.1057 - auc: 0.9925
## 72/603 [==>.....] - ETA: 8s - loss: 0.1090 - auc: 0.9914
## 76/603 [==>.....] - ETA: 8s - loss: 0.1085 - auc: 0.9915
## 80/603 [==>.....] - ETA: 8s - loss: 0.1060 - auc: 0.9919
## 83/603 [===>.....] - ETA: 8s - loss: 0.1072 - auc: 0.9918
## 87/603 [===>.....] - ETA: 8s - loss: 0.1074 - auc: 0.9917
## 90/603 [===>.....] - ETA: 8s - loss: 0.1052 - auc: 0.9920
## 93/603 [===>.....] - ETA: 8s - loss: 0.1049 - auc: 0.9919
## 96/603 [===>.....] - ETA: 8s - loss: 0.1034 - auc: 0.9921
## 100/603 [===>.....] - ETA: 8s - loss: 0.1040 - auc: 0.9920
## 103/603 [====>.....] - ETA: 8s - loss: 0.1054 - auc: 0.9919
## 107/603 [====>.....] - ETA: 8s - loss: 0.1045 - auc: 0.9921
## 111/603 [====>.....] - ETA: 8s - loss: 0.1028 - auc: 0.9923
## 114/603 [====>.....] - ETA: 8s - loss: 0.1020 - auc: 0.9925
## 117/603 [====>.....] - ETA: 8s - loss: 0.1002 - auc: 0.9928
## 121/603 [====>.....] - ETA: 8s - loss: 0.0992 - auc: 0.9928
## 125/603 [====>.....] - ETA: 7s - loss: 0.0983 - auc: 0.9930
## 128/603 [====>.....] - ETA: 7s - loss: 0.0987 - auc: 0.9931
## 132/603 [====>.....] - ETA: 7s - loss: 0.0976 - auc: 0.9933
## 135/603 [====>.....] - ETA: 7s - loss: 0.0979 - auc: 0.9932
## 139/603 [====>.....] - ETA: 7s - loss: 0.0995 - auc: 0.9931
## 142/603 [====>.....] - ETA: 7s - loss: 0.0981 - auc: 0.9933
## 146/603 [====>.....] - ETA: 7s - loss: 0.0981 - auc: 0.9933
## 149/603 [====>.....] - ETA: 7s - loss: 0.0981 - auc: 0.9933
```

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## 152/603 [=====>.....] - ETA: 7s - loss: 0.0985 - auc: 0.9932
## 156/603 [=====>.....] - ETA: 7s - loss: 0.0981 - auc: 0.9933
## 160/603 [=====>.....] - ETA: 7s - loss: 0.1011 - auc: 0.9927
## 164/603 [=====>.....] - ETA: 7s - loss: 0.0999 - auc: 0.9929
## 167/603 [=====>.....] - ETA: 7s - loss: 0.0992 - auc: 0.9930
## 171/603 [=====>.....] - ETA: 7s - loss: 0.0993 - auc: 0.9930
## 174/603 [=====>.....] - ETA: 7s - loss: 0.0992 - auc: 0.9930
## 177/603 [=====>.....] - ETA: 7s - loss: 0.0990 - auc: 0.9930
## 181/603 [=====>.....] - ETA: 7s - loss: 0.0985 - auc: 0.9931
## 185/603 [=====>.....] - ETA: 6s - loss: 0.0976 - auc: 0.9932
## 189/603 [=====>.....] - ETA: 6s - loss: 0.0972 - auc: 0.9932
## 193/603 [=====>.....] - ETA: 6s - loss: 0.0983 - auc: 0.9930
## 197/603 [=====>.....] - ETA: 6s - loss: 0.0976 - auc: 0.9931
## 201/603 [=====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9932
## 204/603 [=====>.....] - ETA: 6s - loss: 0.0980 - auc: 0.9931
## 208/603 [=====>.....] - ETA: 6s - loss: 0.0977 - auc: 0.9931
## 211/603 [=====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9933
## 215/603 [=====>.....] - ETA: 6s - loss: 0.0976 - auc: 0.9931
## 218/603 [=====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9932
## 222/603 [=====>.....] - ETA: 6s - loss: 0.0966 - auc: 0.9933
## 225/603 [=====>.....] - ETA: 6s - loss: 0.0966 - auc: 0.9933
## 228/603 [=====>.....] - ETA: 6s - loss: 0.0963 - auc: 0.9933
## 231/603 [=====>.....] - ETA: 6s - loss: 0.0964 - auc: 0.9933
## 235/603 [=====>.....] - ETA: 6s - loss: 0.0982 - auc: 0.9929
## 239/603 [=====>.....] - ETA: 6s - loss: 0.0980 - auc: 0.9930
## 243/603 [=====>.....] - ETA: 5s - loss: 0.0983 - auc: 0.9930
## 247/603 [=====>.....] - ETA: 5s - loss: 0.0983 - auc: 0.9929
## 250/603 [=====>.....] - ETA: 5s - loss: 0.0981 - auc: 0.9929
## 253/603 [=====>.....] - ETA: 5s - loss: 0.0983 - auc: 0.9929
## 257/603 [=====>.....] - ETA: 5s - loss: 0.0984 - auc: 0.9929
## 260/603 [=====>.....] - ETA: 5s - loss: 0.0989 - auc: 0.9928
## 264/603 [=====>.....] - ETA: 5s - loss: 0.0984 - auc: 0.9929
## 267/603 [=====>.....] - ETA: 5s - loss: 0.0986 - auc: 0.9929
## 271/603 [=====>.....] - ETA: 5s - loss: 0.0997 - auc: 0.9928
## 275/603 [=====>.....] - ETA: 5s - loss: 0.0991 - auc: 0.9929
## 279/603 [=====>.....] - ETA: 5s - loss: 0.0990 - auc: 0.9929
## 282/603 [=====>.....] - ETA: 5s - loss: 0.0999 - auc: 0.9927
## 286/603 [=====>.....] - ETA: 5s - loss: 0.0998 - auc: 0.9928
## 290/603 [=====>.....] - ETA: 5s - loss: 0.1000 - auc: 0.9928
## 293/603 [=====>.....] - ETA: 5s - loss: 0.1005 - auc: 0.9927
## 296/603 [=====>.....] - ETA: 5s - loss: 0.1008 - auc: 0.9926
## 299/603 [=====>.....] - ETA: 5s - loss: 0.1008 - auc: 0.9927
## 303/603 [=====>.....] - ETA: 5s - loss: 0.1015 - auc: 0.9925
## 306/603 [=====>.....] - ETA: 4s - loss: 0.1010 - auc: 0.9925
## 309/603 [=====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9926
## 312/603 [=====>.....] - ETA: 4s - loss: 0.1018 - auc: 0.9924
## 316/603 [=====>.....] - ETA: 4s - loss: 0.1016 - auc: 0.9924
## 319/603 [=====>.....] - ETA: 4s - loss: 0.1009 - auc: 0.9925
## 322/603 [=====>.....] - ETA: 4s - loss: 0.1004 - auc: 0.9926
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1006 - auc: 0.9925
## 329/603 [=====>.....] - ETA: 4s - loss: 0.1005 - auc: 0.9926
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1004 - auc: 0.9926
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1007 - auc: 0.9925
## 340/603 [=====>.....] - ETA: 4s - loss: 0.1004 - auc: 0.9926
## 344/603 [=====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9926
## 347/603 [=====>.....] - ETA: 4s - loss: 0.1008 - auc: 0.9925
```

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## 350/603 [=====>.....] - ETA: 4s - loss: 0.1013 - auc: 0.9924
## 354/603 [=====>.....] - ETA: 4s - loss: 0.1017 - auc: 0.9923
## 357/603 [=====>.....] - ETA: 4s - loss: 0.1016 - auc: 0.9924
## 361/603 [=====>.....] - ETA: 4s - loss: 0.1034 - auc: 0.9921
## 365/603 [=====>.....] - ETA: 3s - loss: 0.1039 - auc: 0.9920
## 368/603 [=====>.....] - ETA: 3s - loss: 0.1035 - auc: 0.9921
## 372/603 [=====>.....] - ETA: 3s - loss: 0.1033 - auc: 0.9921
## 376/603 [=====>.....] - ETA: 3s - loss: 0.1029 - auc: 0.9922
## 380/603 [=====>.....] - ETA: 3s - loss: 0.1038 - auc: 0.9920
## 384/603 [=====>.....] - ETA: 3s - loss: 0.1035 - auc: 0.9921
## 388/603 [=====>.....] - ETA: 3s - loss: 0.1041 - auc: 0.9920
## 392/603 [=====>.....] - ETA: 3s - loss: 0.1041 - auc: 0.9920
## 395/603 [=====>.....] - ETA: 3s - loss: 0.1037 - auc: 0.9921
## 399/603 [=====>.....] - ETA: 3s - loss: 0.1041 - auc: 0.9920
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1045 - auc: 0.9919
## 406/603 [=====>.....] - ETA: 3s - loss: 0.1045 - auc: 0.9919
## 410/603 [=====>.....] - ETA: 3s - loss: 0.1053 - auc: 0.9918
## 414/603 [=====>.....] - ETA: 3s - loss: 0.1064 - auc: 0.9916
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1065 - auc: 0.9916
## 421/603 [=====>.....] - ETA: 3s - loss: 0.1067 - auc: 0.9916
## 424/603 [=====>.....] - ETA: 2s - loss: 0.1064 - auc: 0.9916
## 428/603 [=====>.....] - ETA: 2s - loss: 0.1069 - auc: 0.9915
## 432/603 [=====>.....] - ETA: 2s - loss: 0.1067 - auc: 0.9915
## 435/603 [=====>.....] - ETA: 2s - loss: 0.1067 - auc: 0.9916
## 439/603 [=====>.....] - ETA: 2s - loss: 0.1067 - auc: 0.9916
## 443/603 [=====>.....] - ETA: 2s - loss: 0.1066 - auc: 0.9916
## 447/603 [=====>.....] - ETA: 2s - loss: 0.1074 - auc: 0.9915
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1073 - auc: 0.9915
## 455/603 [=====>.....] - ETA: 2s - loss: 0.1077 - auc: 0.9914
## 459/603 [=====>.....] - ETA: 2s - loss: 0.1075 - auc: 0.9914
## 462/603 [=====>.....] - ETA: 2s - loss: 0.1074 - auc: 0.9915
## 466/603 [=====>.....] - ETA: 2s - loss: 0.1075 - auc: 0.9914
## 469/603 [=====>.....] - ETA: 2s - loss: 0.1072 - auc: 0.9915
## 473/603 [=====>.....] - ETA: 2s - loss: 0.1078 - auc: 0.9914
## 477/603 [=====>.....] - ETA: 2s - loss: 0.1075 - auc: 0.9915
## 480/603 [=====>.....] - ETA: 2s - loss: 0.1072 - auc: 0.9915
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1071 - auc: 0.9915
## 487/603 [=====>.....] - ETA: 1s - loss: 0.1070 - auc: 0.9915
## 491/603 [=====>.....] - ETA: 1s - loss: 0.1068 - auc: 0.9915
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1070 - auc: 0.9915
## 498/603 [=====>.....] - ETA: 1s - loss: 0.1068 - auc: 0.9916
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1067 - auc: 0.9916
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1066 - auc: 0.9916
## 509/603 [=====>.....] - ETA: 1s - loss: 0.1068 - auc: 0.9915
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1065 - auc: 0.9916
## 517/603 [=====>.....] - ETA: 1s - loss: 0.1062 - auc: 0.9916
## 520/603 [=====>.....] - ETA: 1s - loss: 0.1063 - auc: 0.9916
## 523/603 [=====>....] - ETA: 1s - loss: 0.1060 - auc: 0.9916
## 527/603 [=====>....] - ETA: 1s - loss: 0.1058 - auc: 0.9917
## 530/603 [=====>....] - ETA: 1s - loss: 0.1060 - auc: 0.9917
## 534/603 [=====>....] - ETA: 1s - loss: 0.1064 - auc: 0.9916
## 537/603 [=====>....] - ETA: 1s - loss: 0.1063 - auc: 0.9916
## 540/603 [=====>....] - ETA: 1s - loss: 0.1062 - auc: 0.9916
## 543/603 [=====>...] - ETA: 0s - loss: 0.1066 - auc: 0.9916
## 547/603 [=====>...] - ETA: 0s - loss: 0.1067 - auc: 0.9916
## 551/603 [=====>...] - ETA: 0s - loss: 0.1065 - auc: 0.9916
```

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## 555/603 [=====>...] - ETA: 0s - loss: 0.1066 - auc: 0.9916
## 558/603 [=====>...] - ETA: 0s - loss: 0.1065 - auc: 0.9916
## 562/603 [=====>...] - ETA: 0s - loss: 0.1064 - auc: 0.9916
## 566/603 [=====>..] - ETA: 0s - loss: 0.1059 - auc: 0.9917
## 569/603 [=====>..] - ETA: 0s - loss: 0.1058 - auc: 0.9917
## 573/603 [=====>..] - ETA: 0s - loss: 0.1062 - auc: 0.9916
## 577/603 [=====>..] - ETA: 0s - loss: 0.1066 - auc: 0.9915
## 580/603 [=====>..] - ETA: 0s - loss: 0.1065 - auc: 0.9916
## 584/603 [=====>.] - ETA: 0s - loss: 0.1061 - auc: 0.9916
## 588/603 [=====>.] - ETA: 0s - loss: 0.1056 - auc: 0.9917
## 591/603 [=====>.] - ETA: 0s - loss: 0.1062 - auc: 0.9916
## 594/603 [=====>.] - ETA: 0s - loss: 0.1067 - auc: 0.9915
## 598/603 [=====>.] - ETA: 0s - loss: 0.1071 - auc: 0.9915
## 601/603 [=====>.] - ETA: 0s - loss: 0.1069 - auc: 0.9915
## 603/603 [=====] - 10s 17ms/step - loss: 0.1067 - auc: 0.9915 - va
l_loss: 0.1448 - val_auc: 0.9839
## Epoch 15/20
##
## 1/603 [.....] - ETA: 10s - loss: 0.0700 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.1024 - auc: 0.9890
## 9/603 [.....] - ETA: 9s - loss: 0.0987 - auc: 0.9915
## 12/603 [.....] - ETA: 9s - loss: 0.0893 - auc: 0.9932
## 15/603 [.....] - ETA: 9s - loss: 0.0816 - auc: 0.9944
## 19/603 [.....] - ETA: 9s - loss: 0.0823 - auc: 0.9946
## 22/603 [>.....] - ETA: 9s - loss: 0.0795 - auc: 0.9951
## 26/603 [>.....] - ETA: 9s - loss: 0.0868 - auc: 0.9940
## 30/603 [>.....] - ETA: 9s - loss: 0.0856 - auc: 0.9944
## 34/603 [>.....] - ETA: 9s - loss: 0.0845 - auc: 0.9947
## 37/603 [>.....] - ETA: 9s - loss: 0.0814 - auc: 0.9951
## 41/603 [=>.....] - ETA: 9s - loss: 0.0788 - auc: 0.9955
## 45/603 [=>.....] - ETA: 9s - loss: 0.0887 - auc: 0.9939
## 48/603 [=>.....] - ETA: 9s - loss: 0.0904 - auc: 0.9936
## 51/603 [=>.....] - ETA: 9s - loss: 0.0929 - auc: 0.9932
## 54/603 [=>.....] - ETA: 9s - loss: 0.0940 - auc: 0.9929
## 57/603 [=>.....] - ETA: 9s - loss: 0.0945 - auc: 0.9928
## 60/603 [=>.....] - ETA: 9s - loss: 0.0954 - auc: 0.9926
## 63/603 [==>.....] - ETA: 8s - loss: 0.0948 - auc: 0.9927
## 66/603 [==>.....] - ETA: 8s - loss: 0.0976 - auc: 0.9921
## 70/603 [==>.....] - ETA: 8s - loss: 0.0989 - auc: 0.9920
## 73/603 [==>.....] - ETA: 8s - loss: 0.1004 - auc: 0.9919
## 76/603 [==>.....] - ETA: 8s - loss: 0.1002 - auc: 0.9920
## 80/603 [==>.....] - ETA: 8s - loss: 0.1005 - auc: 0.9921
## 83/603 [===>.....] - ETA: 8s - loss: 0.0988 - auc: 0.9923
## 87/603 [===>.....] - ETA: 8s - loss: 0.0970 - auc: 0.9925
## 90/603 [===>.....] - ETA: 8s - loss: 0.0967 - auc: 0.9927
## 94/603 [===>.....] - ETA: 8s - loss: 0.0976 - auc: 0.9926
## 97/603 [===>.....] - ETA: 8s - loss: 0.0960 - auc: 0.9928
## 101/603 [====>.....] - ETA: 8s - loss: 0.0972 - auc: 0.9927
## 104/603 [====>.....] - ETA: 8s - loss: 0.0973 - auc: 0.9927
## 107/603 [====>.....] - ETA: 8s - loss: 0.1034 - auc: 0.9914
## 110/603 [====>.....] - ETA: 8s - loss: 0.1041 - auc: 0.9914
## 113/603 [====>.....] - ETA: 8s - loss: 0.1055 - auc: 0.9911
## 117/603 [====>.....] - ETA: 8s - loss: 0.1052 - auc: 0.9912
## 121/603 [====>.....] - ETA: 8s - loss: 0.1060 - auc: 0.9910
## 124/603 [====>.....] - ETA: 7s - loss: 0.1052 - auc: 0.9911
## 128/603 [====>.....] - ETA: 7s - loss: 0.1053 - auc: 0.9911
```

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## 131/603 [=====>.....] - ETA: 7s - loss: 0.1064 - auc: 0.9910
## 135/603 [=====>.....] - ETA: 7s - loss: 0.1061 - auc: 0.9911
## 138/603 [=====>.....] - ETA: 7s - loss: 0.1066 - auc: 0.9910
## 142/603 [=====>.....] - ETA: 7s - loss: 0.1069 - auc: 0.9910
## 146/603 [=====>.....] - ETA: 7s - loss: 0.1063 - auc: 0.9912
## 150/603 [=====>.....] - ETA: 7s - loss: 0.1093 - auc: 0.9906
## 154/603 [=====>.....] - ETA: 7s - loss: 0.1105 - auc: 0.9905
## 157/603 [=====>.....] - ETA: 7s - loss: 0.1110 - auc: 0.9904
## 161/603 [=====>.....] - ETA: 7s - loss: 0.1123 - auc: 0.9902
## 165/603 [=====>.....] - ETA: 7s - loss: 0.1159 - auc: 0.9895
## 168/603 [=====>.....] - ETA: 7s - loss: 0.1164 - auc: 0.9894
## 172/603 [=====>.....] - ETA: 7s - loss: 0.1169 - auc: 0.9894
## 175/603 [=====>.....] - ETA: 7s - loss: 0.1162 - auc: 0.9895
## 179/603 [=====>.....] - ETA: 7s - loss: 0.1162 - auc: 0.9896
## 183/603 [=====>.....] - ETA: 6s - loss: 0.1170 - auc: 0.9894
## 186/603 [=====>.....] - ETA: 6s - loss: 0.1156 - auc: 0.9896
## 189/603 [=====>.....] - ETA: 6s - loss: 0.1159 - auc: 0.9896
## 192/603 [=====>.....] - ETA: 6s - loss: 0.1149 - auc: 0.9897
## 196/603 [=====>.....] - ETA: 6s - loss: 0.1146 - auc: 0.9898
## 200/603 [=====>.....] - ETA: 6s - loss: 0.1139 - auc: 0.9899
## 204/603 [=====>.....] - ETA: 6s - loss: 0.1134 - auc: 0.9900
## 208/603 [=====>.....] - ETA: 6s - loss: 0.1120 - auc: 0.9902
## 212/603 [=====>.....] - ETA: 6s - loss: 0.1107 - auc: 0.9904
## 215/603 [=====>.....] - ETA: 6s - loss: 0.1106 - auc: 0.9904
## 218/603 [=====>.....] - ETA: 6s - loss: 0.1100 - auc: 0.9905
## 221/603 [=====>.....] - ETA: 6s - loss: 0.1093 - auc: 0.9907
## 225/603 [=====>.....] - ETA: 6s - loss: 0.1104 - auc: 0.9905
## 229/603 [=====>.....] - ETA: 6s - loss: 0.1101 - auc: 0.9906
## 233/603 [=====>.....] - ETA: 6s - loss: 0.1103 - auc: 0.9906
## 236/603 [=====>.....] - ETA: 6s - loss: 0.1096 - auc: 0.9907
## 240/603 [=====>.....] - ETA: 6s - loss: 0.1084 - auc: 0.9909
## 243/603 [=====>.....] - ETA: 5s - loss: 0.1079 - auc: 0.9910
## 247/603 [=====>.....] - ETA: 5s - loss: 0.1078 - auc: 0.9910
## 251/603 [=====>.....] - ETA: 5s - loss: 0.1076 - auc: 0.9911
## 255/603 [=====>.....] - ETA: 5s - loss: 0.1069 - auc: 0.9912
## 259/603 [=====>.....] - ETA: 5s - loss: 0.1065 - auc: 0.9913
## 263/603 [=====>.....] - ETA: 5s - loss: 0.1056 - auc: 0.9914
## 267/603 [=====>.....] - ETA: 5s - loss: 0.1057 - auc: 0.9914
## 270/603 [=====>.....] - ETA: 5s - loss: 0.1052 - auc: 0.9914
## 273/603 [=====>.....] - ETA: 5s - loss: 0.1053 - auc: 0.9914
## 277/603 [=====>.....] - ETA: 5s - loss: 0.1053 - auc: 0.9914
## 280/603 [=====>.....] - ETA: 5s - loss: 0.1059 - auc: 0.9914
## 284/603 [=====>.....] - ETA: 5s - loss: 0.1055 - auc: 0.9915
## 287/603 [=====>.....] - ETA: 5s - loss: 0.1050 - auc: 0.9915
## 291/603 [=====>.....] - ETA: 5s - loss: 0.1045 - auc: 0.9916
## 294/603 [=====>.....] - ETA: 5s - loss: 0.1044 - auc: 0.9917
## 297/603 [=====>.....] - ETA: 5s - loss: 0.1046 - auc: 0.9917
## 301/603 [=====>.....] - ETA: 5s - loss: 0.1039 - auc: 0.9918
## 305/603 [=====>.....] - ETA: 4s - loss: 0.1032 - auc: 0.9919
## 309/603 [=====>.....] - ETA: 4s - loss: 0.1035 - auc: 0.9919
## 313/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9920
## 317/603 [=====>.....] - ETA: 4s - loss: 0.1030 - auc: 0.9919
## 320/603 [=====>.....] - ETA: 4s - loss: 0.1040 - auc: 0.9918
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1040 - auc: 0.9918
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1044 - auc: 0.9917
## 331/603 [=====>.....] - ETA: 4s - loss: 0.1036 - auc: 0.9918
```

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## 335/603 [=====>.....] - ETA: 4s - loss: 0.1032 - auc: 0.9918
## 339/603 [=====>.....] - ETA: 4s - loss: 0.1031 - auc: 0.9918
## 343/603 [=====>.....] - ETA: 4s - loss: 0.1032 - auc: 0.9918
## 347/603 [=====>.....] - ETA: 4s - loss: 0.1030 - auc: 0.9918
## 351/603 [=====>.....] - ETA: 4s - loss: 0.1029 - auc: 0.9918
## 355/603 [=====>.....] - ETA: 4s - loss: 0.1022 - auc: 0.9919
## 358/603 [=====>.....] - ETA: 4s - loss: 0.1033 - auc: 0.9917
## 362/603 [=====>.....] - ETA: 3s - loss: 0.1033 - auc: 0.9917
## 366/603 [=====>.....] - ETA: 3s - loss: 0.1037 - auc: 0.9916
## 370/603 [=====>.....] - ETA: 3s - loss: 0.1039 - auc: 0.9916
## 374/603 [=====>.....] - ETA: 3s - loss: 0.1035 - auc: 0.9917
## 378/603 [=====>.....] - ETA: 3s - loss: 0.1037 - auc: 0.9917
## 382/603 [=====>.....] - ETA: 3s - loss: 0.1038 - auc: 0.9917
## 385/603 [=====>.....] - ETA: 3s - loss: 0.1041 - auc: 0.9916
## 389/603 [=====>.....] - ETA: 3s - loss: 0.1050 - auc: 0.9915
## 393/603 [=====>.....] - ETA: 3s - loss: 0.1048 - auc: 0.9915
## 396/603 [=====>.....] - ETA: 3s - loss: 0.1043 - auc: 0.9916
## 399/603 [=====>.....] - ETA: 3s - loss: 0.1043 - auc: 0.9916
## 402/603 [=====>.....] - ETA: 3s - loss: 0.1042 - auc: 0.9916
## 405/603 [=====>.....] - ETA: 3s - loss: 0.1040 - auc: 0.9916
## 409/603 [=====>.....] - ETA: 3s - loss: 0.1035 - auc: 0.9917
## 413/603 [=====>.....] - ETA: 3s - loss: 0.1038 - auc: 0.9917
## 417/603 [=====>.....] - ETA: 3s - loss: 0.1033 - auc: 0.9918
## 420/603 [=====>.....] - ETA: 3s - loss: 0.1032 - auc: 0.9918
## 423/603 [=====>.....] - ETA: 2s - loss: 0.1032 - auc: 0.9918
## 427/603 [=====>.....] - ETA: 2s - loss: 0.1037 - auc: 0.9918
## 430/603 [=====>.....] - ETA: 2s - loss: 0.1034 - auc: 0.9918
## 434/603 [=====>.....] - ETA: 2s - loss: 0.1037 - auc: 0.9918
## 438/603 [=====>.....] - ETA: 2s - loss: 0.1034 - auc: 0.9919
## 441/603 [=====>.....] - ETA: 2s - loss: 0.1030 - auc: 0.9919
## 444/603 [=====>.....] - ETA: 2s - loss: 0.1028 - auc: 0.9920
## 447/603 [=====>.....] - ETA: 2s - loss: 0.1029 - auc: 0.9919
## 451/603 [=====>.....] - ETA: 2s - loss: 0.1028 - auc: 0.9919
## 454/603 [=====>.....] - ETA: 2s - loss: 0.1031 - auc: 0.9919
## 458/603 [=====>.....] - ETA: 2s - loss: 0.1037 - auc: 0.9917
## 461/603 [=====>.....] - ETA: 2s - loss: 0.1038 - auc: 0.9917
## 465/603 [=====>.....] - ETA: 2s - loss: 0.1041 - auc: 0.9917
## 469/603 [=====>.....] - ETA: 2s - loss: 0.1040 - auc: 0.9918
## 472/603 [=====>.....] - ETA: 2s - loss: 0.1044 - auc: 0.9917
## 475/603 [=====>.....] - ETA: 2s - loss: 0.1042 - auc: 0.9917
## 479/603 [=====>.....] - ETA: 2s - loss: 0.1038 - auc: 0.9918
## 483/603 [=====>.....] - ETA: 1s - loss: 0.1038 - auc: 0.9918
## 486/603 [=====>.....] - ETA: 1s - loss: 0.1039 - auc: 0.9918
## 489/603 [=====>.....] - ETA: 1s - loss: 0.1036 - auc: 0.9919
## 492/603 [=====>.....] - ETA: 1s - loss: 0.1038 - auc: 0.9918
## 495/603 [=====>.....] - ETA: 1s - loss: 0.1040 - auc: 0.9918
## 499/603 [=====>.....] - ETA: 1s - loss: 0.1036 - auc: 0.9918
## 502/603 [=====>.....] - ETA: 1s - loss: 0.1036 - auc: 0.9918
## 506/603 [=====>.....] - ETA: 1s - loss: 0.1031 - auc: 0.9919
## 509/603 [=====>.....] - ETA: 1s - loss: 0.1030 - auc: 0.9919
## 513/603 [=====>.....] - ETA: 1s - loss: 0.1031 - auc: 0.9919
## 516/603 [=====>.....] - ETA: 1s - loss: 0.1029 - auc: 0.9919
## 520/603 [=====>.....] - ETA: 1s - loss: 0.1026 - auc: 0.9920
## 524/603 [=====>....] - ETA: 1s - loss: 0.1025 - auc: 0.9920
## 527/603 [=====>....] - ETA: 1s - loss: 0.1023 - auc: 0.9920
## 531/603 [=====>....] - ETA: 1s - loss: 0.1023 - auc: 0.9920
```

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## 535/603 [=====>....] - ETA: 1s - loss: 0.1023 - auc: 0.9920
## 539/603 [=====>....] - ETA: 1s - loss: 0.1026 - auc: 0.9920
## 543/603 [=====>...] - ETA: 0s - loss: 0.1033 - auc: 0.9919
## 546/603 [=====>...] - ETA: 0s - loss: 0.1034 - auc: 0.9918
## 550/603 [=====>...] - ETA: 0s - loss: 0.1034 - auc: 0.9919
## 554/603 [=====>...] - ETA: 0s - loss: 0.1034 - auc: 0.9919
## 558/603 [=====>...] - ETA: 0s - loss: 0.1039 - auc: 0.9918
## 562/603 [=====>...] - ETA: 0s - loss: 0.1038 - auc: 0.9918
## 566/603 [=====>..] - ETA: 0s - loss: 0.1039 - auc: 0.9918
## 569/603 [=====>..] - ETA: 0s - loss: 0.1036 - auc: 0.9919
## 573/603 [=====>..] - ETA: 0s - loss: 0.1036 - auc: 0.9919
## 576/603 [=====>..] - ETA: 0s - loss: 0.1034 - auc: 0.9919
## 580/603 [=====>..] - ETA: 0s - loss: 0.1035 - auc: 0.9919
## 583/603 [=====>.] - ETA: 0s - loss: 0.1031 - auc: 0.9920
## 586/603 [=====>.] - ETA: 0s - loss: 0.1036 - auc: 0.9919
## 589/603 [=====>.] - ETA: 0s - loss: 0.1037 - auc: 0.9919
## 593/603 [=====>.] - ETA: 0s - loss: 0.1039 - auc: 0.9918
## 597/603 [=====>.] - ETA: 0s - loss: 0.1039 - auc: 0.9918
## 600/603 [=====>.] - ETA: 0s - loss: 0.1039 - auc: 0.9918
## 603/603 [=====] - ETA: 0s - loss: 0.1040 - auc: 0.9918
## 603/603 [=====] - 10s 17ms/step - loss: 0.1040 - auc: 0.9918 - va
l_loss: 0.1593 - val_auc: 0.9859
## Epoch 16/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0340 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.0867 - auc: 0.9925
## 9/603 [.....] - ETA: 9s - loss: 0.0916 - auc: 0.9922
## 13/603 [.....] - ETA: 9s - loss: 0.0920 - auc: 0.9929
## 17/603 [.....] - ETA: 9s - loss: 0.0821 - auc: 0.9942
## 21/603 [>.....] - ETA: 9s - loss: 0.0865 - auc: 0.9940
## 24/603 [>.....] - ETA: 9s - loss: 0.0979 - auc: 0.9927
## 27/603 [>.....] - ETA: 9s - loss: 0.1009 - auc: 0.9928
## 31/603 [>.....] - ETA: 9s - loss: 0.1011 - auc: 0.9927
## 34/603 [>.....] - ETA: 9s - loss: 0.1003 - auc: 0.9929
## 38/603 [>.....] - ETA: 9s - loss: 0.1122 - auc: 0.9906
## 41/603 [=>.....] - ETA: 9s - loss: 0.1102 - auc: 0.9911
## 45/603 [=>.....] - ETA: 9s - loss: 0.1128 - auc: 0.9905
## 49/603 [=>.....] - ETA: 9s - loss: 0.1081 - auc: 0.9912
## 53/603 [=>.....] - ETA: 9s - loss: 0.1138 - auc: 0.9906
## 57/603 [=>.....] - ETA: 9s - loss: 0.1093 - auc: 0.9912
## 61/603 [==>.....] - ETA: 8s - loss: 0.1101 - auc: 0.9913
## 65/603 [==>.....] - ETA: 8s - loss: 0.1075 - auc: 0.9917
## 69/603 [==>.....] - ETA: 8s - loss: 0.1055 - auc: 0.9920
## 72/603 [==>.....] - ETA: 8s - loss: 0.1040 - auc: 0.9923
## 75/603 [==>.....] - ETA: 8s - loss: 0.1030 - auc: 0.9924
## 78/603 [==>.....] - ETA: 8s - loss: 0.1033 - auc: 0.9922
## 81/603 [===>.....] - ETA: 8s - loss: 0.1014 - auc: 0.9925
## 85/603 [===>.....] - ETA: 8s - loss: 0.1006 - auc: 0.9926
## 89/603 [===>.....] - ETA: 8s - loss: 0.1032 - auc: 0.9924
## 92/603 [===>.....] - ETA: 8s - loss: 0.1025 - auc: 0.9925
## 95/603 [===>.....] - ETA: 8s - loss: 0.1027 - auc: 0.9925
## 98/603 [===>.....] - ETA: 8s - loss: 0.1020 - auc: 0.9926
## 102/603 [===>.....] - ETA: 8s - loss: 0.1002 - auc: 0.9928
## 106/603 [===>.....] - ETA: 8s - loss: 0.0980 - auc: 0.9931
## 109/603 [===>.....] - ETA: 8s - loss: 0.0986 - auc: 0.9931
## 113/603 [===>.....] - ETA: 8s - loss: 0.0986 - auc: 0.9931
```

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## 117/603 [====>.....] - ETA: 8s - loss: 0.0979 - auc: 0.9933
## 120/603 [====>.....] - ETA: 8s - loss: 0.0963 - auc: 0.9935
## 124/603 [====>.....] - ETA: 7s - loss: 0.0950 - auc: 0.9936
## 127/603 [====>.....] - ETA: 7s - loss: 0.0948 - auc: 0.9937
## 131/603 [====>.....] - ETA: 7s - loss: 0.0952 - auc: 0.9937
## 134/603 [====>.....] - ETA: 7s - loss: 0.0943 - auc: 0.9938
## 138/603 [====>.....] - ETA: 7s - loss: 0.0936 - auc: 0.9939
## 142/603 [====>.....] - ETA: 7s - loss: 0.0937 - auc: 0.9939
## 145/603 [====>.....] - ETA: 7s - loss: 0.0940 - auc: 0.9939
## 149/603 [====>.....] - ETA: 7s - loss: 0.0946 - auc: 0.9937
## 153/603 [====>.....] - ETA: 7s - loss: 0.0938 - auc: 0.9938
## 157/603 [====>.....] - ETA: 7s - loss: 0.0938 - auc: 0.9938
## 161/603 [====>.....] - ETA: 7s - loss: 0.0939 - auc: 0.9937
## 164/603 [====>.....] - ETA: 7s - loss: 0.0936 - auc: 0.9938
## 167/603 [====>.....] - ETA: 7s - loss: 0.0931 - auc: 0.9939
## 171/603 [====>.....] - ETA: 7s - loss: 0.0945 - auc: 0.9937
## 175/603 [====>.....] - ETA: 7s - loss: 0.0952 - auc: 0.9936
## 179/603 [====>.....] - ETA: 7s - loss: 0.0955 - auc: 0.9936
## 183/603 [====>.....] - ETA: 6s - loss: 0.0960 - auc: 0.9936
## 187/603 [====>.....] - ETA: 6s - loss: 0.0957 - auc: 0.9936
## 191/603 [====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9933
## 194/603 [====>.....] - ETA: 6s - loss: 0.0962 - auc: 0.9934
## 198/603 [====>.....] - ETA: 6s - loss: 0.0957 - auc: 0.9935
## 202/603 [====>.....] - ETA: 6s - loss: 0.0956 - auc: 0.9935
## 206/603 [====>.....] - ETA: 6s - loss: 0.0967 - auc: 0.9933
## 210/603 [====>.....] - ETA: 6s - loss: 0.0968 - auc: 0.9932
## 213/603 [====>.....] - ETA: 6s - loss: 0.0966 - auc: 0.9932
## 216/603 [====>.....] - ETA: 6s - loss: 0.0984 - auc: 0.9930
## 220/603 [====>.....] - ETA: 6s - loss: 0.0975 - auc: 0.9931
## 224/603 [====>.....] - ETA: 6s - loss: 0.0973 - auc: 0.9932
## 227/603 [====>.....] - ETA: 6s - loss: 0.0972 - auc: 0.9932
## 231/603 [====>.....] - ETA: 6s - loss: 0.0978 - auc: 0.9932
## 234/603 [====>.....] - ETA: 6s - loss: 0.0979 - auc: 0.9930
## 237/603 [====>.....] - ETA: 6s - loss: 0.0973 - auc: 0.9931
## 240/603 [====>.....] - ETA: 6s - loss: 0.0968 - auc: 0.9932
## 244/603 [====>.....] - ETA: 5s - loss: 0.0972 - auc: 0.9932
## 248/603 [====>.....] - ETA: 5s - loss: 0.0971 - auc: 0.9932
## 251/603 [====>.....] - ETA: 5s - loss: 0.0974 - auc: 0.9932
## 255/603 [====>.....] - ETA: 5s - loss: 0.0976 - auc: 0.9931
## 259/603 [====>.....] - ETA: 5s - loss: 0.0973 - auc: 0.9932
## 263/603 [====>.....] - ETA: 5s - loss: 0.0985 - auc: 0.9930
## 267/603 [====>.....] - ETA: 5s - loss: 0.0993 - auc: 0.9928
## 270/603 [====>.....] - ETA: 5s - loss: 0.0998 - auc: 0.9927
## 273/603 [====>.....] - ETA: 5s - loss: 0.0998 - auc: 0.9927
## 277/603 [====>.....] - ETA: 5s - loss: 0.0991 - auc: 0.9928
## 280/603 [====>.....] - ETA: 5s - loss: 0.0992 - auc: 0.9928
## 283/603 [====>.....] - ETA: 5s - loss: 0.0992 - auc: 0.9928
## 287/603 [====>.....] - ETA: 5s - loss: 0.0995 - auc: 0.9927
## 290/603 [====>.....] - ETA: 5s - loss: 0.0993 - auc: 0.9927
## 294/603 [====>.....] - ETA: 5s - loss: 0.0991 - auc: 0.9928
## 298/603 [====>.....] - ETA: 5s - loss: 0.0994 - auc: 0.9928
## 301/603 [====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9926
## 305/603 [====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9927
## 309/603 [====>.....] - ETA: 4s - loss: 0.0996 - auc: 0.9928
## 313/603 [====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9927
## 317/603 [====>.....] - ETA: 4s - loss: 0.0995 - auc: 0.9928
```

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## 321/603 [=====>.....] - ETA: 4s - loss: 0.0993 - auc: 0.9928
## 325/603 [=====>.....] - ETA: 4s - loss: 0.1001 - auc: 0.9927
## 329/603 [=====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9927
## 333/603 [=====>.....] - ETA: 4s - loss: 0.0994 - auc: 0.9928
## 337/603 [=====>.....] - ETA: 4s - loss: 0.1000 - auc: 0.9927
## 341/603 [=====>.....] - ETA: 4s - loss: 0.0996 - auc: 0.9928
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1002 - auc: 0.9927
## 349/603 [=====>.....] - ETA: 4s - loss: 0.0997 - auc: 0.9927
## 352/603 [=====>.....] - ETA: 4s - loss: 0.0999 - auc: 0.9927
## 356/603 [=====>.....] - ETA: 4s - loss: 0.1001 - auc: 0.9927
## 360/603 [=====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9927
## 364/603 [=====>.....] - ETA: 3s - loss: 0.0992 - auc: 0.9928
## 368/603 [=====>.....] - ETA: 3s - loss: 0.0985 - auc: 0.9929
## 372/603 [=====>.....] - ETA: 3s - loss: 0.0982 - auc: 0.9930
## 376/603 [=====>.....] - ETA: 3s - loss: 0.0977 - auc: 0.9930
## 380/603 [=====>.....] - ETA: 3s - loss: 0.0975 - auc: 0.9931
## 384/603 [=====>.....] - ETA: 3s - loss: 0.0970 - auc: 0.9931
## 387/603 [=====>.....] - ETA: 3s - loss: 0.0964 - auc: 0.9932
## 390/603 [=====>.....] - ETA: 3s - loss: 0.0965 - auc: 0.9932
## 394/603 [=====>.....] - ETA: 3s - loss: 0.0967 - auc: 0.9932
## 398/603 [=====>.....] - ETA: 3s - loss: 0.0973 - auc: 0.9930
## 401/603 [=====>.....] - ETA: 3s - loss: 0.0970 - auc: 0.9930
## 404/603 [=====>.....] - ETA: 3s - loss: 0.0966 - auc: 0.9931
## 408/603 [=====>.....] - ETA: 3s - loss: 0.0966 - auc: 0.9931
## 411/603 [=====>.....] - ETA: 3s - loss: 0.0962 - auc: 0.9931
## 414/603 [=====>.....] - ETA: 3s - loss: 0.0961 - auc: 0.9931
## 418/603 [=====>.....] - ETA: 3s - loss: 0.0955 - auc: 0.9932
## 421/603 [=====>.....] - ETA: 3s - loss: 0.0956 - auc: 0.9932
## 424/603 [=====>.....] - ETA: 2s - loss: 0.0955 - auc: 0.9932
## 428/603 [=====>.....] - ETA: 2s - loss: 0.0954 - auc: 0.9932
## 432/603 [=====>.....] - ETA: 2s - loss: 0.0949 - auc: 0.9933
## 436/603 [=====>.....] - ETA: 2s - loss: 0.0947 - auc: 0.9933
## 440/603 [=====>.....] - ETA: 2s - loss: 0.0949 - auc: 0.9933
## 444/603 [=====>.....] - ETA: 2s - loss: 0.0953 - auc: 0.9933
## 448/603 [=====>.....] - ETA: 2s - loss: 0.0955 - auc: 0.9932
## 452/603 [=====>.....] - ETA: 2s - loss: 0.0953 - auc: 0.9932
## 456/603 [=====>.....] - ETA: 2s - loss: 0.0963 - auc: 0.9931
## 460/603 [=====>.....] - ETA: 2s - loss: 0.0961 - auc: 0.9931
## 463/603 [=====>.....] - ETA: 2s - loss: 0.0966 - auc: 0.9930
## 466/603 [=====>.....] - ETA: 2s - loss: 0.0965 - auc: 0.9930
## 470/603 [=====>.....] - ETA: 2s - loss: 0.0964 - auc: 0.9931
## 474/603 [=====>.....] - ETA: 2s - loss: 0.0961 - auc: 0.9931
## 477/603 [=====>.....] - ETA: 2s - loss: 0.0960 - auc: 0.9931
## 481/603 [=====>.....] - ETA: 2s - loss: 0.0964 - auc: 0.9930
## 485/603 [=====>.....] - ETA: 1s - loss: 0.0964 - auc: 0.9930
## 489/603 [=====>.....] - ETA: 1s - loss: 0.0968 - auc: 0.9930
## 493/603 [=====>.....] - ETA: 1s - loss: 0.0968 - auc: 0.9930
## 497/603 [=====>.....] - ETA: 1s - loss: 0.0970 - auc: 0.9929
## 500/603 [=====>.....] - ETA: 1s - loss: 0.0968 - auc: 0.9930
## 504/603 [=====>.....] - ETA: 1s - loss: 0.0964 - auc: 0.9930
## 507/603 [=====>.....] - ETA: 1s - loss: 0.0962 - auc: 0.9930
## 511/603 [=====>.....] - ETA: 1s - loss: 0.0959 - auc: 0.9931
## 515/603 [=====>.....] - ETA: 1s - loss: 0.0961 - auc: 0.9930
## 519/603 [=====>.....] - ETA: 1s - loss: 0.0961 - auc: 0.9931
## 522/603 [=====>.....] - ETA: 1s - loss: 0.0960 - auc: 0.9931
## 525/603 [=====>....] - ETA: 1s - loss: 0.0960 - auc: 0.9931
```

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## 529/603 [=====>...] - ETA: 1s - loss: 0.0963 - auc: 0.9931
## 533/603 [=====>...] - ETA: 1s - loss: 0.0970 - auc: 0.9930
## 536/603 [=====>...] - ETA: 1s - loss: 0.0973 - auc: 0.9929
## 539/603 [=====>...] - ETA: 1s - loss: 0.0971 - auc: 0.9929
## 543/603 [=====>...] - ETA: 0s - loss: 0.0972 - auc: 0.9929
## 547/603 [=====>...] - ETA: 0s - loss: 0.0971 - auc: 0.9929
## 551/603 [=====>...] - ETA: 0s - loss: 0.0974 - auc: 0.9929
## 555/603 [=====>...] - ETA: 0s - loss: 0.0973 - auc: 0.9929
## 559/603 [=====>...] - ETA: 0s - loss: 0.0969 - auc: 0.9929
## 563/603 [=====>..] - ETA: 0s - loss: 0.0974 - auc: 0.9929
## 567/603 [=====>..] - ETA: 0s - loss: 0.0978 - auc: 0.9929
## 570/603 [=====>..] - ETA: 0s - loss: 0.0984 - auc: 0.9928
## 573/603 [=====>..] - ETA: 0s - loss: 0.0986 - auc: 0.9927
## 577/603 [=====>..] - ETA: 0s - loss: 0.0989 - auc: 0.9927
## 581/603 [=====>..] - ETA: 0s - loss: 0.0990 - auc: 0.9927
## 584/603 [=====>.] - ETA: 0s - loss: 0.0993 - auc: 0.9927
## 587/603 [=====>.] - ETA: 0s - loss: 0.0993 - auc: 0.9927
## 591/603 [=====>.] - ETA: 0s - loss: 0.0991 - auc: 0.9927
## 595/603 [=====>.] - ETA: 0s - loss: 0.0990 - auc: 0.9928
## 598/603 [=====>.] - ETA: 0s - loss: 0.0993 - auc: 0.9927
## 601/603 [=====>.] - ETA: 0s - loss: 0.0992 - auc: 0.9927
## 603/603 [=====>] - 10s 17ms/step - loss: 0.0992 - auc: 0.9927 - va
l_loss: 0.1561 - val_auc: 0.9861
## Epoch 17/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.1436 - auc: 0.9902
## 5/603 [.....] - ETA: 9s - loss: 0.0628 - auc: 0.9932
## 8/603 [.....] - ETA: 9s - loss: 0.0673 - auc: 0.9950
## 12/603 [.....] - ETA: 9s - loss: 0.0729 - auc: 0.9960
## 16/603 [.....] - ETA: 9s - loss: 0.1025 - auc: 0.9904
## 20/603 [.....] - ETA: 9s - loss: 0.0973 - auc: 0.9915
## 23/603 [>.....] - ETA: 9s - loss: 0.1007 - auc: 0.9918
## 27/603 [>.....] - ETA: 9s - loss: 0.1021 - auc: 0.9920
## 30/603 [>.....] - ETA: 9s - loss: 0.0994 - auc: 0.9927
## 34/603 [>.....] - ETA: 9s - loss: 0.0974 - auc: 0.9930
## 38/603 [>.....] - ETA: 9s - loss: 0.0998 - auc: 0.9929
## 41/603 [=>.....] - ETA: 9s - loss: 0.1010 - auc: 0.9925
## 45/603 [=>.....] - ETA: 9s - loss: 0.0974 - auc: 0.9931
## 49/603 [=>.....] - ETA: 9s - loss: 0.0943 - auc: 0.9935
## 53/603 [=>.....] - ETA: 9s - loss: 0.0938 - auc: 0.9934
## 57/603 [=>.....] - ETA: 8s - loss: 0.0950 - auc: 0.9933
## 60/603 [=>.....] - ETA: 8s - loss: 0.0971 - auc: 0.9929
## 64/603 [==>.....] - ETA: 8s - loss: 0.1004 - auc: 0.9920
## 68/603 [==>.....] - ETA: 8s - loss: 0.0977 - auc: 0.9925
## 72/603 [==>.....] - ETA: 8s - loss: 0.0983 - auc: 0.9920
## 76/603 [==>.....] - ETA: 8s - loss: 0.0969 - auc: 0.9923
## 80/603 [==>.....] - ETA: 8s - loss: 0.0963 - auc: 0.9924
## 83/603 [===>.....] - ETA: 8s - loss: 0.0958 - auc: 0.9926
## 86/603 [===>.....] - ETA: 8s - loss: 0.0957 - auc: 0.9927
## 89/603 [===>.....] - ETA: 8s - loss: 0.0947 - auc: 0.9929
## 93/603 [===>.....] - ETA: 8s - loss: 0.0944 - auc: 0.9930
## 97/603 [===>.....] - ETA: 8s - loss: 0.0927 - auc: 0.9933
## 100/603 [===>.....] - ETA: 8s - loss: 0.0932 - auc: 0.9931
## 104/603 [====>.....] - ETA: 8s - loss: 0.0916 - auc: 0.9934
## 107/603 [====>.....] - ETA: 8s - loss: 0.0916 - auc: 0.9934
## 111/603 [====>.....] - ETA: 8s - loss: 0.0924 - auc: 0.9934
```

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## 114/603 [====>.....] - ETA: 8s - loss: 0.0919 - auc: 0.9935
## 118/603 [====>.....] - ETA: 7s - loss: 0.0939 - auc: 0.9933
## 122/603 [====>.....] - ETA: 7s - loss: 0.0963 - auc: 0.9930
## 125/603 [====>.....] - ETA: 7s - loss: 0.0965 - auc: 0.9930
## 129/603 [====>.....] - ETA: 7s - loss: 0.0953 - auc: 0.9931
## 133/603 [====>.....] - ETA: 7s - loss: 0.0954 - auc: 0.9932
## 137/603 [====>.....] - ETA: 7s - loss: 0.1002 - auc: 0.9926
## 141/603 [====>.....] - ETA: 7s - loss: 0.1012 - auc: 0.9923
## 145/603 [====>.....] - ETA: 7s - loss: 0.1016 - auc: 0.9922
## 149/603 [====>.....] - ETA: 7s - loss: 0.1001 - auc: 0.9923
## 152/603 [====>.....] - ETA: 7s - loss: 0.1011 - auc: 0.9923
## 156/603 [====>.....] - ETA: 7s - loss: 0.1020 - auc: 0.9922
## 159/603 [====>.....] - ETA: 7s - loss: 0.1011 - auc: 0.9923
## 162/603 [====>.....] - ETA: 7s - loss: 0.1022 - auc: 0.9922
## 165/603 [====>.....] - ETA: 7s - loss: 0.1017 - auc: 0.9922
## 169/603 [====>.....] - ETA: 7s - loss: 0.1006 - auc: 0.9924
## 172/603 [====>.....] - ETA: 7s - loss: 0.1016 - auc: 0.9923
## 176/603 [====>.....] - ETA: 7s - loss: 0.1006 - auc: 0.9925
## 180/603 [====>.....] - ETA: 6s - loss: 0.1005 - auc: 0.9925
## 184/603 [====>.....] - ETA: 6s - loss: 0.1005 - auc: 0.9925
## 188/603 [====>.....] - ETA: 6s - loss: 0.0996 - auc: 0.9926
## 191/603 [====>.....] - ETA: 6s - loss: 0.0987 - auc: 0.9928
## 195/603 [====>.....] - ETA: 6s - loss: 0.0997 - auc: 0.9927
## 199/603 [====>.....] - ETA: 6s - loss: 0.0990 - auc: 0.9928
## 202/603 [====>.....] - ETA: 6s - loss: 0.0984 - auc: 0.9929
## 205/603 [====>.....] - ETA: 6s - loss: 0.0977 - auc: 0.9930
## 208/603 [====>.....] - ETA: 6s - loss: 0.0984 - auc: 0.9930
## 212/603 [====>.....] - ETA: 6s - loss: 0.0998 - auc: 0.9925
## 215/603 [====>.....] - ETA: 6s - loss: 0.0998 - auc: 0.9925
## 218/603 [====>.....] - ETA: 6s - loss: 0.0997 - auc: 0.9926
## 222/603 [====>.....] - ETA: 6s - loss: 0.1003 - auc: 0.9926
## 225/603 [====>.....] - ETA: 6s - loss: 0.1012 - auc: 0.9923
## 228/603 [====>.....] - ETA: 6s - loss: 0.1011 - auc: 0.9924
## 231/603 [====>.....] - ETA: 6s - loss: 0.1006 - auc: 0.9924
## 234/603 [====>.....] - ETA: 6s - loss: 0.1013 - auc: 0.9924
## 237/603 [====>.....] - ETA: 6s - loss: 0.1007 - auc: 0.9924
## 240/603 [====>.....] - ETA: 6s - loss: 0.1004 - auc: 0.9925
## 243/603 [====>.....] - ETA: 5s - loss: 0.1005 - auc: 0.9925
## 246/603 [====>.....] - ETA: 5s - loss: 0.1008 - auc: 0.9924
## 249/603 [====>.....] - ETA: 5s - loss: 0.1000 - auc: 0.9925
## 253/603 [====>.....] - ETA: 5s - loss: 0.1002 - auc: 0.9925
## 257/603 [====>.....] - ETA: 5s - loss: 0.0990 - auc: 0.9926
## 260/603 [====>.....] - ETA: 5s - loss: 0.0990 - auc: 0.9926
## 264/603 [====>.....] - ETA: 5s - loss: 0.0985 - auc: 0.9927
## 267/603 [====>.....] - ETA: 5s - loss: 0.0980 - auc: 0.9928
## 270/603 [====>.....] - ETA: 5s - loss: 0.0977 - auc: 0.9928
## 273/603 [====>.....] - ETA: 5s - loss: 0.0977 - auc: 0.9928
## 276/603 [====>.....] - ETA: 5s - loss: 0.0978 - auc: 0.9928
## 279/603 [====>.....] - ETA: 5s - loss: 0.0981 - auc: 0.9928
## 282/603 [====>.....] - ETA: 5s - loss: 0.0976 - auc: 0.9928
## 285/603 [====>.....] - ETA: 5s - loss: 0.0981 - auc: 0.9928
## 288/603 [====>.....] - ETA: 5s - loss: 0.0989 - auc: 0.9926
## 291/603 [====>.....] - ETA: 5s - loss: 0.0984 - auc: 0.9927
## 294/603 [====>.....] - ETA: 5s - loss: 0.0982 - auc: 0.9928
## 298/603 [====>.....] - ETA: 5s - loss: 0.0988 - auc: 0.9926
## 301/603 [====>.....] - ETA: 5s - loss: 0.0998 - auc: 0.9925
```

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## 305/603 [=====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9925
## 308/603 [=====>.....] - ETA: 4s - loss: 0.0994 - auc: 0.9925
## 311/603 [=====>.....] - ETA: 4s - loss: 0.1007 - auc: 0.9923
## 314/603 [=====>.....] - ETA: 4s - loss: 0.1011 - auc: 0.9922
## 317/603 [=====>.....] - ETA: 4s - loss: 0.1009 - auc: 0.9923
## 321/603 [=====>.....] - ETA: 4s - loss: 0.1005 - auc: 0.9924
## 324/603 [=====>.....] - ETA: 4s - loss: 0.1007 - auc: 0.9924
## 327/603 [=====>.....] - ETA: 4s - loss: 0.1006 - auc: 0.9924
## 330/603 [=====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9924
## 333/603 [=====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9924
## 336/603 [=====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9925
## 339/603 [=====>.....] - ETA: 4s - loss: 0.0998 - auc: 0.9925
## 342/603 [=====>.....] - ETA: 4s - loss: 0.1000 - auc: 0.9925
## 345/603 [=====>.....] - ETA: 4s - loss: 0.1003 - auc: 0.9924
## 348/603 [=====>.....] - ETA: 4s - loss: 0.1001 - auc: 0.9924
## 351/603 [=====>.....] - ETA: 4s - loss: 0.1004 - auc: 0.9923
## 354/603 [=====>.....] - ETA: 4s - loss: 0.1000 - auc: 0.9924
## 357/603 [=====>.....] - ETA: 4s - loss: 0.0996 - auc: 0.9924
## 360/603 [=====>.....] - ETA: 4s - loss: 0.0995 - auc: 0.9925
## 363/603 [=====>.....] - ETA: 4s - loss: 0.0993 - auc: 0.9925
## 366/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9924
## 370/603 [=====>.....] - ETA: 3s - loss: 0.1001 - auc: 0.9924
## 373/603 [=====>.....] - ETA: 3s - loss: 0.0998 - auc: 0.9924
## 376/603 [=====>.....] - ETA: 3s - loss: 0.0993 - auc: 0.9925
## 379/603 [=====>.....] - ETA: 3s - loss: 0.0989 - auc: 0.9925
## 382/603 [=====>.....] - ETA: 3s - loss: 0.0985 - auc: 0.9926
## 386/603 [=====>.....] - ETA: 3s - loss: 0.0997 - auc: 0.9924
## 390/603 [=====>.....] - ETA: 3s - loss: 0.0998 - auc: 0.9924
## 393/603 [=====>.....] - ETA: 3s - loss: 0.0998 - auc: 0.9924
## 396/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9924
## 400/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9925
## 403/603 [=====>.....] - ETA: 3s - loss: 0.1000 - auc: 0.9925
## 406/603 [=====>.....] - ETA: 3s - loss: 0.0999 - auc: 0.9925
## 409/603 [=====>.....] - ETA: 3s - loss: 0.0995 - auc: 0.9925
## 412/603 [=====>.....] - ETA: 3s - loss: 0.0997 - auc: 0.9925
## 416/603 [=====>.....] - ETA: 3s - loss: 0.0994 - auc: 0.9926
## 419/603 [=====>.....] - ETA: 3s - loss: 0.0990 - auc: 0.9926
## 422/603 [=====>.....] - ETA: 3s - loss: 0.0991 - auc: 0.9926
## 425/603 [=====>.....] - ETA: 2s - loss: 0.0990 - auc: 0.9926
## 429/603 [=====>.....] - ETA: 2s - loss: 0.0990 - auc: 0.9926
## 432/603 [=====>.....] - ETA: 2s - loss: 0.0992 - auc: 0.9926
## 435/603 [=====>.....] - ETA: 2s - loss: 0.0993 - auc: 0.9926
## 438/603 [=====>.....] - ETA: 2s - loss: 0.0989 - auc: 0.9926
## 441/603 [=====>.....] - ETA: 2s - loss: 0.0989 - auc: 0.9926
## 445/603 [=====>.....] - ETA: 2s - loss: 0.0989 - auc: 0.9926
## 448/603 [=====>.....] - ETA: 2s - loss: 0.0989 - auc: 0.9927
## 451/603 [=====>.....] - ETA: 2s - loss: 0.0987 - auc: 0.9927
## 454/603 [=====>.....] - ETA: 2s - loss: 0.0986 - auc: 0.9927
## 457/603 [=====>.....] - ETA: 2s - loss: 0.0984 - auc: 0.9927
## 461/603 [=====>.....] - ETA: 2s - loss: 0.0996 - auc: 0.9925
## 464/603 [=====>.....] - ETA: 2s - loss: 0.0996 - auc: 0.9925
## 467/603 [=====>.....] - ETA: 2s - loss: 0.0992 - auc: 0.9926
## 470/603 [=====>.....] - ETA: 2s - loss: 0.0989 - auc: 0.9926
## 474/603 [=====>.....] - ETA: 2s - loss: 0.0987 - auc: 0.9926
## 477/603 [=====>.....] - ETA: 2s - loss: 0.0987 - auc: 0.9926
## 480/603 [=====>.....] - ETA: 2s - loss: 0.0985 - auc: 0.9926
```

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## 484/603 [=====>.....] - ETA: 1s - loss: 0.0989 - auc: 0.9926
## 488/603 [=====>.....] - ETA: 1s - loss: 0.0991 - auc: 0.9925
## 492/603 [=====>.....] - ETA: 1s - loss: 0.0991 - auc: 0.9926
## 495/603 [=====>.....] - ETA: 1s - loss: 0.0992 - auc: 0.9925
## 498/603 [=====>.....] - ETA: 1s - loss: 0.0991 - auc: 0.9926
## 501/603 [=====>.....] - ETA: 1s - loss: 0.0997 - auc: 0.9925
## 504/603 [=====>.....] - ETA: 1s - loss: 0.0994 - auc: 0.9925
## 507/603 [=====>.....] - ETA: 1s - loss: 0.0994 - auc: 0.9925
## 511/603 [=====>.....] - ETA: 1s - loss: 0.0993 - auc: 0.9925
## 515/603 [=====>.....] - ETA: 1s - loss: 0.0991 - auc: 0.9926
## 518/603 [=====>.....] - ETA: 1s - loss: 0.0991 - auc: 0.9926
## 522/603 [=====>.....] - ETA: 1s - loss: 0.0994 - auc: 0.9925
## 525/603 [=====>....] - ETA: 1s - loss: 0.0994 - auc: 0.9925
## 529/603 [=====>....] - ETA: 1s - loss: 0.0993 - auc: 0.9926
## 533/603 [=====>....] - ETA: 1s - loss: 0.0988 - auc: 0.9926
## 536/603 [=====>....] - ETA: 1s - loss: 0.0990 - auc: 0.9926
## 539/603 [=====>....] - ETA: 1s - loss: 0.0986 - auc: 0.9927
## 542/603 [=====>....] - ETA: 1s - loss: 0.0987 - auc: 0.9927
## 546/603 [=====>...] - ETA: 0s - loss: 0.0986 - auc: 0.9927
## 549/603 [=====>...] - ETA: 0s - loss: 0.0983 - auc: 0.9927
## 552/603 [=====>...] - ETA: 0s - loss: 0.0984 - auc: 0.9927
## 556/603 [=====>...] - ETA: 0s - loss: 0.0985 - auc: 0.9927
## 559/603 [=====>...] - ETA: 0s - loss: 0.0983 - auc: 0.9927
## 562/603 [=====>...] - ETA: 0s - loss: 0.0981 - auc: 0.9927
## 565/603 [=====>..] - ETA: 0s - loss: 0.0978 - auc: 0.9928
## 568/603 [=====>..] - ETA: 0s - loss: 0.0983 - auc: 0.9927
## 571/603 [=====>..] - ETA: 0s - loss: 0.0982 - auc: 0.9927
## 575/603 [=====>..] - ETA: 0s - loss: 0.0978 - auc: 0.9927
## 578/603 [=====>..] - ETA: 0s - loss: 0.0979 - auc: 0.9927
## 582/603 [=====>..] - ETA: 0s - loss: 0.0983 - auc: 0.9926
## 585/603 [=====>.] - ETA: 0s - loss: 0.0984 - auc: 0.9926
## 588/603 [=====>.] - ETA: 0s - loss: 0.0981 - auc: 0.9926
## 592/603 [=====>.] - ETA: 0s - loss: 0.0980 - auc: 0.9927
## 595/603 [=====>.] - ETA: 0s - loss: 0.0977 - auc: 0.9927
## 598/603 [=====>.] - ETA: 0s - loss: 0.0976 - auc: 0.9927
## 602/603 [=====>.] - ETA: 0s - loss: 0.0980 - auc: 0.9927
## 603/603 [=====] - 11s 18ms/step - loss: 0.0979 - auc: 0.9927 - va
l_loss: 0.1541 - val_auc: 0.9869
## Epoch 18/20
##
## 1/603 [.....] - ETA: 10s - loss: 0.1571 - auc: 0.9902
## 5/603 [.....] - ETA: 9s - loss: 0.1105 - auc: 0.9919
## 8/603 [.....] - ETA: 9s - loss: 0.1151 - auc: 0.9922
## 12/603 [.....] - ETA: 9s - loss: 0.1036 - auc: 0.9932
## 15/603 [.....] - ETA: 9s - loss: 0.1053 - auc: 0.9921
## 19/603 [.....] - ETA: 9s - loss: 0.1137 - auc: 0.9904
## 23/603 [>.....] - ETA: 9s - loss: 0.1132 - auc: 0.9906
## 26/603 [>.....] - ETA: 9s - loss: 0.1186 - auc: 0.9896
## 29/603 [>.....] - ETA: 9s - loss: 0.1262 - auc: 0.9867
## 32/603 [>.....] - ETA: 9s - loss: 0.1192 - auc: 0.9879
## 35/603 [>.....] - ETA: 9s - loss: 0.1196 - auc: 0.9883
## 38/603 [>.....] - ETA: 9s - loss: 0.1145 - auc: 0.9890
## 41/603 [=>.....] - ETA: 9s - loss: 0.1115 - auc: 0.9895
## 44/603 [=>.....] - ETA: 9s - loss: 0.1128 - auc: 0.9898
## 47/603 [=>.....] - ETA: 9s - loss: 0.1088 - auc: 0.9905
## 51/603 [=>.....] - ETA: 9s - loss: 0.1052 - auc: 0.9910
```

```
## 54/603 [=>.....] - ETA: 9s - loss: 0.1022 - auc: 0.9914
## 57/603 [=>.....] - ETA: 9s - loss: 0.1026 - auc: 0.9913
## 61/603 [==>.....] - ETA: 9s - loss: 0.0989 - auc: 0.9918
## 65/603 [==>.....] - ETA: 9s - loss: 0.0968 - auc: 0.9922
## 69/603 [==>.....] - ETA: 8s - loss: 0.0984 - auc: 0.9919
## 73/603 [==>.....] - ETA: 8s - loss: 0.0980 - auc: 0.9921
## 76/603 [==>.....] - ETA: 8s - loss: 0.0975 - auc: 0.9921
## 80/603 [==>.....] - ETA: 8s - loss: 0.0979 - auc: 0.9920
## 84/603 [===>.....] - ETA: 8s - loss: 0.0970 - auc: 0.9922
## 87/603 [===>.....] - ETA: 8s - loss: 0.0949 - auc: 0.9925
## 90/603 [===>.....] - ETA: 8s - loss: 0.0956 - auc: 0.9925
## 93/603 [===>.....] - ETA: 8s - loss: 0.0947 - auc: 0.9926
## 97/603 [===>.....] - ETA: 8s - loss: 0.0934 - auc: 0.9928
## 101/603 [====>.....] - ETA: 8s - loss: 0.0934 - auc: 0.9930
## 104/603 [====>.....] - ETA: 8s - loss: 0.0947 - auc: 0.9929
## 108/603 [====>.....] - ETA: 8s - loss: 0.0966 - auc: 0.9924
## 112/603 [====>.....] - ETA: 8s - loss: 0.0960 - auc: 0.9925
## 116/603 [====>.....] - ETA: 8s - loss: 0.0975 - auc: 0.9923
## 119/603 [====>.....] - ETA: 8s - loss: 0.0968 - auc: 0.9924
## 123/603 [=====>.....] - ETA: 8s - loss: 0.0976 - auc: 0.9923
## 127/603 [=====>.....] - ETA: 7s - loss: 0.0991 - auc: 0.9921
## 130/603 [=====>.....] - ETA: 7s - loss: 0.0988 - auc: 0.9922
## 134/603 [=====>.....] - ETA: 7s - loss: 0.0992 - auc: 0.9922
## 137/603 [=====>.....] - ETA: 7s - loss: 0.0995 - auc: 0.9922
## 140/603 [=====>.....] - ETA: 7s - loss: 0.0996 - auc: 0.9922
## 144/603 [=====>.....] - ETA: 7s - loss: 0.0979 - auc: 0.9924
## 147/603 [=====>.....] - ETA: 7s - loss: 0.0986 - auc: 0.9923
## 150/603 [=====>.....] - ETA: 7s - loss: 0.0981 - auc: 0.9924
## 154/603 [=====>.....] - ETA: 7s - loss: 0.0978 - auc: 0.9925
## 157/603 [=====>.....] - ETA: 7s - loss: 0.0972 - auc: 0.9926
## 161/603 [=====>.....] - ETA: 7s - loss: 0.0956 - auc: 0.9928
## 165/603 [=====>.....] - ETA: 7s - loss: 0.0968 - auc: 0.9926
## 168/603 [=====>.....] - ETA: 7s - loss: 0.0968 - auc: 0.9927
## 171/603 [=====>.....] - ETA: 7s - loss: 0.0965 - auc: 0.9927
## 175/603 [=====>.....] - ETA: 7s - loss: 0.0959 - auc: 0.9929
## 179/603 [=====>.....] - ETA: 7s - loss: 0.0969 - auc: 0.9928
## 183/603 [=====>.....] - ETA: 7s - loss: 0.0967 - auc: 0.9929
## 186/603 [=====>.....] - ETA: 6s - loss: 0.0974 - auc: 0.9929
## 190/603 [=====>.....] - ETA: 6s - loss: 0.0964 - auc: 0.9930
## 193/603 [=====>.....] - ETA: 6s - loss: 0.0965 - auc: 0.9930
## 196/603 [=====>.....] - ETA: 6s - loss: 0.0962 - auc: 0.9930
## 199/603 [=====>.....] - ETA: 6s - loss: 0.0968 - auc: 0.9929
## 203/603 [=====>.....] - ETA: 6s - loss: 0.0961 - auc: 0.9930
## 207/603 [=====>.....] - ETA: 6s - loss: 0.0966 - auc: 0.9928
## 210/603 [=====>.....] - ETA: 6s - loss: 0.0962 - auc: 0.9929
## 213/603 [=====>.....] - ETA: 6s - loss: 0.0961 - auc: 0.9930
## 216/603 [=====>.....] - ETA: 6s - loss: 0.0955 - auc: 0.9930
## 219/603 [=====>.....] - ETA: 6s - loss: 0.0959 - auc: 0.9930
## 223/603 [=====>.....] - ETA: 6s - loss: 0.0967 - auc: 0.9929
## 227/603 [=====>.....] - ETA: 6s - loss: 0.0966 - auc: 0.9930
## 230/603 [=====>.....] - ETA: 6s - loss: 0.0960 - auc: 0.9930
## 233/603 [=====>.....] - ETA: 6s - loss: 0.0971 - auc: 0.9928
## 237/603 [=====>.....] - ETA: 6s - loss: 0.0972 - auc: 0.9929
## 240/603 [=====>.....] - ETA: 6s - loss: 0.0962 - auc: 0.9930
## 244/603 [=====>.....] - ETA: 6s - loss: 0.0954 - auc: 0.9931
## 248/603 [=====>.....] - ETA: 5s - loss: 0.0953 - auc: 0.9931
```

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## 252/603 [=====>.....] - ETA: 5s - loss: 0.0955 - auc: 0.9930
## 256/603 [=====>.....] - ETA: 5s - loss: 0.0955 - auc: 0.9930
## 259/603 [=====>.....] - ETA: 5s - loss: 0.0953 - auc: 0.9931
## 262/603 [=====>.....] - ETA: 5s - loss: 0.0957 - auc: 0.9930
## 266/603 [=====>.....] - ETA: 5s - loss: 0.0955 - auc: 0.9930
## 269/603 [=====>.....] - ETA: 5s - loss: 0.0955 - auc: 0.9930
## 273/603 [=====>.....] - ETA: 5s - loss: 0.0955 - auc: 0.9931
## 277/603 [=====>.....] - ETA: 5s - loss: 0.0963 - auc: 0.9929
## 280/603 [=====>.....] - ETA: 5s - loss: 0.0962 - auc: 0.9929
## 284/603 [=====>.....] - ETA: 5s - loss: 0.0962 - auc: 0.9929
## 287/603 [=====>.....] - ETA: 5s - loss: 0.0970 - auc: 0.9928
## 290/603 [=====>.....] - ETA: 5s - loss: 0.0967 - auc: 0.9928
## 293/603 [=====>.....] - ETA: 5s - loss: 0.0975 - auc: 0.9926
## 297/603 [=====>.....] - ETA: 5s - loss: 0.0970 - auc: 0.9927
## 300/603 [=====>.....] - ETA: 5s - loss: 0.0969 - auc: 0.9926
## 303/603 [=====>.....] - ETA: 5s - loss: 0.0967 - auc: 0.9927
## 306/603 [=====>.....] - ETA: 4s - loss: 0.0964 - auc: 0.9927
## 310/603 [=====>.....] - ETA: 4s - loss: 0.0962 - auc: 0.9927
## 314/603 [=====>.....] - ETA: 4s - loss: 0.0957 - auc: 0.9928
## 318/603 [=====>.....] - ETA: 4s - loss: 0.0950 - auc: 0.9929
## 322/603 [=====>.....] - ETA: 4s - loss: 0.0959 - auc: 0.9927
## 325/603 [=====>.....] - ETA: 4s - loss: 0.0960 - auc: 0.9928
## 329/603 [=====>.....] - ETA: 4s - loss: 0.0957 - auc: 0.9928
## 333/603 [=====>.....] - ETA: 4s - loss: 0.0970 - auc: 0.9926
## 337/603 [=====>.....] - ETA: 4s - loss: 0.0966 - auc: 0.9927
## 340/603 [=====>.....] - ETA: 4s - loss: 0.0965 - auc: 0.9927
## 343/603 [=====>.....] - ETA: 4s - loss: 0.0963 - auc: 0.9927
## 347/603 [=====>.....] - ETA: 4s - loss: 0.0962 - auc: 0.9928
## 350/603 [=====>.....] - ETA: 4s - loss: 0.0968 - auc: 0.9927
## 354/603 [=====>.....] - ETA: 4s - loss: 0.0970 - auc: 0.9926
## 358/603 [=====>.....] - ETA: 4s - loss: 0.0971 - auc: 0.9926
## 361/603 [=====>.....] - ETA: 4s - loss: 0.0965 - auc: 0.9927
## 364/603 [=====>.....] - ETA: 3s - loss: 0.0966 - auc: 0.9927
## 368/603 [=====>.....] - ETA: 3s - loss: 0.0960 - auc: 0.9928
## 371/603 [=====>.....] - ETA: 3s - loss: 0.0957 - auc: 0.9928
## 375/603 [=====>.....] - ETA: 3s - loss: 0.0955 - auc: 0.9929
## 379/603 [=====>.....] - ETA: 3s - loss: 0.0955 - auc: 0.9929
## 383/603 [=====>.....] - ETA: 3s - loss: 0.0954 - auc: 0.9929
## 386/603 [=====>.....] - ETA: 3s - loss: 0.0960 - auc: 0.9928
## 389/603 [=====>.....] - ETA: 3s - loss: 0.0958 - auc: 0.9929
## 393/603 [=====>.....] - ETA: 3s - loss: 0.0955 - auc: 0.9929
## 397/603 [=====>.....] - ETA: 3s - loss: 0.0960 - auc: 0.9929
## 401/603 [=====>.....] - ETA: 3s - loss: 0.0961 - auc: 0.9929
## 404/603 [=====>.....] - ETA: 3s - loss: 0.0957 - auc: 0.9929
## 407/603 [=====>.....] - ETA: 3s - loss: 0.0956 - auc: 0.9929
## 411/603 [=====>.....] - ETA: 3s - loss: 0.0957 - auc: 0.9929
## 414/603 [=====>.....] - ETA: 3s - loss: 0.0954 - auc: 0.9930
## 418/603 [=====>.....] - ETA: 3s - loss: 0.0950 - auc: 0.9930
## 421/603 [=====>.....] - ETA: 3s - loss: 0.0946 - auc: 0.9931
## 424/603 [=====>.....] - ETA: 2s - loss: 0.0947 - auc: 0.9931
## 427/603 [=====>.....] - ETA: 2s - loss: 0.0946 - auc: 0.9931
## 430/603 [=====>.....] - ETA: 2s - loss: 0.0953 - auc: 0.9929
## 433/603 [=====>.....] - ETA: 2s - loss: 0.0953 - auc: 0.9929
## 436/603 [=====>.....] - ETA: 2s - loss: 0.0951 - auc: 0.9929
## 440/603 [=====>.....] - ETA: 2s - loss: 0.0956 - auc: 0.9929
## 444/603 [=====>.....] - ETA: 2s - loss: 0.0959 - auc: 0.9929
```

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## 447/603 [=====>.....] - ETA: 2s - loss: 0.0962 - auc: 0.9928
## 450/603 [=====>.....] - ETA: 2s - loss: 0.0960 - auc: 0.9928
## 454/603 [=====>.....] - ETA: 2s - loss: 0.0959 - auc: 0.9929
## 458/603 [=====>.....] - ETA: 2s - loss: 0.0968 - auc: 0.9927
## 462/603 [=====>.....] - ETA: 2s - loss: 0.0968 - auc: 0.9927
## 465/603 [=====>.....] - ETA: 2s - loss: 0.0968 - auc: 0.9927
## 468/603 [=====>.....] - ETA: 2s - loss: 0.0965 - auc: 0.9927
## 471/603 [=====>.....] - ETA: 2s - loss: 0.0961 - auc: 0.9928
## 474/603 [=====>.....] - ETA: 2s - loss: 0.0961 - auc: 0.9928
## 477/603 [=====>.....] - ETA: 2s - loss: 0.0957 - auc: 0.9929
## 481/603 [=====>.....] - ETA: 2s - loss: 0.0955 - auc: 0.9929
## 484/603 [=====>.....] - ETA: 1s - loss: 0.0957 - auc: 0.9929
## 487/603 [=====>.....] - ETA: 1s - loss: 0.0961 - auc: 0.9927
## 491/603 [=====>.....] - ETA: 1s - loss: 0.0965 - auc: 0.9927
## 495/603 [=====>.....] - ETA: 1s - loss: 0.0964 - auc: 0.9927
## 499/603 [=====>.....] - ETA: 1s - loss: 0.0967 - auc: 0.9926
## 502/603 [=====>.....] - ETA: 1s - loss: 0.0967 - auc: 0.9926
## 506/603 [=====>.....] - ETA: 1s - loss: 0.0966 - auc: 0.9927
## 509/603 [=====>.....] - ETA: 1s - loss: 0.0968 - auc: 0.9926
## 512/603 [=====>.....] - ETA: 1s - loss: 0.0969 - auc: 0.9926
## 515/603 [=====>.....] - ETA: 1s - loss: 0.0972 - auc: 0.9926
## 519/603 [=====>.....] - ETA: 1s - loss: 0.0971 - auc: 0.9926
## 523/603 [=====>....] - ETA: 1s - loss: 0.0967 - auc: 0.9927
## 527/603 [=====>....] - ETA: 1s - loss: 0.0973 - auc: 0.9926
## 530/603 [=====>....] - ETA: 1s - loss: 0.0973 - auc: 0.9926
## 533/603 [=====>....] - ETA: 1s - loss: 0.0975 - auc: 0.9925
## 537/603 [=====>....] - ETA: 1s - loss: 0.0978 - auc: 0.9925
## 540/603 [=====>....] - ETA: 1s - loss: 0.0976 - auc: 0.9925
## 543/603 [=====>...] - ETA: 1s - loss: 0.0975 - auc: 0.9925
## 546/603 [=====>...] - ETA: 0s - loss: 0.0973 - auc: 0.9926
## 550/603 [=====>...] - ETA: 0s - loss: 0.0971 - auc: 0.9926
## 554/603 [=====>...] - ETA: 0s - loss: 0.0970 - auc: 0.9926
## 558/603 [=====>...] - ETA: 0s - loss: 0.0967 - auc: 0.9927
## 562/603 [=====>...] - ETA: 0s - loss: 0.0969 - auc: 0.9926
## 566/603 [=====>..] - ETA: 0s - loss: 0.0971 - auc: 0.9926
## 569/603 [=====>..] - ETA: 0s - loss: 0.0969 - auc: 0.9926
## 573/603 [=====>..] - ETA: 0s - loss: 0.0965 - auc: 0.9927
## 576/603 [=====>..] - ETA: 0s - loss: 0.0965 - auc: 0.9927
## 579/603 [=====>..] - ETA: 0s - loss: 0.0963 - auc: 0.9928
## 582/603 [=====>..] - ETA: 0s - loss: 0.0967 - auc: 0.9927
## 585/603 [=====>.] - ETA: 0s - loss: 0.0969 - auc: 0.9927
## 588/603 [=====>.] - ETA: 0s - loss: 0.0971 - auc: 0.9926
## 591/603 [=====>.] - ETA: 0s - loss: 0.0970 - auc: 0.9926
## 594/603 [=====>.] - ETA: 0s - loss: 0.0969 - auc: 0.9927
## 597/603 [=====>.] - ETA: 0s - loss: 0.0966 - auc: 0.9927
## 601/603 [=====>.] - ETA: 0s - loss: 0.0963 - auc: 0.9928
## 603/603 [=====] - 11s 18ms/step - loss: 0.0962 - auc: 0.9928 - va
l_loss: 0.1394 - val_auc: 0.9877
## Epoch 19/20
##
## 1/603 [.....] - ETA: 10s - loss: 0.0335 - auc: 1.0000
## 5/603 [.....] - ETA: 10s - loss: 0.0454 - auc: 1.0000
## 8/603 [.....] - ETA: 10s - loss: 0.0365 - auc: 0.9998
## 11/603 [.....] - ETA: 10s - loss: 0.0484 - auc: 0.9990
## 14/603 [.....] - ETA: 10s - loss: 0.0557 - auc: 0.9985
## 17/603 [.....] - ETA: 10s - loss: 0.0698 - auc: 0.9975
```

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## 21/603 [>.....] - ETA: 9s - loss: 0.0700 - auc: 0.9972
## 24/603 [>.....] - ETA: 9s - loss: 0.0674 - auc: 0.9975
## 27/603 [>.....] - ETA: 9s - loss: 0.0678 - auc: 0.9972
## 30/603 [>.....] - ETA: 9s - loss: 0.0674 - auc: 0.9973
## 33/603 [>.....] - ETA: 9s - loss: 0.0689 - auc: 0.9971
## 37/603 [>.....] - ETA: 9s - loss: 0.0778 - auc: 0.9961
## 40/603 [>.....] - ETA: 9s - loss: 0.0757 - auc: 0.9964
## 44/603 [=>.....] - ETA: 9s - loss: 0.0818 - auc: 0.9956
## 48/603 [=>.....] - ETA: 9s - loss: 0.0854 - auc: 0.9952
## 51/603 [=>.....] - ETA: 9s - loss: 0.0819 - auc: 0.9955
## 54/603 [=>.....] - ETA: 9s - loss: 0.0813 - auc: 0.9956
## 57/603 [=>.....] - ETA: 9s - loss: 0.0813 - auc: 0.9957
## 60/603 [=>.....] - ETA: 9s - loss: 0.0854 - auc: 0.9952
## 64/603 [==>.....] - ETA: 9s - loss: 0.0889 - auc: 0.9943
## 68/603 [==>.....] - ETA: 9s - loss: 0.0880 - auc: 0.9945
## 71/603 [==>.....] - ETA: 8s - loss: 0.0884 - auc: 0.9945
## 75/603 [==>.....] - ETA: 8s - loss: 0.0882 - auc: 0.9943
## 79/603 [==>.....] - ETA: 8s - loss: 0.0898 - auc: 0.9942
## 83/603 [===>.....] - ETA: 8s - loss: 0.0930 - auc: 0.9936
## 86/603 [===>.....] - ETA: 8s - loss: 0.0940 - auc: 0.9936
## 90/603 [===>.....] - ETA: 8s - loss: 0.0972 - auc: 0.9931
## 94/603 [===>.....] - ETA: 8s - loss: 0.0968 - auc: 0.9932
## 97/603 [===>.....] - ETA: 8s - loss: 0.0956 - auc: 0.9934
## 101/603 [====>.....] - ETA: 8s - loss: 0.0950 - auc: 0.9935
## 105/603 [====>.....] - ETA: 8s - loss: 0.0926 - auc: 0.9938
## 108/603 [====>.....] - ETA: 8s - loss: 0.0917 - auc: 0.9939
## 112/603 [====>.....] - ETA: 8s - loss: 0.0952 - auc: 0.9935
## 116/603 [====>.....] - ETA: 8s - loss: 0.0937 - auc: 0.9937
## 119/603 [====>.....] - ETA: 8s - loss: 0.0929 - auc: 0.9938
## 122/603 [====>.....] - ETA: 8s - loss: 0.0925 - auc: 0.9939
## 126/603 [====>.....] - ETA: 8s - loss: 0.0907 - auc: 0.9941
## 129/603 [====>.....] - ETA: 7s - loss: 0.0925 - auc: 0.9938
## 132/603 [====>.....] - ETA: 7s - loss: 0.0924 - auc: 0.9939
## 135/603 [====>.....] - ETA: 7s - loss: 0.0915 - auc: 0.9940
## 139/603 [====>.....] - ETA: 7s - loss: 0.0912 - auc: 0.9941
## 142/603 [====>.....] - ETA: 7s - loss: 0.0922 - auc: 0.9939
## 146/603 [====>.....] - ETA: 7s - loss: 0.0917 - auc: 0.9939
## 150/603 [====>.....] - ETA: 7s - loss: 0.0916 - auc: 0.9939
## 153/603 [====>.....] - ETA: 7s - loss: 0.0919 - auc: 0.9939
## 156/603 [====>.....] - ETA: 7s - loss: 0.0927 - auc: 0.9937
## 159/603 [====>.....] - ETA: 7s - loss: 0.0926 - auc: 0.9937
## 163/603 [====>.....] - ETA: 7s - loss: 0.0921 - auc: 0.9938
## 166/603 [====>.....] - ETA: 7s - loss: 0.0911 - auc: 0.9939
## 170/603 [====>.....] - ETA: 7s - loss: 0.0906 - auc: 0.9940
## 174/603 [====>.....] - ETA: 7s - loss: 0.0942 - auc: 0.9936
## 177/603 [====>.....] - ETA: 7s - loss: 0.0942 - auc: 0.9936
## 180/603 [====>.....] - ETA: 7s - loss: 0.0938 - auc: 0.9936
## 183/603 [====>.....] - ETA: 7s - loss: 0.0934 - auc: 0.9937
## 186/603 [====>.....] - ETA: 7s - loss: 0.0939 - auc: 0.9936
## 190/603 [====>.....] - ETA: 6s - loss: 0.0950 - auc: 0.9935
## 194/603 [====>.....] - ETA: 6s - loss: 0.0943 - auc: 0.9936
## 197/603 [====>.....] - ETA: 6s - loss: 0.0939 - auc: 0.9936
## 200/603 [====>.....] - ETA: 6s - loss: 0.0941 - auc: 0.9936
## 203/603 [====>.....] - ETA: 6s - loss: 0.0934 - auc: 0.9937
## 206/603 [====>.....] - ETA: 6s - loss: 0.0927 - auc: 0.9938
## 209/603 [====>.....] - ETA: 6s - loss: 0.0919 - auc: 0.9939
```

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## 213/603 [=====>.....] - ETA: 6s - loss: 0.0920 - auc: 0.9939
## 216/603 [=====>.....] - ETA: 6s - loss: 0.0926 - auc: 0.9938
## 220/603 [=====>.....] - ETA: 6s - loss: 0.0935 - auc: 0.9936
## 224/603 [=====>.....] - ETA: 6s - loss: 0.0925 - auc: 0.9938
## 227/603 [=====>.....] - ETA: 6s - loss: 0.0924 - auc: 0.9937
## 230/603 [=====>.....] - ETA: 6s - loss: 0.0931 - auc: 0.9937
## 233/603 [=====>.....] - ETA: 6s - loss: 0.0933 - auc: 0.9937
## 237/603 [=====>.....] - ETA: 6s - loss: 0.0942 - auc: 0.9936
## 240/603 [=====>.....] - ETA: 6s - loss: 0.0940 - auc: 0.9936
## 243/603 [=====>.....] - ETA: 6s - loss: 0.0939 - auc: 0.9937
## 246/603 [=====>.....] - ETA: 6s - loss: 0.0937 - auc: 0.9937
## 249/603 [=====>.....] - ETA: 5s - loss: 0.0929 - auc: 0.9938
## 252/603 [=====>.....] - ETA: 5s - loss: 0.0924 - auc: 0.9939
## 255/603 [=====>.....] - ETA: 5s - loss: 0.0929 - auc: 0.9937
## 259/603 [=====>.....] - ETA: 5s - loss: 0.0925 - auc: 0.9938
## 263/603 [=====>.....] - ETA: 5s - loss: 0.0932 - auc: 0.9936
## 266/603 [=====>.....] - ETA: 5s - loss: 0.0932 - auc: 0.9936
## 269/603 [=====>.....] - ETA: 5s - loss: 0.0945 - auc: 0.9934
## 273/603 [=====>.....] - ETA: 5s - loss: 0.0946 - auc: 0.9934
## 277/603 [=====>.....] - ETA: 5s - loss: 0.0952 - auc: 0.9932
## 281/603 [=====>.....] - ETA: 5s - loss: 0.0963 - auc: 0.9930
## 285/603 [=====>.....] - ETA: 5s - loss: 0.0970 - auc: 0.9929
## 288/603 [=====>.....] - ETA: 5s - loss: 0.0975 - auc: 0.9928
## 292/603 [=====>.....] - ETA: 5s - loss: 0.0973 - auc: 0.9929
## 295/603 [=====>.....] - ETA: 5s - loss: 0.0968 - auc: 0.9929
## 298/603 [=====>.....] - ETA: 5s - loss: 0.0968 - auc: 0.9929
## 301/603 [=====>.....] - ETA: 5s - loss: 0.0970 - auc: 0.9929
## 305/603 [=====>.....] - ETA: 5s - loss: 0.0962 - auc: 0.9930
## 308/603 [=====>.....] - ETA: 4s - loss: 0.0964 - auc: 0.9930
## 311/603 [=====>.....] - ETA: 4s - loss: 0.0967 - auc: 0.9930
## 314/603 [=====>.....] - ETA: 4s - loss: 0.0964 - auc: 0.9930
## 317/603 [=====>.....] - ETA: 4s - loss: 0.0975 - auc: 0.9928
## 321/603 [=====>.....] - ETA: 4s - loss: 0.0971 - auc: 0.9928
## 324/603 [=====>.....] - ETA: 4s - loss: 0.0968 - auc: 0.9929
## 327/603 [=====>.....] - ETA: 4s - loss: 0.0965 - auc: 0.9930
## 330/603 [=====>.....] - ETA: 4s - loss: 0.0962 - auc: 0.9930
## 333/603 [=====>.....] - ETA: 4s - loss: 0.0963 - auc: 0.9930
## 336/603 [=====>.....] - ETA: 4s - loss: 0.0963 - auc: 0.9930
## 340/603 [=====>.....] - ETA: 4s - loss: 0.0959 - auc: 0.9931
## 343/603 [=====>.....] - ETA: 4s - loss: 0.0967 - auc: 0.9929
## 346/603 [=====>.....] - ETA: 4s - loss: 0.0966 - auc: 0.9929
## 350/603 [=====>.....] - ETA: 4s - loss: 0.0965 - auc: 0.9929
## 354/603 [=====>.....] - ETA: 4s - loss: 0.0963 - auc: 0.9929
## 357/603 [=====>.....] - ETA: 4s - loss: 0.0959 - auc: 0.9930
## 361/603 [=====>.....] - ETA: 4s - loss: 0.0955 - auc: 0.9930
## 364/603 [=====>.....] - ETA: 4s - loss: 0.0953 - auc: 0.9931
## 367/603 [=====>.....] - ETA: 3s - loss: 0.0950 - auc: 0.9931
## 370/603 [=====>.....] - ETA: 3s - loss: 0.0956 - auc: 0.9931
## 374/603 [=====>.....] - ETA: 3s - loss: 0.0957 - auc: 0.9930
## 377/603 [=====>.....] - ETA: 3s - loss: 0.0954 - auc: 0.9931
## 380/603 [=====>.....] - ETA: 3s - loss: 0.0953 - auc: 0.9931
## 384/603 [=====>.....] - ETA: 3s - loss: 0.0949 - auc: 0.9931
## 387/603 [=====>.....] - ETA: 3s - loss: 0.0946 - auc: 0.9932
## 390/603 [=====>.....] - ETA: 3s - loss: 0.0944 - auc: 0.9932
## 393/603 [=====>.....] - ETA: 3s - loss: 0.0944 - auc: 0.9932
## 396/603 [=====>.....] - ETA: 3s - loss: 0.0948 - auc: 0.9931
```

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## 400/603 [=====>.....] - ETA: 3s - loss: 0.0949 - auc: 0.9931
## 403/603 [=====>.....] - ETA: 3s - loss: 0.0953 - auc: 0.9931
## 406/603 [=====>.....] - ETA: 3s - loss: 0.0951 - auc: 0.9931
## 410/603 [=====>.....] - ETA: 3s - loss: 0.0951 - auc: 0.9931
## 414/603 [=====>.....] - ETA: 3s - loss: 0.0951 - auc: 0.9931
## 418/603 [=====>.....] - ETA: 3s - loss: 0.0947 - auc: 0.9932
## 421/603 [=====>.....] - ETA: 3s - loss: 0.0945 - auc: 0.9932
## 424/603 [=====>.....] - ETA: 3s - loss: 0.0942 - auc: 0.9932
## 428/603 [=====>.....] - ETA: 2s - loss: 0.0940 - auc: 0.9933
## 431/603 [=====>.....] - ETA: 2s - loss: 0.0946 - auc: 0.9932
## 434/603 [=====>.....] - ETA: 2s - loss: 0.0944 - auc: 0.9932
## 437/603 [=====>.....] - ETA: 2s - loss: 0.0943 - auc: 0.9932
## 440/603 [=====>.....] - ETA: 2s - loss: 0.0940 - auc: 0.9933
## 444/603 [=====>.....] - ETA: 2s - loss: 0.0940 - auc: 0.9933
## 447/603 [=====>.....] - ETA: 2s - loss: 0.0938 - auc: 0.9933
## 451/603 [=====>.....] - ETA: 2s - loss: 0.0941 - auc: 0.9933
## 455/603 [=====>.....] - ETA: 2s - loss: 0.0944 - auc: 0.9933
## 458/603 [=====>.....] - ETA: 2s - loss: 0.0946 - auc: 0.9932
## 462/603 [=====>.....] - ETA: 2s - loss: 0.0943 - auc: 0.9932
## 465/603 [=====>.....] - ETA: 2s - loss: 0.0941 - auc: 0.9933
## 469/603 [=====>.....] - ETA: 2s - loss: 0.0942 - auc: 0.9933
## 473/603 [=====>.....] - ETA: 2s - loss: 0.0937 - auc: 0.9933
## 476/603 [=====>.....] - ETA: 2s - loss: 0.0936 - auc: 0.9933
## 479/603 [=====>.....] - ETA: 2s - loss: 0.0936 - auc: 0.9933
## 482/603 [=====>.....] - ETA: 2s - loss: 0.0932 - auc: 0.9933
## 485/603 [=====>.....] - ETA: 1s - loss: 0.0931 - auc: 0.9934
## 488/603 [=====>.....] - ETA: 1s - loss: 0.0934 - auc: 0.9933
## 491/603 [=====>.....] - ETA: 1s - loss: 0.0931 - auc: 0.9933
## 494/603 [=====>.....] - ETA: 1s - loss: 0.0930 - auc: 0.9933
## 498/603 [=====>.....] - ETA: 1s - loss: 0.0928 - auc: 0.9934
## 501/603 [=====>.....] - ETA: 1s - loss: 0.0931 - auc: 0.9934
## 505/603 [=====>.....] - ETA: 1s - loss: 0.0939 - auc: 0.9932
## 509/603 [=====>.....] - ETA: 1s - loss: 0.0939 - auc: 0.9932
## 512/603 [=====>.....] - ETA: 1s - loss: 0.0937 - auc: 0.9932
## 515/603 [=====>.....] - ETA: 1s - loss: 0.0944 - auc: 0.9932
## 518/603 [=====>.....] - ETA: 1s - loss: 0.0944 - auc: 0.9932
## 521/603 [=====>.....] - ETA: 1s - loss: 0.0947 - auc: 0.9931
## 524/603 [=====>....] - ETA: 1s - loss: 0.0945 - auc: 0.9932
## 527/603 [=====>....] - ETA: 1s - loss: 0.0945 - auc: 0.9932
## 531/603 [=====>....] - ETA: 1s - loss: 0.0947 - auc: 0.9932
## 534/603 [=====>....] - ETA: 1s - loss: 0.0945 - auc: 0.9932
## 538/603 [=====>....] - ETA: 1s - loss: 0.0943 - auc: 0.9932
## 542/603 [=====>....] - ETA: 1s - loss: 0.0946 - auc: 0.9932
## 545/603 [=====>...] - ETA: 0s - loss: 0.0945 - auc: 0.9932
## 548/603 [=====>...] - ETA: 0s - loss: 0.0946 - auc: 0.9932
## 551/603 [=====>...] - ETA: 0s - loss: 0.0944 - auc: 0.9932
## 554/603 [=====>...] - ETA: 0s - loss: 0.0946 - auc: 0.9932
## 557/603 [=====>...] - ETA: 0s - loss: 0.0947 - auc: 0.9932
## 560/603 [=====>...] - ETA: 0s - loss: 0.0946 - auc: 0.9932
## 564/603 [=====>..] - ETA: 0s - loss: 0.0944 - auc: 0.9932
## 567/603 [=====>..] - ETA: 0s - loss: 0.0943 - auc: 0.9932
## 570/603 [=====>..] - ETA: 0s - loss: 0.0943 - auc: 0.9932
## 574/603 [=====>..] - ETA: 0s - loss: 0.0943 - auc: 0.9932
## 578/603 [=====>..] - ETA: 0s - loss: 0.0940 - auc: 0.9933
## 581/603 [=====>..] - ETA: 0s - loss: 0.0940 - auc: 0.9933
## 584/603 [=====>.] - ETA: 0s - loss: 0.0942 - auc: 0.9933
```

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## 587/603 [=====>.] - ETA: 0s - loss: 0.0943 - auc: 0.9932
## 590/603 [=====>.] - ETA: 0s - loss: 0.0942 - auc: 0.9933
## 593/603 [=====>.] - ETA: 0s - loss: 0.0944 - auc: 0.9932
## 597/603 [=====>.] - ETA: 0s - loss: 0.0941 - auc: 0.9933
## 601/603 [=====>.] - ETA: 0s - loss: 0.0942 - auc: 0.9933
## 603/603 [=====] - 11s 18ms/step - loss: 0.0943 - auc: 0.9932 - va
l_loss: 0.1522 - val_auc: 0.9877
## Epoch 20/20
##
## 1/603 [.....] - ETA: 9s - loss: 0.0569 - auc: 1.0000
## 5/603 [.....] - ETA: 9s - loss: 0.0925 - auc: 0.9953
## 8/603 [.....] - ETA: 10s - loss: 0.1017 - auc: 0.9959
## 11/603 [.....] - ETA: 10s - loss: 0.0929 - auc: 0.9965
## 14/603 [.....] - ETA: 10s - loss: 0.1095 - auc: 0.9949
## 17/603 [.....] - ETA: 9s - loss: 0.1060 - auc: 0.9953
## 21/603 [>.....] - ETA: 9s - loss: 0.1097 - auc: 0.9933
## 24/603 [>.....] - ETA: 9s - loss: 0.1245 - auc: 0.9896
## 27/603 [>.....] - ETA: 9s - loss: 0.1173 - auc: 0.9906
## 31/603 [>.....] - ETA: 9s - loss: 0.1116 - auc: 0.9915
## 34/603 [>.....] - ETA: 9s - loss: 0.1106 - auc: 0.9914
## 37/603 [>.....] - ETA: 9s - loss: 0.1073 - auc: 0.9915
## 40/603 [>.....] - ETA: 9s - loss: 0.1080 - auc: 0.9912
## 43/603 [=>.....] - ETA: 9s - loss: 0.1060 - auc: 0.9917
## 46/603 [=>.....] - ETA: 9s - loss: 0.1102 - auc: 0.9905
## 50/603 [=>.....] - ETA: 9s - loss: 0.1062 - auc: 0.9911
## 53/603 [=>.....] - ETA: 9s - loss: 0.1033 - auc: 0.9914
## 57/603 [=>.....] - ETA: 9s - loss: 0.0988 - auc: 0.9921
## 60/603 [=>.....] - ETA: 9s - loss: 0.0975 - auc: 0.9921
## 64/603 [==>.....] - ETA: 9s - loss: 0.0975 - auc: 0.9923
## 68/603 [==>.....] - ETA: 9s - loss: 0.0957 - auc: 0.9926
## 71/603 [==>.....] - ETA: 9s - loss: 0.0957 - auc: 0.9927
## 75/603 [==>.....] - ETA: 8s - loss: 0.0955 - auc: 0.9925
## 78/603 [==>.....] - ETA: 8s - loss: 0.0982 - auc: 0.9920
## 82/603 [===>.....] - ETA: 8s - loss: 0.0958 - auc: 0.9923
## 85/603 [===>.....] - ETA: 8s - loss: 0.0950 - auc: 0.9925
## 89/603 [===>.....] - ETA: 8s - loss: 0.0956 - auc: 0.9925
## 92/603 [===>.....] - ETA: 8s - loss: 0.0945 - auc: 0.9927
## 96/603 [===>.....] - ETA: 8s - loss: 0.0941 - auc: 0.9928
## 99/603 [===>.....] - ETA: 8s - loss: 0.0953 - auc: 0.9927
## 102/603 [====>.....] - ETA: 8s - loss: 0.0969 - auc: 0.9924
## 106/603 [====>.....] - ETA: 8s - loss: 0.0951 - auc: 0.9926
## 109/603 [====>.....] - ETA: 8s - loss: 0.0966 - auc: 0.9924
## 112/603 [====>.....] - ETA: 8s - loss: 0.0961 - auc: 0.9925
## 115/603 [====>.....] - ETA: 8s - loss: 0.0978 - auc: 0.9922
## 119/603 [====>.....] - ETA: 8s - loss: 0.0988 - auc: 0.9919
## 122/603 [====>.....] - ETA: 8s - loss: 0.0979 - auc: 0.9921
## 125/603 [====>.....] - ETA: 8s - loss: 0.0982 - auc: 0.9921
## 129/603 [====>.....] - ETA: 7s - loss: 0.0969 - auc: 0.9924
## 132/603 [====>.....] - ETA: 7s - loss: 0.0971 - auc: 0.9924
## 135/603 [====>.....] - ETA: 7s - loss: 0.0979 - auc: 0.9924
## 138/603 [====>.....] - ETA: 7s - loss: 0.0982 - auc: 0.9923
## 142/603 [====>.....] - ETA: 7s - loss: 0.0972 - auc: 0.9925
## 146/603 [====>.....] - ETA: 7s - loss: 0.0979 - auc: 0.9924
## 149/603 [====>.....] - ETA: 7s - loss: 0.0983 - auc: 0.9923
## 152/603 [====>.....] - ETA: 7s - loss: 0.0996 - auc: 0.9921
## 155/603 [====>.....] - ETA: 7s - loss: 0.0992 - auc: 0.9921
```

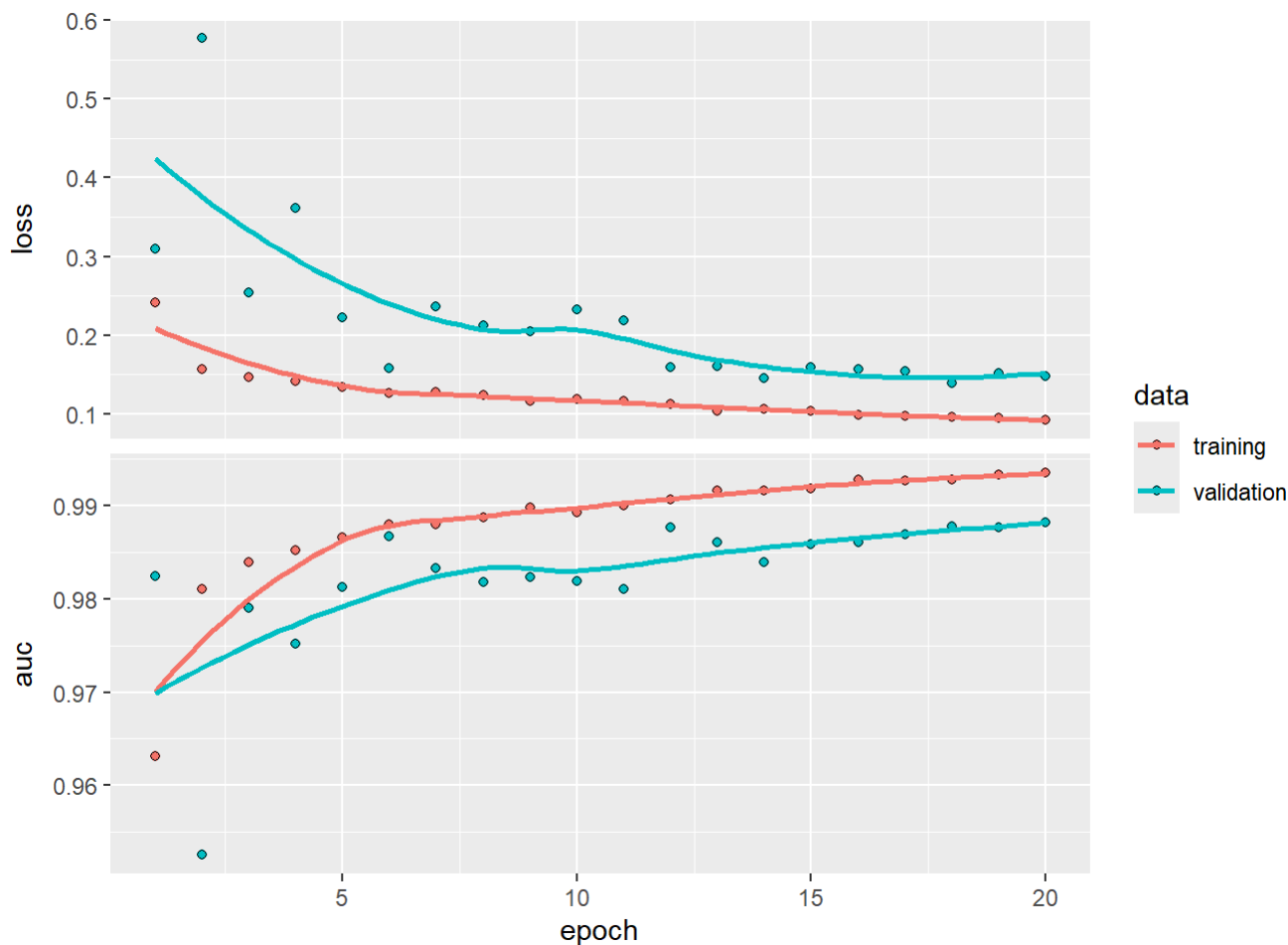
```
## 158/603 [=====>.....] - ETA: 7s - loss: 0.0990 - auc: 0.9922
## 161/603 [=====>.....] - ETA: 7s - loss: 0.0983 - auc: 0.9923
## 164/603 [=====>.....] - ETA: 7s - loss: 0.0982 - auc: 0.9923
## 168/603 [=====>.....] - ETA: 7s - loss: 0.0987 - auc: 0.9923
## 171/603 [=====>.....] - ETA: 7s - loss: 0.0978 - auc: 0.9924
## 174/603 [=====>.....] - ETA: 7s - loss: 0.0983 - auc: 0.9923
## 178/603 [=====>.....] - ETA: 7s - loss: 0.0975 - auc: 0.9924
## 181/603 [=====>.....] - ETA: 7s - loss: 0.0985 - auc: 0.9923
## 185/603 [=====>.....] - ETA: 7s - loss: 0.0975 - auc: 0.9924
## 189/603 [=====>.....] - ETA: 6s - loss: 0.0963 - auc: 0.9926
## 192/603 [=====>.....] - ETA: 6s - loss: 0.0953 - auc: 0.9927
## 196/603 [=====>.....] - ETA: 6s - loss: 0.0954 - auc: 0.9927
## 199/603 [=====>.....] - ETA: 6s - loss: 0.0949 - auc: 0.9928
## 202/603 [=====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9926
## 205/603 [=====>.....] - ETA: 6s - loss: 0.0972 - auc: 0.9926
## 209/603 [=====>.....] - ETA: 6s - loss: 0.0968 - auc: 0.9927
## 213/603 [=====>.....] - ETA: 6s - loss: 0.0969 - auc: 0.9927
## 217/603 [=====>.....] - ETA: 6s - loss: 0.0974 - auc: 0.9927
## 220/603 [=====>.....] - ETA: 6s - loss: 0.0975 - auc: 0.9926
## 223/603 [=====>.....] - ETA: 6s - loss: 0.0990 - auc: 0.9924
## 226/603 [=====>.....] - ETA: 6s - loss: 0.0988 - auc: 0.9924
## 229/603 [=====>.....] - ETA: 6s - loss: 0.0987 - auc: 0.9925
## 233/603 [=====>.....] - ETA: 6s - loss: 0.0980 - auc: 0.9926
## 236/603 [=====>.....] - ETA: 6s - loss: 0.0978 - auc: 0.9926
## 239/603 [=====>.....] - ETA: 6s - loss: 0.0979 - auc: 0.9926
## 242/603 [=====>.....] - ETA: 6s - loss: 0.0973 - auc: 0.9926
## 245/603 [=====>.....] - ETA: 6s - loss: 0.0988 - auc: 0.9923
## 248/603 [=====>.....] - ETA: 5s - loss: 0.0986 - auc: 0.9923
## 251/603 [=====>.....] - ETA: 5s - loss: 0.0987 - auc: 0.9924
## 255/603 [=====>.....] - ETA: 5s - loss: 0.0985 - auc: 0.9923
## 258/603 [=====>.....] - ETA: 5s - loss: 0.0999 - auc: 0.9921
## 262/603 [=====>.....] - ETA: 5s - loss: 0.0993 - auc: 0.9922
## 265/603 [=====>.....] - ETA: 5s - loss: 0.0989 - auc: 0.9923
## 268/603 [=====>.....] - ETA: 5s - loss: 0.0996 - auc: 0.9922
## 272/603 [=====>.....] - ETA: 5s - loss: 0.0994 - auc: 0.9922
## 275/603 [=====>.....] - ETA: 5s - loss: 0.1000 - auc: 0.9920
## 279/603 [=====>.....] - ETA: 5s - loss: 0.0993 - auc: 0.9921
## 282/603 [=====>.....] - ETA: 5s - loss: 0.0990 - auc: 0.9922
## 285/603 [=====>.....] - ETA: 5s - loss: 0.0988 - auc: 0.9922
## 289/603 [=====>.....] - ETA: 5s - loss: 0.0990 - auc: 0.9922
## 292/603 [=====>.....] - ETA: 5s - loss: 0.0987 - auc: 0.9922
## 295/603 [=====>.....] - ETA: 5s - loss: 0.0984 - auc: 0.9923
## 299/603 [=====>.....] - ETA: 5s - loss: 0.0979 - auc: 0.9923
## 302/603 [=====>.....] - ETA: 5s - loss: 0.0973 - auc: 0.9924
## 305/603 [=====>.....] - ETA: 5s - loss: 0.0979 - auc: 0.9923
## 309/603 [=====>.....] - ETA: 4s - loss: 0.0981 - auc: 0.9923
## 312/603 [=====>.....] - ETA: 4s - loss: 0.0981 - auc: 0.9923
## 315/603 [=====>.....] - ETA: 4s - loss: 0.0978 - auc: 0.9923
## 319/603 [=====>.....] - ETA: 4s - loss: 0.0972 - auc: 0.9924
## 322/603 [=====>.....] - ETA: 4s - loss: 0.0974 - auc: 0.9924
## 325/603 [=====>.....] - ETA: 4s - loss: 0.0972 - auc: 0.9924
## 328/603 [=====>.....] - ETA: 4s - loss: 0.0977 - auc: 0.9924
## 331/603 [=====>.....] - ETA: 4s - loss: 0.0975 - auc: 0.9924
## 335/603 [=====>.....] - ETA: 4s - loss: 0.0971 - auc: 0.9925
## 339/603 [=====>.....] - ETA: 4s - loss: 0.0969 - auc: 0.9925
## 342/603 [=====>.....] - ETA: 4s - loss: 0.0963 - auc: 0.9926
```

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## 346/603 [=====>.....] - ETA: 4s - loss: 0.0958 - auc: 0.9927
## 349/603 [=====>.....] - ETA: 4s - loss: 0.0955 - auc: 0.9927
## 352/603 [=====>.....] - ETA: 4s - loss: 0.0953 - auc: 0.9928
## 356/603 [=====>.....] - ETA: 4s - loss: 0.0949 - auc: 0.9928
## 359/603 [=====>.....] - ETA: 4s - loss: 0.0946 - auc: 0.9929
## 363/603 [=====>.....] - ETA: 4s - loss: 0.0944 - auc: 0.9929
## 366/603 [=====>.....] - ETA: 3s - loss: 0.0941 - auc: 0.9930
## 369/603 [=====>.....] - ETA: 3s - loss: 0.0941 - auc: 0.9930
## 373/603 [=====>.....] - ETA: 3s - loss: 0.0938 - auc: 0.9930
## 376/603 [=====>.....] - ETA: 3s - loss: 0.0936 - auc: 0.9931
## 379/603 [=====>.....] - ETA: 3s - loss: 0.0941 - auc: 0.9930
## 382/603 [=====>.....] - ETA: 3s - loss: 0.0944 - auc: 0.9930
## 385/603 [=====>.....] - ETA: 3s - loss: 0.0946 - auc: 0.9930
## 388/603 [=====>.....] - ETA: 3s - loss: 0.0943 - auc: 0.9930
## 391/603 [=====>.....] - ETA: 3s - loss: 0.0942 - auc: 0.9930
## 394/603 [=====>.....] - ETA: 3s - loss: 0.0938 - auc: 0.9931
## 397/603 [=====>.....] - ETA: 3s - loss: 0.0939 - auc: 0.9931
## 400/603 [=====>.....] - ETA: 3s - loss: 0.0942 - auc: 0.9930
## 403/603 [=====>.....] - ETA: 3s - loss: 0.0940 - auc: 0.9930
## 406/603 [=====>.....] - ETA: 3s - loss: 0.0939 - auc: 0.9930
## 409/603 [=====>.....] - ETA: 3s - loss: 0.0937 - auc: 0.9931
## 412/603 [=====>.....] - ETA: 3s - loss: 0.0936 - auc: 0.9931
## 415/603 [=====>.....] - ETA: 3s - loss: 0.0937 - auc: 0.9931
## 418/603 [=====>.....] - ETA: 3s - loss: 0.0938 - auc: 0.9931
## 421/603 [=====>.....] - ETA: 3s - loss: 0.0935 - auc: 0.9931
## 425/603 [=====>.....] - ETA: 3s - loss: 0.0932 - auc: 0.9932
## 428/603 [=====>.....] - ETA: 2s - loss: 0.0934 - auc: 0.9931
## 432/603 [=====>.....] - ETA: 2s - loss: 0.0929 - auc: 0.9932
## 435/603 [=====>.....] - ETA: 2s - loss: 0.0935 - auc: 0.9931
## 438/603 [=====>.....] - ETA: 2s - loss: 0.0936 - auc: 0.9931
## 442/603 [=====>.....] - ETA: 2s - loss: 0.0932 - auc: 0.9932
## 445/603 [=====>.....] - ETA: 2s - loss: 0.0929 - auc: 0.9932
## 448/603 [=====>.....] - ETA: 2s - loss: 0.0930 - auc: 0.9932
## 451/603 [=====>.....] - ETA: 2s - loss: 0.0928 - auc: 0.9932
## 455/603 [=====>.....] - ETA: 2s - loss: 0.0924 - auc: 0.9933
## 458/603 [=====>.....] - ETA: 2s - loss: 0.0921 - auc: 0.9934
## 461/603 [=====>.....] - ETA: 2s - loss: 0.0919 - auc: 0.9934
## 464/603 [=====>.....] - ETA: 2s - loss: 0.0922 - auc: 0.9933
## 467/603 [=====>.....] - ETA: 2s - loss: 0.0920 - auc: 0.9934
## 471/603 [=====>.....] - ETA: 2s - loss: 0.0923 - auc: 0.9933
## 474/603 [=====>.....] - ETA: 2s - loss: 0.0925 - auc: 0.9933
## 477/603 [=====>.....] - ETA: 2s - loss: 0.0924 - auc: 0.9933
## 481/603 [=====>.....] - ETA: 2s - loss: 0.0922 - auc: 0.9934
## 484/603 [=====>.....] - ETA: 2s - loss: 0.0920 - auc: 0.9934
## 488/603 [=====>.....] - ETA: 1s - loss: 0.0920 - auc: 0.9934
## 492/603 [=====>.....] - ETA: 1s - loss: 0.0922 - auc: 0.9934
## 496/603 [=====>.....] - ETA: 1s - loss: 0.0922 - auc: 0.9934
## 500/603 [=====>.....] - ETA: 1s - loss: 0.0918 - auc: 0.9934
## 503/603 [=====>.....] - ETA: 1s - loss: 0.0916 - auc: 0.9935
## 507/603 [=====>.....] - ETA: 1s - loss: 0.0920 - auc: 0.9933
## 511/603 [=====>.....] - ETA: 1s - loss: 0.0920 - auc: 0.9933
## 514/603 [=====>.....] - ETA: 1s - loss: 0.0920 - auc: 0.9934
## 517/603 [=====>.....] - ETA: 1s - loss: 0.0921 - auc: 0.9933
## 520/603 [=====>.....] - ETA: 1s - loss: 0.0922 - auc: 0.9933
## 523/603 [=====>....] - ETA: 1s - loss: 0.0918 - auc: 0.9934
## 526/603 [=====>....] - ETA: 1s - loss: 0.0916 - auc: 0.9934
```

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## 530/603 [=====>...] - ETA: 1s - loss: 0.0918 - auc: 0.9934
## 534/603 [=====>...] - ETA: 1s - loss: 0.0920 - auc: 0.9934
## 538/603 [=====>...] - ETA: 1s - loss: 0.0916 - auc: 0.9934
## 541/603 [=====>...] - ETA: 1s - loss: 0.0919 - auc: 0.9934
## 544/603 [=====>...] - ETA: 0s - loss: 0.0921 - auc: 0.9934
## 547/603 [=====>...] - ETA: 0s - loss: 0.0921 - auc: 0.9934
## 551/603 [=====>...] - ETA: 0s - loss: 0.0922 - auc: 0.9934
## 554/603 [=====>...] - ETA: 0s - loss: 0.0922 - auc: 0.9934
## 557/603 [=====>...] - ETA: 0s - loss: 0.0920 - auc: 0.9934
## 560/603 [=====>...] - ETA: 0s - loss: 0.0921 - auc: 0.9934
## 563/603 [=====>..] - ETA: 0s - loss: 0.0922 - auc: 0.9934
## 567/603 [=====>..] - ETA: 0s - loss: 0.0925 - auc: 0.9934
## 570/603 [=====>..] - ETA: 0s - loss: 0.0922 - auc: 0.9934
## 574/603 [=====>..] - ETA: 0s - loss: 0.0923 - auc: 0.9934
## 577/603 [=====>..] - ETA: 0s - loss: 0.0925 - auc: 0.9934
## 581/603 [=====>..] - ETA: 0s - loss: 0.0926 - auc: 0.9934
## 584/603 [=====>.] - ETA: 0s - loss: 0.0926 - auc: 0.9934
## 588/603 [=====>.] - ETA: 0s - loss: 0.0923 - auc: 0.9934
## 592/603 [=====>.] - ETA: 0s - loss: 0.0921 - auc: 0.9935
## 595/603 [=====>.] - ETA: 0s - loss: 0.0920 - auc: 0.9935
## 599/603 [=====>.] - ETA: 0s - loss: 0.0919 - auc: 0.9935
## 602/603 [=====>.] - ETA: 0s - loss: 0.0919 - auc: 0.9935
## 603/603 [=====] - 11s 18ms/step - loss: 0.0920 - auc: 0.9935 - va
l_loss: 0.1485 - val_auc: 0.9882
```

12. Plot training and validation AUC:

```
##### Turn to comments from the second run onwards #####
# Plot training and validation loss and AUC:
plot(history)
```



13. Using the saved best model weights, evaluate the model's AUC on training and test set:

```
# Load the saved best model weights to the model
weights_filepath <- "malaria_best_model_weights.h5"
my_model %>% load_model_weights_tf(weights_filepath)

### Evaluate the model on training set:
my_model %>% evaluate(X_train_, y_train, verbose = 0) # Compute training Loss and AUC
```

```
##      loss      auc
## 0.08284641 0.99569279
```

```
### Evaluate the model on test set:
y_test_pred <- predict(my_model, X_test_, verbose = 0)
y_test_pred <- as.vector(y_test_pred)

my_model %>% evaluate(X_test_, y_test, verbose = 0) # Compute test Loss and AUC
```

```
##      loss      auc
## 0.1156334 0.9911484
```

B/ Choosing decision threshold by ROC curve and decision curve:

1. Criterias to determine the decision threshold for the CNN model:

- **Youden index criterion:**

```
# Optimal threshold obtained by maximizing youden index
j_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  j <- tpr - fpr
  j <- sort(j, decreasing = TRUE, index.return = TRUE)
  j_ordered <- data.frame(j = j[1], thresholds = thresholds[rank(j$ix)])
  for (k in 1:nrow(j_ordered)){
    if (is.finite(j_ordered$thresholds[k])) {
      return(j_ordered$thresholds[k])
      break
    }
  }
}

cj <- j_threshold(y_test, y_test_pred)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
cj
```

```
## [1] 0.3763034
```

- **Closest to (0,1) criterion:**

```
# Optimal threshold obtained by maximizing sensitivity and specificity to the extent that bot
h are closest to 100%
d_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  d <- sapply(1:length(fpr), function(i) sqrt((1 - tpr[i])^2 + (fpr[i])^2))
  d <- sort(d, decreasing = FALSE, index.return = TRUE)
  d_ordered <- data.frame(d = d[1], thresholds = thresholds[rank(d$ix)])
  for (k in 1:nrow(d_ordered)){
    if (is.finite(d_ordered$thresholds[k])) {
      return(d_ordered$thresholds[k])
      break
    }
  }
}

cd <- d_threshold(y_test, y_test_pred)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
cd
```

```
## [1] 0.2996681
```

- **Symetric point criterion:**

```
# Optimal threshold obtained by finding the point where sensitivity and specificity are equivalent
s_threshold <- function(y_true, y_pred) {
  roc_obj <- pROC::roc(y_true, y_pred)
  fpr <- 1 - roc_obj$specificities
  tpr <- roc_obj$sensitivities
  thresholds <- roc_obj$thresholds
  s <- abs(fpr + tpr - 1)
  s <- sort(s, decreasing = FALSE, index.return = TRUE)
  s_ordered <- data.frame(s = s[1], thresholds = thresholds[rank(s$ix)])
  for (k in 1:nrow(s_ordered)){
    if (is.finite(s_ordered$thresholds[k])) {
      return(s_ordered$thresholds[k])
      break
    }
  }
}

cs <- s_threshold(y_test, y_test_pred)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
cs
```

```
## [1] 0.2412464
```

2. Plot ROC curve, decision curve and optimal thresholds for the CNN model:

```
# Plot ROC curve:
roc_obj <- pROC::roc(y_test, y_test_pred)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```

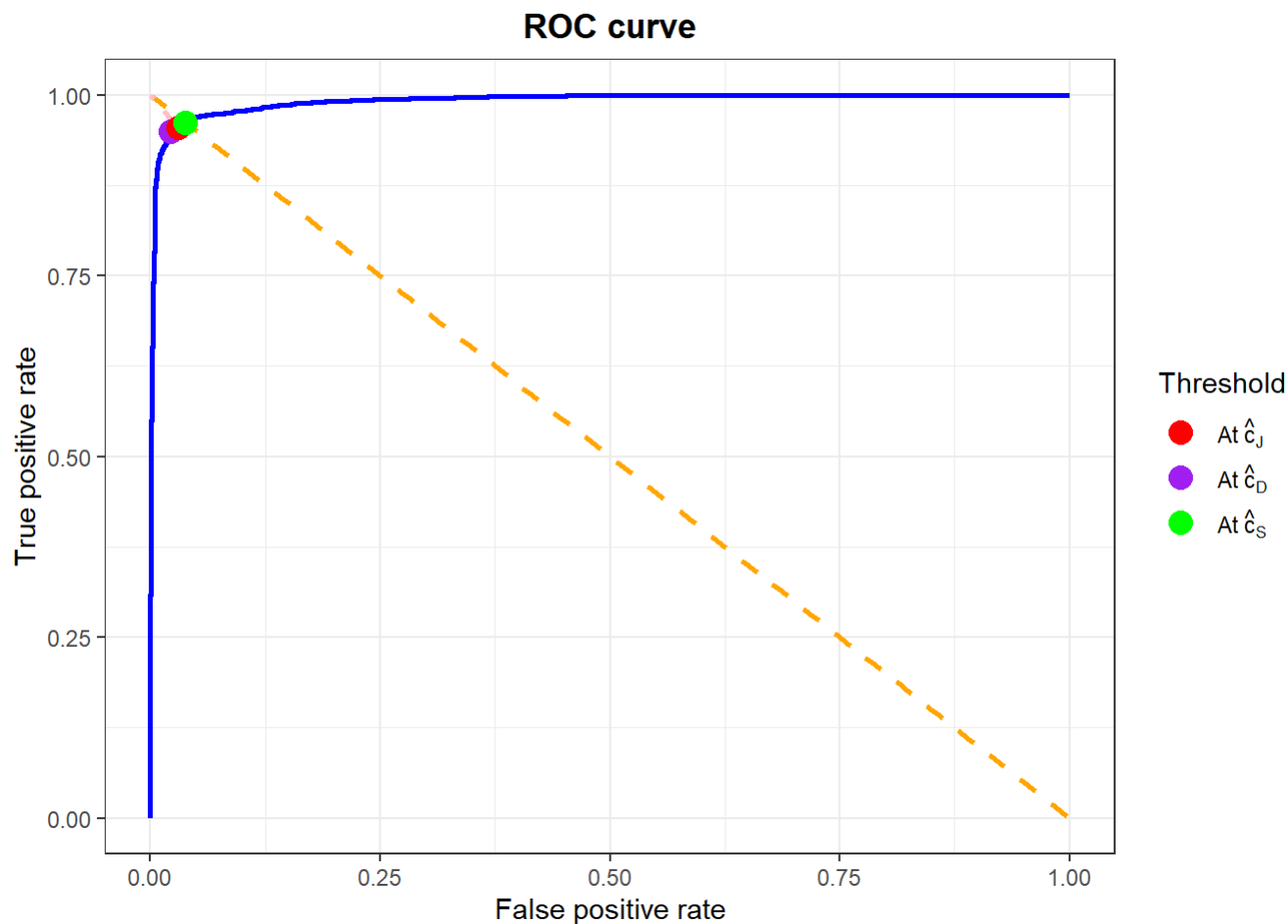
thresholds_cnn <- roc_obj$thresholds
roc_data <- data.frame(fpr = rev(1 - roc_obj$specificities), tpr = rev(roc_obj$sensitivitie
s))

for (i in 1:length(roc_obj$thresholds)){
  if (roc_obj$thresholds[i] == cj){
    specificity_j <- roc_obj$specificities[i]
    sensitivity_j <- roc_obj$sensitivities[i]
  }
  if (roc_obj$thresholds[i] == cd){
    specificity_d <- roc_obj$specificities[i]
    sensitivity_d <- roc_obj$sensitivities[i]
  }
  if (roc_obj$thresholds[i] == cs){
    specificity_s <- roc_obj$specificities[i]
    sensitivity_s <- roc_obj$sensitivities[i]
  }
}

diagonal_data <- data.frame(x = seq(0, 1, length.out = 100), y = 1 - seq(0, 1, length.out = 1
00))
thresholds_data <- data.frame(x1 = 0, x2 = 1 - specificity_d, y1 = 1, y2 = sensitivity_d)
sp_se_j <- data.frame(sp = specificity_j, se = sensitivity_j)
sp_se_d <- data.frame(sp = specificity_d, se = sensitivity_d)
sp_se_s <- data.frame(sp = specificity_s, se = sensitivity_s)

ggplot(data = roc_data, aes(x = fpr, y = tpr)) +
  geom_line(linetype = "solid", linewidth = 1, col = "blue") +
  geom_line(data = diagonal_data, aes(x = x, y = y), linetype = "dashed", linewidth = 1, col
= "orange") +
  geom_segment(data = thresholds_data, aes(x = x1, y = y1, xend = x2, yend = y2), linetype =
"dotdash", linewidth = 1, col = "pink") +
  geom_point(data = sp_se_j, aes(x = 1 - sp, y = se, col = "c_j"), size = 4, show.legend = TR
UE) +
  geom_point(data = sp_se_d, aes(x = 1 - sp, y = se, col = "c_d"), size = 4, show.legend = TR
UE) +
  geom_point(data = sp_se_s, aes(x = 1 - sp, y = se, col = "c_s"), size = 4, show.legend = TR
UE) +
  labs(title = "ROC curve", x = "False positive rate", y = "True positive rate") +
  scale_color_manual(name = "Threshold", labels = unname(TeX(c("At  $\hat{c}_{J}$ ", "At  $\hat{c}_{D}$ ", "At  $\hat{c}_{S}$ "))), values=c("red", "purple", "green")) +
  theme_bw() +
  theme(plot.title = element_text(face="bold", hjust = 0.5))

```



```
# Plot decision curve

# Net benefit function
nb <- function(y_true, y_pred, c, prev){
  roc_obj <- pROC::roc(y_true, y_pred)
  for (i in 1:length(roc_obj$thresholds)){
    if (roc_obj$thresholds[i] == c){
      se_c <- roc_obj$sensitivities[i]
      sp_c <- roc_obj$specificities[i]
      break
    }
  }
  phi_c <- se_c * prev - (1 - sp_c) * (1 - prev) * c/(1 - c)
  return(phi_c)
}

# Calculate disease prevalence in the population:
p = sum(y_test)/length(y_test)

df_cnn <- data.frame(y_test_pred, y_test)
nb_j <- data.frame(cj = cj, nbj = nb(y_test, y_test_pred, cj, p))
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
nb_d <- data.frame(cd = cd, nbd = nb(y_test, y_test_pred, cd, p))
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

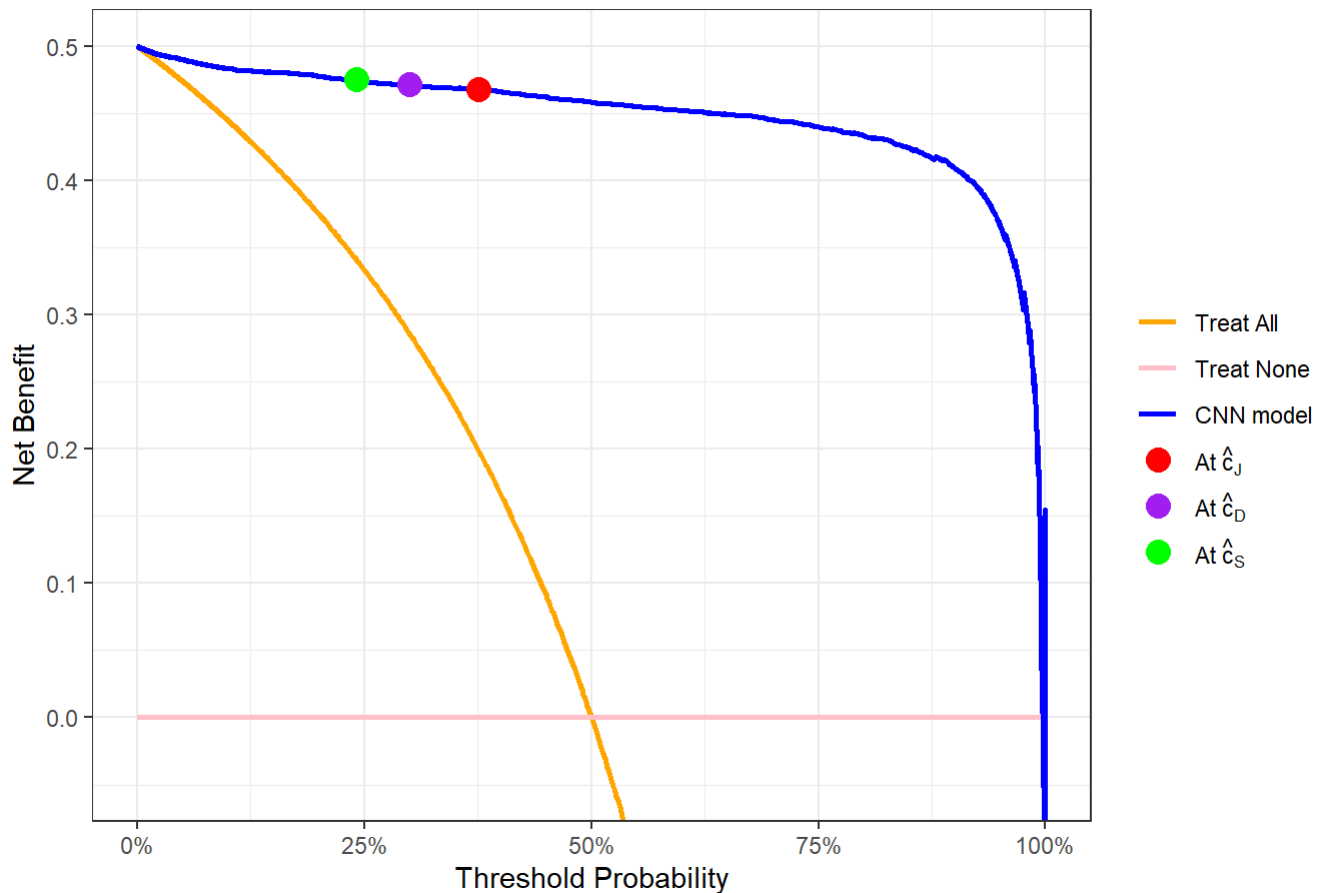
```
nb_s <- data.frame(cs = cs, nbs = nb(y_test, y_test_pred, cs, p))
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```
# For the Decision curve plot, use the thresholds values from ROC curve plot to make sure that the optimal thresholds show on the decision curve correctly
dcurves::dca(y_test ~ y_test_pred, data = df_cnn, label = list(y_test_pred = "CNN model"), thresholds = thresholds_cnn, prevalence = p) %>%
  plot(smooth = FALSE) + geom_line(linewidth = 1) + labs(title = "Decision curve") +
  geom_point(data = nb_j, aes(x = cj, y = nbj, col = "c_j"), size = 4, show.legend = TRUE) +
  geom_point(data = nb_d, aes(x = cd, y = nbd, col = "c_d"), size = 4, show.legend = TRUE) +
  geom_point(data = nb_s, aes(x = cs, y = nbs, col = "c_s"), size = 4, show.legend = TRUE) +
  scale_color_manual(labels = unname(c("Treat All", "Treat None", "CNN model", "At  $\hat{c}_J$ ", "At  $\hat{c}_D$ ", "At  $\hat{c}_S$ ")), values=c("orange", "pink", "blue", "red", "purple", "green")) +
  guides(colour = guide_legend(override.aes = list(linetype = c("solid", "solid", "solid", "solid", "solid", "solid"), shape = c(NA, NA, NA, rep(16, 3)))))) +
  theme(plot.title = element_text(face="bold", hjust = 0.5))
```

```
## Assuming '1' is [Event] and '0' is [non-Event]
```

Decision curve



```
# Finding the optimal threshold
nb_j <- nb(y_test, y_test_pred, cj, p)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
nb_d <- nb(y_test, y_test_pred, cd, p)
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```
nb_s <- nb(y_test, y_test_pred, cs, p)
```

```
## Setting levels: control = 0, case = 1
## Setting direction: controls < cases
```

```
nb_values <- c(cj = nb_j, cd = nb_d, cs = nb_s)

best_threshold <- names(which.max(nb_values)) # Name of the threshold with highest net bene
fit
threshold_map <- list(cj = cj, cd = cd, cs = cs)
c_res <- threshold_map[[best_threshold]] # Get the optimal threshold's value
cat("Optimal threshold:", best_threshold, "=", c_res)
```

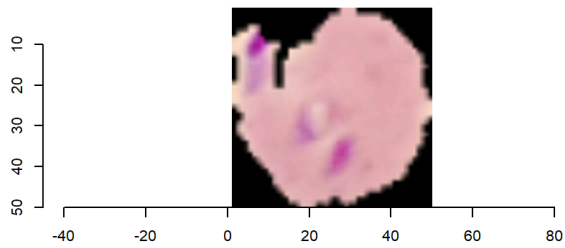
```
## Optimal threshold: cs = 0.2412464
```

```
## Example: Predicting the class of a new data point with CNN model
new_obs_ <- X_test[2:7] # Some observations from the test set
new_obs <- do.call(rbind, new_obs_)
new_obs <- array_reshape(new_obs, c(dim(new_obs)[1], 50, 50, 3), order = "F")
# In the procedure: You need to preprocess (Convert into the correct format, Resize, Normalize, Reshape, ... in the same method and with the same values calculated from the preprocessing step of the training set) the new data if needed
actual_classes <- y_test[2:7]
class_preds <- c()

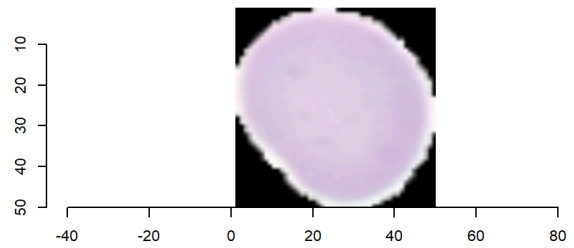
y_pred <- predict(my_model, new_obs, verbose = 0)
y_pred = round(y_pred, digits = 7)
class_preds <- ifelse(y_pred >= c_res, "Parasitized", "Uninfected")
actual_classes <- ifelse(actual_classes == 1, "Parasitized", "Uninfected")

# Final result
par(mfrow = c(3, 2)) # Create subplots with 2 rows and 3 columns

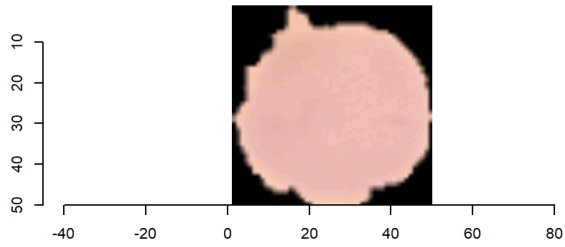
for (i in 1:nrow(new_obs)){
  suppressWarnings(plot(as.cimg(new_obs[i,,,])))
  title(main = paste0("Observation ", i, "\n Predicted class: ", class_preds[i], "; Actual class: ", actual_classes[i]), sub = paste0("Predicted probability: ", y_pred[i]), cex.main = 1.4, cex.sub = 1.4)
}
```

Observation 1**Predicted class: Parasitized; Actual class: Parasitized**

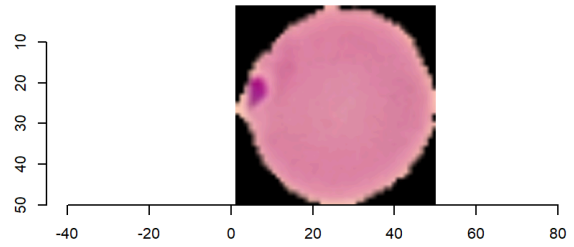
Predicted probability: 0.9999936

Observation 2**Predicted class: Uninfected; Actual class: Uninfected**

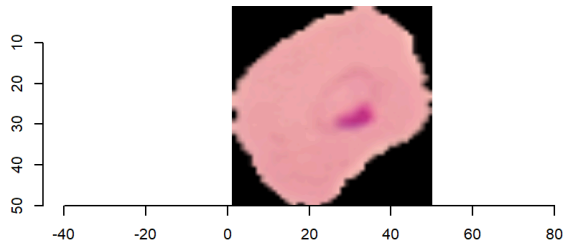
Predicted probability: 0.0019037

Observation 3**Predicted class: Uninfected; Actual class: Uninfected**

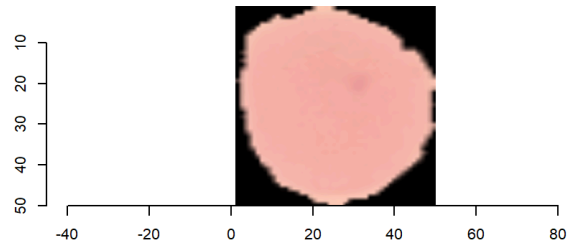
Predicted probability: 0.0031326

Observation 4**Predicted class: Parasitized; Actual class: Parasitized**

Predicted probability: 0.9999188

Observation 5**Predicted class: Parasitized; Actual class: Parasitized**

Predicted probability: 0.9383187

Observation 6**Predicted class: Uninfected; Actual class: Uninfected**

Predicted probability: 0.005273